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Electric Vehicle System Engineering

UNIT 4: EV Modelling and Simulation

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Unit 4: EV Modelling and Simulation

4.1 Introduction to EV Modelling and Simulation

4.1.1 Introduction to EV System Modelling

An electric vehicle is a coupled system. Electrical energy leaves the battery, passes through an inverter and motor, turns into torque at the wheels and finally appears as motion, aerodynamic heat, tyre heat and a small amount of stored kinetic and potential energy. A change anywhere in this chain affects everything else. Heavier batteries add range but also add road load; a higher gear ratio improves launch torque but raises motor speed at cruise and may push the motor into a less efficient region. A **model** is a set of equations and data that captures these interactions, and a **simulation** is the numerical solution of that model for a chosen scenario [1, 2].

Units 2 and 3 described the battery, power converters and machines individually. This unit joins them into complete vehicle models, and uses those models to answer questions such as: how much energy does a car need on a test cycle, how much of the braking energy can be recovered, what size of motor and battery will meet a specification, and how far can the car travel?

4.1.2 Importance of EV Modelling and Simulation

Prototype testing is slow and expensive, and a chassis dynamometer can only test hardware that already exists. Simulation is valuable for these reasons:

- **Cost and time.** Thousands of battery, motor and gear combinations can be screened in the time it takes to build one prototype.
- **Early decisions.** The most expensive choices, such as pack voltage, cell format and drive layout, are made before hardware exists.
- **Safe testing.** Faults, thermal runaway precursors and extreme braking can be explored without risk to people or equipment.
- **Repeatability.** A simulated test is exactly repeatable, so the effect of one parameter can be isolated.
- **Control development.** Algorithms can be tested against a virtual vehicle before they are loaded on an ECU.

4.1.3 Objectives of EV Simulation

A simulation study always starts with a stated objective, because that objective decides which model is adequate. Typical objectives are to predict energy consumption and range over a

drive cycle; to size the battery, motor and inverter; to evaluate acceleration, top speed and gradeability; to quantify regenerative energy recovery; to develop and tune control strategies; and to estimate component stresses such as peak current, temperature and duty.

4.1.4 Modelling Terms: Definitions and Differences

Six terms are often confused. Their meanings are distinct, and Table 4.1 summarises them.

1. **Mathematical modelling** expresses system behaviour as equations, for example $\delta m\dot{v} = F_t - F_{\text{res}}$. It is an abstraction: the equations hold only under stated assumptions.
2. **Physical modelling** builds a model by connecting components that exchange energy, such as a battery connected to an inverter through electrical ports. The tool writes the equations from the topology. Simscape is a common example [34].
3. **Simulation** is the numerical solution of a model over time for specified inputs and initial conditions.
4. **System identification** estimates model parameters from measured data, for example fitting R_0, R_1 and C_1 to a current pulse test (Unit 2) or fitting road-load coefficients to a coast-down test.
5. **Model verification** asks: *did we solve the equations correctly?* It checks code, units, solver accuracy and energy balance.
6. **Model validation** asks: *did we build the right model?* It compares predictions with independent measurements over the intended operating range.

Table 4.1: Modelling terms and the question each answers

Term	Question answered	EV example
Mathematical modelling	Which equations describe it?	Road-load force equation
Physical modelling	How are components connected?	Simscape battery–inverter–motor network
Simulation	What happens for this input?	Battery SoC over WLTC
System identification	What are the parameter values?	C_dA from a coast-down test
Verification	Is the model solved correctly?	Energy balance closes to 0.1%
Validation	Does it match reality?	Measured Wh/km within 3% of prediction

The table shows why verification and validation must not be merged. A perfectly verified model can still be invalid if an important loss was left out, and a model can match one test by accident while containing a coding error that cancels another error.

4.1.5 Physical, Mathematical and Dynamic System Modelling

Figure 4.1 places each physical subsystem beside its mathematical representation. Reading across each row shows the act of idealisation: a pack containing hundreds of cells becomes a voltage source behind a resistance, a gear train becomes a ratio and an efficiency, and the whole vehicle body becomes a single point mass. These equations reappear throughout this unit. The figure is important because it shows that every model is a deliberate simplification, chosen so that it keeps the effects that matter for the question and discards the rest.

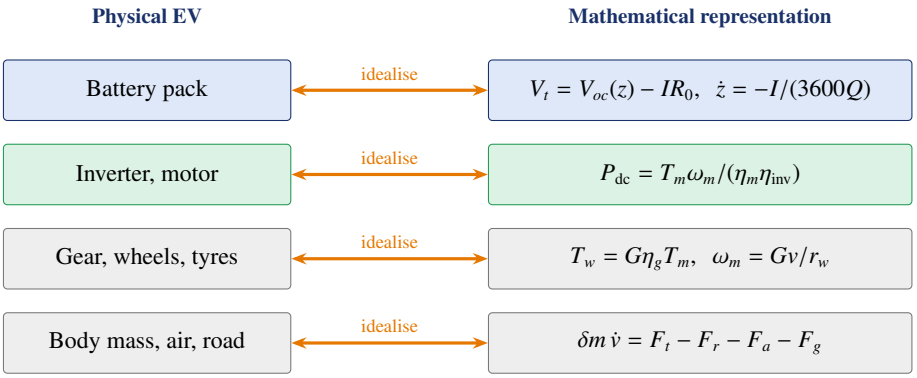


Figure 4.1: Physical EV system and its mathematical representation.

A model is **static** if its output depends only on the present input, for example a motor efficiency map. It is **dynamic** if it has memory, written as differential equations with state variables. In general form,

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t), \mathbf{p}), \qquad \mathbf{y}(t) = \mathbf{g}(\mathbf{x}(t), \mathbf{u}(t), \mathbf{p}), \tag{4.1}$$

where $\mathbf{x}$ is the state vector (vehicle speed v in m/s, state of charge z, polarisation voltage V_1 in V and so on), $\mathbf{u}$ is the input vector, $\mathbf{p}$ is the parameter vector and $\mathbf{y}$ is the output vector. The states hold the system's energy storage, which is why a model of an EV usually needs at least v (kinetic energy) and z (electrochemical energy). In a numerical solution, (4.1) is stepped forward with a solver. The simplest is forward Euler,

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \Delta t\, \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k, \mathbf{p}), \tag{4.2}$$

with step Δt in s. It is accurate and stable only if Δt is small relative to the fastest time constant.

4.1.6 Simulation-Based Design and the Simulation Workflow

Simulation-based design uses models as the main design tool, with hardware tests reserved for confirmation. Figure 4.2 shows the workflow. The objective is defined first, parameters are collected, the model is built and its code verified. It is then validated against data. A mismatch sends the work back to parameter identification, rather than forward to conclusions. Scenarios are run only after that, and the analysis either refines the design (returning to the model) or releases it for prototype test. The loops in the figure are the essential point: a simulation is a process of iteration, not a single calculation.

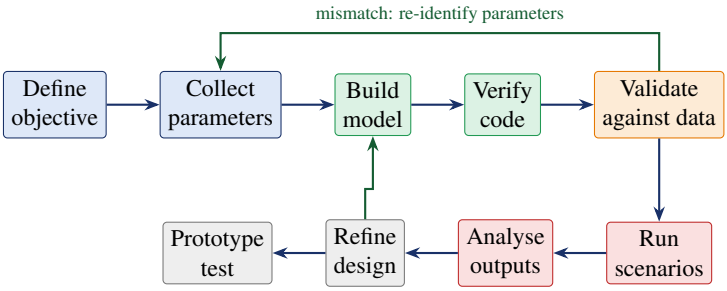


Figure 4.2: EV simulation workflow.

4.1.7 Model Complexity and Computational Requirements

More detail is not automatically better. Model fidelity should match the objective, since each extra level of detail needs more parameters and more computing time. Table 4.2 shows typical fidelity levels with the time scale each can resolve.

Table 4.2: Model fidelity levels for an EV inverter and motor

Level	Time step	Speed	Used for
Efficiency map	0.1–1 s	Much faster than real time	Energy, range, sizing
Averaged dq model	10–100 μs	Near real time	Torque control tuning
Switching model	10–100 ns	Far slower than real time	Ripple, losses, EMI

The table shows a range of six to seven orders of magnitude in time step. A 1800 s range simulation with a switching model would be impractical and would give no better energy answer, while an efficiency map cannot show current ripple. Stiff systems, which mix fast and slow time constants, should use a variable-step implicit solver or separate fast dynamics from slow ones.

4.1.8 System-Level, Component-Level and Vehicle-Level Modelling

Figure 4.3 shows the modelling hierarchy. **Component-level** models describe one device in detail, such as a cell or a machine. **System-level** models join these devices into subsystems such as the powertrain, thermal loop or brake system. A **vehicle-level** model joins all subsystems to predict quantities only the whole vehicle has, such as range and 0–100 km/h time. The vertical arrows carry parameters upward (for example, a cell's resistance becoming a pack resistance) and demands downward (for example, a torque request becoming a current in each cell). Good practice is to validate each level separately before validating the assembled vehicle.

4.1.9 Energy-Based, Power-Based and Control-Oriented Modelling

Three modelling viewpoints appear repeatedly.

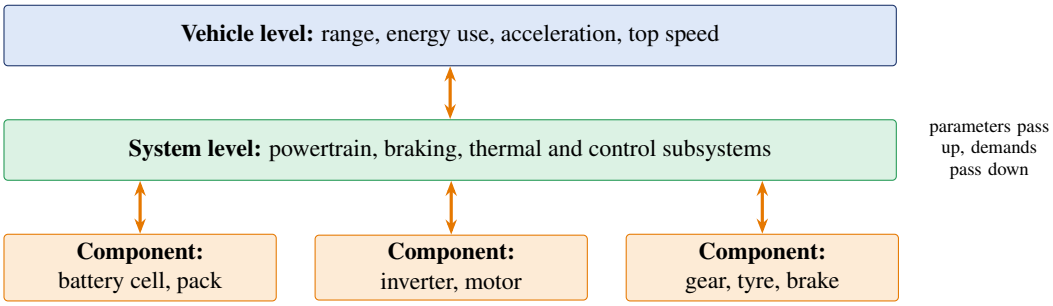


Figure 4.3: Component-level, system-level and vehicle-level modelling.

- **Energy-based modelling** integrates power over a whole cycle and works with stored and dissipated energy, such as Wh/km. It is suited to range and battery sizing.
- **Power-based modelling** works with instantaneous power at each interface, $P = T\omega$ or $P = VI$, which reveals peak demands and suits component ratings.
- **Control-oriented modelling** keeps only the dynamics that matter for a controller and puts the model in a compact form such as a transfer function or linear state-space form. A speed-loop model of the vehicle might be just $1/(\delta ms)$.

These views are consistent, as they are related by $E = \int P\,dt$, and they differ only in what they emphasise.

4.1.10 The Complete EV Modelling Framework

Figure 4.4 assembles the blocks developed in this unit. The cycle or driver sets the demand. The control strategy divides that demand between motor and brake models. The inverter and motor models draw power from the battery model, then pass torque through the transmission to the chassis. Green feedback lines return speed to the controller, while energy, range and SoC come out as results. Each block has its own section: control in Section 4.7, brakes in Section 4.6, chassis in Section 4.8 and battery in Section 4.10. The figure therefore serves as a map of the unit.

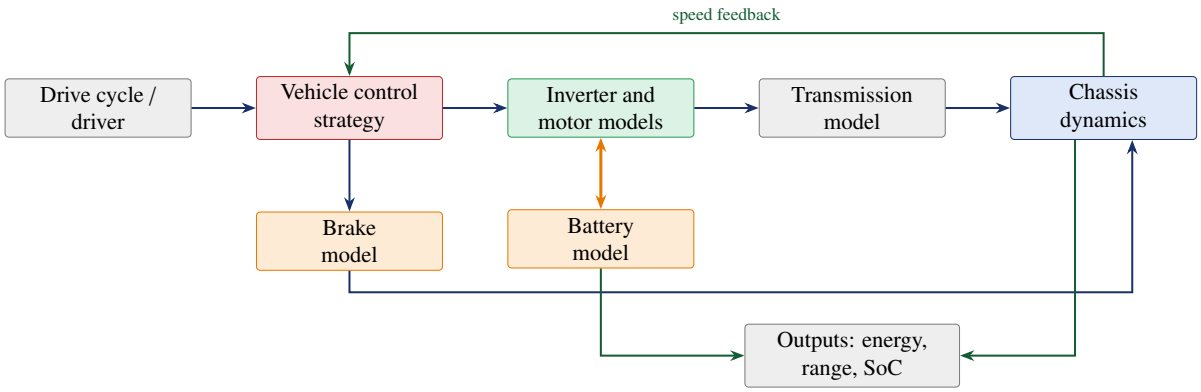


Figure 4.4: Complete EV modelling framework.

4.1.11 Model Validation and Verification

Verification methods include unit tests of each block, dimensional checks, comparison of solver step sizes, and the energy-balance test: battery energy must equal the sum of road-work energy, all losses and auxiliary consumption. Validation methods compare predictions with measured speed, current, voltage and SoC on a dynamometer or road. A common error metric is the root-mean-square error,

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}(y_k - \hat{y}_k)^2}, \tag{4.3}$$

with y_k the measurement and $\hat{y}_k$ the prediction, in the units of y. Validation data must be independent of the data used in system identification, otherwise the test merely repeats the fit.

4.1.12 Simulation Tools Used in EV Engineering

Several tools are in common use. MATLAB and Simulink are the most widespread for model-based design [32]. Powertrain Blockset provides pre-built drive-cycle and vehicle reference models [33], Simscape Electrical provides physical network models, and Stateflow provides state machines. NREL's ADVISOR was an early public vehicle simulator that popularised the combined backward/forward approach [11, 12], and NREL's FASTSim is a fast open tool for energy and range studies [13]. Commercial platforms such as GT-SUITE, AVL Cruise and Altair tools provide integrated multi-physics environments.

4.1.13 Introduction to MATLAB and Simulink for EV Modelling

MATLAB is a text-based numerical environment, suitable for parameter scripts, post-processing and calculations over arrays of samples, as in a backward-looking model. Simulink is a graphical block-diagram environment, in which signals flow between blocks and a solver advances the states. It suits closed-loop dynamic models such as a forward-looking model. Practice usually combines the two, as in the conceptual architecture of Figure 4.5: a script loads vehicle parameters and the drive cycle into the workspace, the Simulink model runs subsystems for driver and VCU, powertrain, battery and vehicle, and logged signals return to MATLAB for plots and reports. All Simulink diagrams in these notes are conceptual block diagrams drawn by hand; they are not screenshots of working software models.

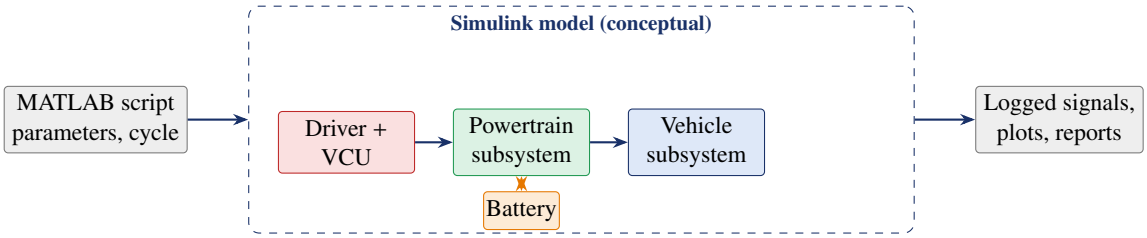


Figure 4.5: MATLAB/Simulink-based EV modelling architecture (conceptual).

4.1.14 Integrated EV Simulation Platforms

An integrated platform connects several domains in one environment: electrical, mechanical, thermal and control. Its value lies in capturing cross-coupling, such as a hot battery reducing the power limit, which in turn slows the car, which in turn changes the energy used. Co-simulation links separate specialist tools through an interface such as the Functional Mock-up Interface. As a design matures, the same models move to software-in-the-loop and hardware-in-the-loop testing, with the real ECU code running against a real-time vehicle model.

4.1.15 Applications of EV Simulation

- **Vehicle design.** Trade-offs between mass, drag and rolling resistance can be expressed in Wh/km.
- **Battery sizing.** The cycle consumption and range target fix usable energy; peak power fixes C-rate (Sections 4.9 and 4.11).
- **Motor selection.** Operating points on the torque-speed plane show which machine suits the duty (Section 4.11).
- **Inverter selection.** Peak and rms current follow from torque demand and voltage.
- **Energy consumption prediction.** This is the main output of a drive-cycle simulation (Section 4.3).
- **Range estimation.** Consumption and usable energy together give range (Section 4.15).
- **Regenerative braking analysis.** Simulation shows how much braking energy reaches the battery after limits and losses (Section 4.6).
- **Control strategy development.** Controllers are tuned against a virtual vehicle (Sections 4.7 and 4.13).
- **Performance optimisation.** Parameters are swept, or optimised, against a cost such as lifetime energy cost.

4.1.16 Reference Vehicle Used in This Unit

For consistency, all simulations and most worked examples use the **hypothetical** reference BEV of Table 4.3. Its values are realistic for a compact car, but it is not a commercial model, and none of the numbers are measured specifications. Simulated results in later sections come from the author's backward-looking and forward-looking calculations using these parameters and constant efficiencies. They are illustrative.

Two sign conventions hold in every section. Battery current I and battery power P_b are positive for **discharge**, as in Unit 2, so charging during regeneration makes them negative. Vehicle speed v is in m/s in all equations, and km/h values must be divided by 3.6 first.

Table 4.3: Hypothetical reference BEV parameters used in this unit

Parameter	Symbol	Value
Test mass	m	1500 kg
Rotational mass factor	δ	1.05
Rolling resistance coefficient	C_{rr}	0.012
Drag coefficient, frontal area	C_d, A	0.30, 2.2 m^2
Air density, gravity	ρ, g	1.2 kg/m^3, 9.81 m/s^2
Wheel radius, gear ratio	r_w, G	0.31 m, 9
Gear, motor, inverter efficiency	$\eta_g, \eta_m, \eta_{\text{inv}}$	0.97, 0.93, 0.97 (so $\eta_{\text{dt}} = 0.875$)
Auxiliary load	P_{aux}	600 W unless stated
Motor peak (30 s)	$T_{\text{pk}}, P_{\text{pk}}$	280 N m, 110 kW
Motor continuous	$T_{\text{c}}, P_{\text{c}}$	150 N m, 60 kW
Pack configuration	96s3p	48 Ah, 3.65 V, 1.5 mΩ cells
Pack capacity, nominal energy	Q, E_{nom}	144 Ah, 50.46 kWh
Pack resistance	R_0	48 mΩ
Open-circuit voltage	$V_{oc}(z)$	$96(3.30 + 0.70z)$ V (linear, illustrative)
Usable window	E_u	90% of nominal, 45.4 kWh
Regeneration limit (DC side)	P_{chg}	50 kW, fading to zero between 15 and 5 km/h

4.2 Modelling of BEV: Forward-Looking Model

4.2.1 Introduction to Forward-Looking Modelling

A **forward-looking model** simulates a vehicle in the same order that cause follows effect in a real car. A driver sees a target speed and moves the pedals. The pedals produce a torque command. The motor produces torque, the torque accelerates the vehicle and the resulting speed is compared with the target. The approach is also called a *driver-oriented* or *causal* model [2, 11].

The approach is widely used in EV engineering texts [4, 6]. Its defining features are:

- the input is a *target*, not the actual speed;
- the actual speed is an *output* obtained by integrating the equation of motion;
- a driver model is needed to close the loop;
- component limits act naturally, so the vehicle can fail to follow the target.

4.2.2 Forward-Looking Model Architecture

Figure 4.6 shows the complete model. A drive cycle supplies the reference speed, the driver model compares it with measured speed, and the VCU converts pedal commands into a motor torque command and a friction brake command. The motor controller, inverter and motor then

produce torque from battery power. Torque passes through the transmission to the vehicle dynamics block, which outputs speed, fed back along the top of the diagram. The battery also feeds SoC and power limits to the VCU, so that a weak battery reduces performance just as in a real car. The figure is the central picture of this section, because every equation below belongs to one of its blocks.

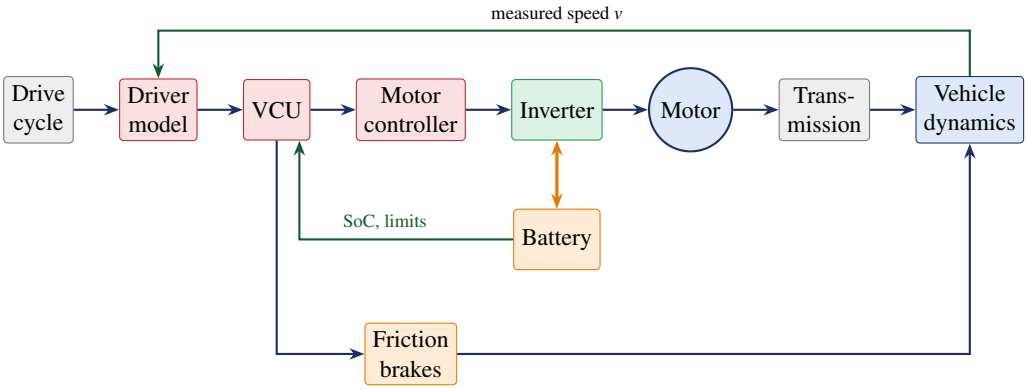


Figure 4.6: Complete forward-looking BEV model.

4.2.3 Signal Flow

The signal path is

$$\text{Driver Reference} \rightarrow \text{Driver Model} \rightarrow \text{Vehicle Controller} \rightarrow \text{Motor Controller} \rightarrow \text{Inverter} \rightarrow \text{Motor} \rightarrow \text{Transmission} \rightarrow \text{Vehicle Dynamics},$$

with actual speed returned to the driver model. Figure 4.7 labels each block with what it does. The driver turns the error e_v into a pedal signal u. The VCU turns u into torque T_m^*. The motor controller turns torque into current commands, as in Unit 3, and the inverter applies voltages. The motor produces T_m, the gear gives wheel torque T_w, Newton's law gives acceleration a and an integrator gives actual speed v. Since one integrator lies in the forward path, the loop naturally tracks a constant reference with zero steady-state error once a controller with integral action is added.

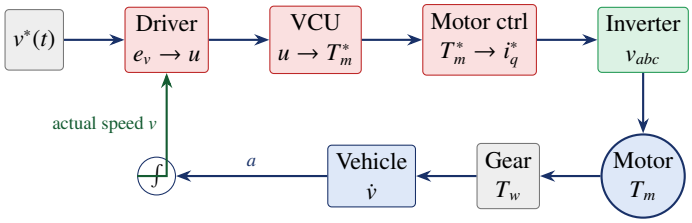


Figure 4.7: Forward-looking model signal flow.

4.2.4 Driver and Vehicle Control Architecture

Figure 4.8 separates the two layers of decision making. The *driver layer* acts on speed error and produces pedal positions u_a and u_b. The *vehicle control layer* interprets those pedals through a

map and arbitrates limits, producing the torque command for the plant. This separation mirrors a real car: a person controls speed, while software controls torque. Keeping it in a model allows a new pedal map or limit strategy to be tested without retuning the driver.

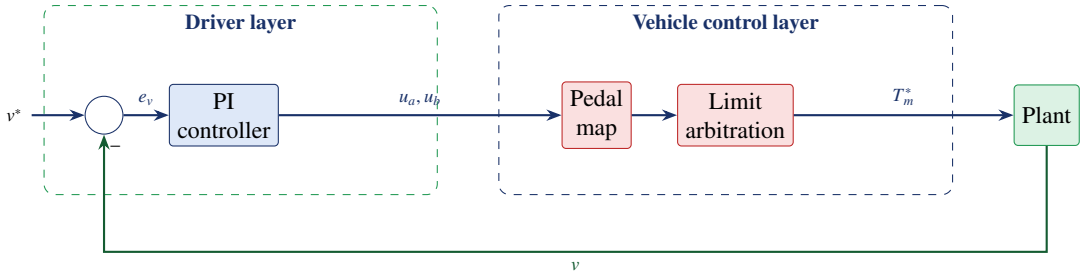


Figure 4.8: Driver and vehicle control architecture.

4.2.5 Driver Model and Reference Speed Generation

The **reference speed** $v^*(t)$ normally comes from a drive-cycle table interpolated at the simulation time. It may also come from a route planner or a measured trace. The **speed error**, in m/s, is

$$e_v(t) = v^*(t) - v(t). \tag{4.4}$$

A positive e_v means the car is too slow and needs accelerating. The driver is usually modelled as a PI controller producing a normalised demand $u \in [-1, 1]$,

$$u(t) = \mathrm{sat}\left[K_p e_v(t) + K_i \int_0^t e_v(\tau)\, d\tau\right], \tag{4.5}$$

where K_p has units of s/m and K_i has units of 1/m, so that u is dimensionless. The positive and negative parts become the accelerator and brake demands,

$$u_a = \max(u, 0), \qquad u_b = \max(-u, 0), \qquad 0 \le u_a, u_b \le 1. \tag{4.6}$$

The definition guarantees that the two pedals are never applied together. Section 4.5 develops this model in detail.

4.2.6 Vehicle Controller and Powertrain Controller

The accelerator demand scales the torque available at the present speed. In its simplest form, the **motor torque reference** in N m is

$$T_m^* = u_a\, T_{\max}(\omega_m), \qquad T_{\max}(\omega_m) = \min\left(T_{\mathrm{pk}},\ \frac{P_{\mathrm{pk}}}{\omega_m}\right), \tag{4.7}$$

where ω_m is motor speed in rad/s and P_{pk} is peak power in W. The first term is the constant-torque limit and the second the constant-power limit. The brake demand requests a total braking force,

$$F_b^* = u_b F_{b,\max}, \tag{4.8}$$

which Section 4.6 splits into regenerative and friction parts. Section 4.7 adds battery, thermal and rate limits.

4.2.7 Motor and Inverter Interface

Figure 4.9 shows the interface between controller, inverter, motor and shaft. In a *detailed* model the torque command becomes i_d^* and i_q^*, current loops generate voltages, and a dq machine model gives torque, as in Unit 3. Feedback of phase current and rotor position closes the inner loop. In a *simplified* system-level model, which is sufficient here because the current loop is about a thousand times faster than vehicle dynamics, the whole block reduces to a first-order lag and an efficiency,

$$\tau_T \frac{dT_m}{dt} + T_m = T_m^*, \tag{4.9}$$

where τ_T (s) is the torque response time, typically tens of milliseconds to a few tenths of a second once software rate limits are included. The electrical power then follows from the sign of the mechanical power $P_m = T_m \omega_m$ in W:

$$P_{\text{dc}} = \begin{cases} \dfrac{P_m}{\eta_m \eta_{\text{inv}}}, & P_m \geq 0 \quad \text{(motoring)}, \\ P_m \, \eta_m \eta_{\text{inv}}, & P_m < 0 \quad \text{(generating)}. \end{cases} \tag{4.10}$$

The two branches differ because losses always reduce the power reaching the destination: efficiencies divide when power flows from the battery to the wheels, and multiply when it flows the other way. Using the same form for both is a classic mistake.

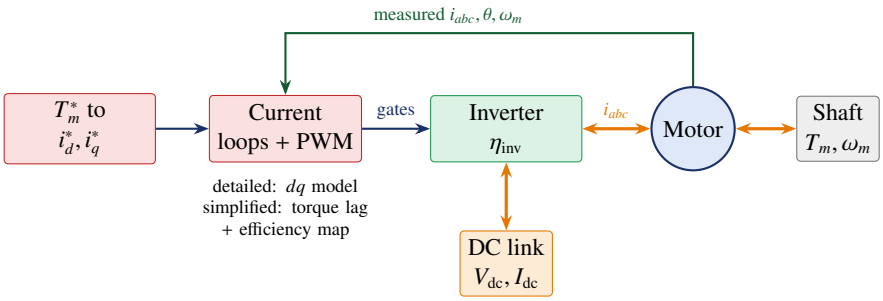


Figure 4.9: Motor and inverter interface in the forward-looking model.

4.2.8 Transmission Model and Wheel Torque

A single-speed gear with ratio G links motor and wheel. Speed and torque relations are

$$\omega_m = G\omega_w = \frac{Gv}{r_w}, \qquad T_w = \begin{cases} G\,\eta_g\,T_m, & T_m \geq 0, \\ G\,T_m/\eta_g, & T_m < 0, \end{cases} \tag{4.11}$$

with r_w the wheel radius in m and ω_w wheel speed in rad/s. When braking, power flows from wheel to motor, so the gear loss means the wheel must supply more torque than the motor absorbs, which is why η_g divides. The tractive force at the road is $F_t = T_w / r_w$.

4.2.9 Vehicle Longitudinal Dynamics, Acceleration and Speed Response

Newton's second law, developed fully in Section 4.8, gives the vehicle acceleration in m/s^2:

$$\frac{dv}{dt} = a = \frac{F_t - F_b - F_r - F_a - F_g}{\delta m}, \tag{4.12}$$

where F_b is the friction braking force, F_r rolling resistance, F_a aerodynamic drag and F_g gradient force, all in N, and δ is the rotational mass factor. Integrating gives the vehicle speed and position:

$$v(t) = v(0) + \int_0^t a(\tau)\, d\tau, \qquad x(t) = x(0) + \int_0^t v(\tau)\, d\tau. \tag{4.13}$$

Acceleration occurs when the net force is positive; deceleration occurs when drag, rolling resistance, grade and braking together exceed tractive force.

4.2.10 Actual Speed versus Reference Speed

Figure 4.10 shows the outer speed control loop in compact form. It looks like any feedback system: the driver is the controller, the powertrain is the actuator, $1/(\delta m s)$ is the plant, and the road load is a disturbance. Actual speed lags the reference because of the driver and actuator dynamics, and it may saturate if the reference demands more force than is available. That saturation is a key result, since it is precisely what a forward-looking model reveals and a backward-looking model hides.

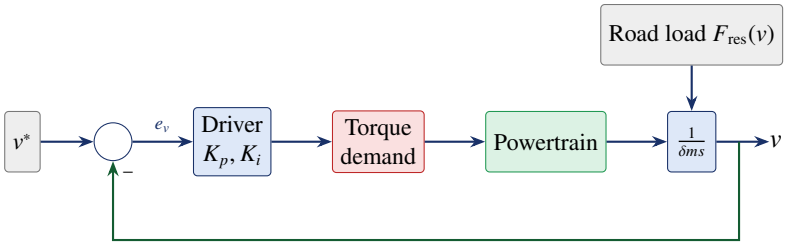


Figure 4.10: Speed reference versus actual speed control architecture.

4.2.11 Battery-to-Wheel Power Flow, Battery Demand and State Variation

Figure 4.11 traces motoring power from battery to wheel. The auxiliaries are fed directly from the battery, and each conversion stage rejects some heat, shown in red. Reversing the arrows gives the regenerative case. The **battery power demand** in W is the DC-side power plus the auxiliary load:

$$P_b = P_{\text{dc}} + P_{\text{aux}}. \tag{4.14}$$

With the Rint model $V_t = V_{oc} - IR_0$ and $P_b = V_t I$, the battery current is the solution of $R_0 I^2 - V_{oc} I + P_b = 0$ that is continuous at $P_b = 0$,

$$I = \frac{V_{oc} - \sqrt{V_{oc}^2 - 4R_0 P_b}}{2R_0}. \tag{4.15}$$

It is positive for discharge and negative for charge, and the square root is real only if $P_b \le V_{oc}^2/(4R_0)$, the maximum power the source can deliver. State of charge then evolves according to

$$\frac{dz}{dt} = -\frac{I}{3600\,Q}, \tag{4.16}$$

where Q is capacity in Ah. Over a drive, z falls during traction and rises slightly in regeneration.

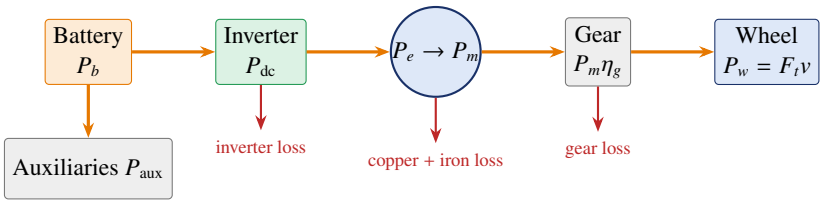


Figure 4.11: Battery-to-wheel power flow in the motoring direction.

4.2.12 Energy Consumption Calculation

Battery energy in J is the time integral of terminal power, and energy consumption per unit distance follows:

$$E_b(t) = \int_0^t P_b(\tau)\,d\tau, \qquad e = \frac{E_b(T)}{3600\,D}\ \text{Wh/m}, \tag{4.17}$$

with D the distance in m. Multiplying by 1000 gives Wh/km. Regenerative intervals subtract automatically because $P_b < 0$ there.

4.2.13 Assumptions of Simplified and Detailed Models

Table 4.4 sets out the choices. A simplified model suits most energy and driveability studies, and a detailed model is reserved for controller and component design.

Table 4.4: Assumptions in simplified and detailed forward-looking models

Block	Simplified	Detailed
Motor, inverter	Lag (4.9) and constant efficiency	dq model, loss maps, field weakening
Battery	Rint with fixed R_0	2-RC with $R(z,\Theta)$, thermal state
Transmission	Rigid, constant η_g	Backlash, shaft compliance, speed-dependent loss
Tyres	No slip	Slip curve and wheel inertia
Chassis	Point mass, flat road	Load transfer, grade profile, wind
Driver	PI on speed	Preview, reaction delay

Example 1: Forward-looking BEV driver model

Problem. The reference BEV follows a ramp reference of 10.0, 10.2, 10.4 m/s at steps $k = 0, 1, 2$. Find the driver commands, torque and speed for three steps, and then the brake command for a later moment at which the car is too fast.

Given (assumed). Table 4.3 data, level road, $K_p = 0.6$ s/m, $K_i = 0.1$ /m, $\Delta t = 0.1$ s, $v_0 = 9.5$ m/s, $\xi_0 = 0.05$, where ξ is the integrator state. Torque lag is ignored. Maximum brake force $F_{b,\max} = 0.8mg = 11\,772$ N.

Required. e_v, u, T_m^*, a, v, P_b.

Governing equations. (4.4), (4.5), (4.7), (4.11), (4.12), (4.14).

Solution. Step 0: $e_v = 10 - 9.5 = 0.5$ m/s and $u = 0.6(0.5) + 0.05 = 0.35$. Motor speed is $\omega_m = 9(9.5)/0.31 = 275.8$ rad/s, below base speed, so $T_{\max} = 280$ N m and $T_m^* = 98.0$ N m.
$F_t = 98.0(9)(0.97)/0.31 = 2760$ N. Road load is $F_r + F_a = 176.6 + 35.7 = 212.3$ N.
$a = (2760 - 212.3)/1575 = 1.62$ m/s^2, so $v_1 = 9.5 + 0.162 = 9.66$ m/s.
Power: $P_m = 98(275.8) = 27.03$ kW, $P_b = 27.03/0.9021 + 0.6 = 30.56$ kW.
The integrator becomes $\xi_1 = 0.05 + 0.1(0.5)(0.1) = 0.055$. Continuing gives:

k	e_v (m/s)	u	T_m^* (N m)	a (m/s^2)	v_{k+1} (m/s)	P_b (kW)
0	0.500	0.350	98.0	1.62	9.66	30.6
1	0.538	0.378	105.8	1.76	9.84	33.5
2	0.563	0.398	111.4	1.86	10.02	35.9

Brake case: suppose $v = 9.5$ m/s, $v^* = 9.0$ m/s, $\xi = 0.05$. Then $e_v = -0.5$ m/s and $u = -0.25$, so $u_a = 0$, $u_b = 0.25$ and $F_b^* = 2943$ N. With road load, deceleration $= (2943 + 212)/1575 = 2.00$ m/s^2.

Result. The driver moves from 35% to 40% accelerator as the ramp pulls away, and switches to 25% brake when the car is 0.5 m/s too fast.

Physical interpretation. A ramp of 2 m/s^2 requires positive error, so e_v grows until the integrator supplies the steady force.

Engineering significance. The output u is a demand rather than a physical torque: the same code will work for any vehicle, though the gains will need retuning. This is the calculation loop inside every forward-looking simulator.

4.2.14 Simulation Algorithm

Algorithm 1 and the flowchart in Figure 4.12 set out the forward-looking procedure. At each step the driver acts on the current error, the limits are applied, the power flows are evaluated and only then are the states advanced. Note that the loop uses the *previous* speed to compute the error, which breaks the algebraic loop around the driver and vehicle.

4.2.15 Advantages, Disadvantages and Computational Requirements

Advantages. The model behaves like the real vehicle, respects physical limits, reveals whether the cycle can actually be followed, and allows controllers to be tested. Hardware-in-the-loop tests need this causal structure.

Algorithm 1 Forward-looking BEV simulation

Input: Parameters of Table 4.3; cycle (t_j, v_j^*); gains K_p, K_i; step Δt
Output: $v(t)$, $T_m(t)$, $I(t)$, $z(t)$, e in Wh/km, RMSE of e_v

Initialise $v \leftarrow 0$, $T_m \leftarrow 0$, $\xi \leftarrow 0$, $z \leftarrow z_0$, $E \leftarrow 0$, $D \leftarrow 0$
for $t = 0$ to t_{end} in steps of Δt **do**
  $v^* \leftarrow$ interpolate cycle at t; $e_v \leftarrow v^* - v$
  $u \leftarrow \text{saturate}(K_p e_v + \xi)$; update ξ with anti-windup
  $u_a \leftarrow \max(u, 0)$; $u_b \leftarrow \max(-u, 0)$
  $\omega_m \leftarrow G v / r_w$; $T^* \leftarrow$ apply (4.7) and limits
  if $u_b > 0$ **then** split F_b^* into T_{reg} and F_b
  end if
  $T_m \leftarrow T_m + (\Delta t/\tau_T)(T^* - T_m)$
  $P_{\text{dc}} \leftarrow$ (4.10); $P_b \leftarrow P_{\text{dc}} + P_{\text{aux}}$; $I \leftarrow$ (4.15)
  $a \leftarrow$ (4.12); $v \leftarrow \max(0, v + a\Delta t)$
  $z \leftarrow z - I\Delta t/(3600Q)$; $E \leftarrow E + P_b \Delta t$; $D \leftarrow D + v\Delta t$
end for
$e \leftarrow E/(3.6D)$ ▷ E in J, D in m, e in Wh/km

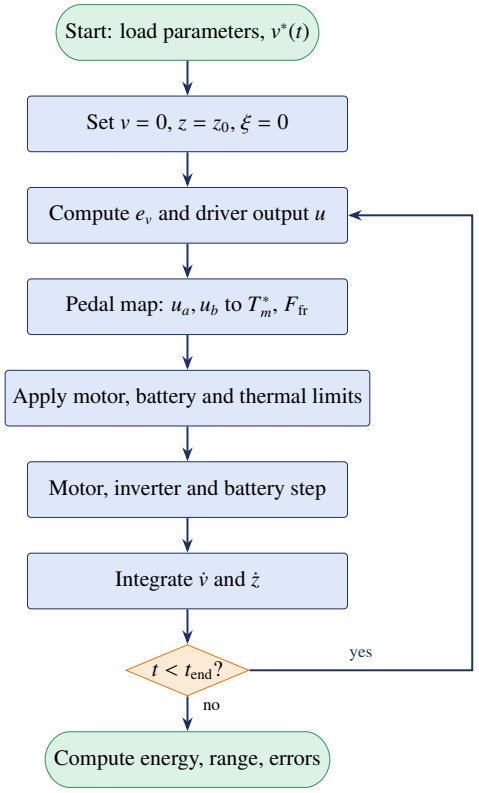


Figure 4.12: Forward-looking simulation flowchart.

Disadvantages. It needs a tuned driver, more states and a small time step, typically 1–100 ms. The simulated trace never matches the reference exactly, so two runs with different driver gains consume slightly different energy, which complicates comparison.

Computational requirements. Solving ordinary differential equations with feedback typically takes tens to hundreds of times the cost of a backward calculation. It is still faster than real time on a laptop if switching detail is excluded.

4.2.16 MATLAB/Simulink Implementation of the Forward-Looking Model

Figure 4.13 is a conceptual Simulink layout. Reference speed enters from a workspace table, passes through driver and VCU subsystems into the motor and vehicle subsystems, and speed returns through a `Unit Delay`, which prevents an algebraic loop if the driver has direct feedthrough. Table 4.5 lists the details.

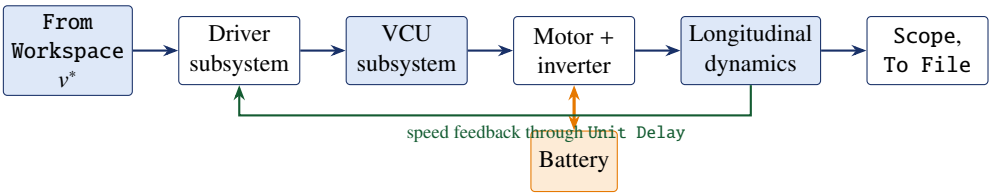


Figure 4.13: Conceptual Simulink diagram of the forward-looking BEV model.

Table 4.5: Forward-looking BEV model: conceptual Simulink implementation

Inputs	Reference speed $v^*(t)$, grade $\theta(t)$, ambient, parameters
Outputs	v, T_m, ω_m, I, V_t, P_b, z, energy
States	v, x, ξ, T_m, z (and V_1, V_2 if RC model used)
Equations	(4.4)–(4.16)
Blocks	`From Workspace`, `PID Controller`, `Saturation`, `Lookup Table`, `Integrator`, `Unit Delay`
Feedback	Speed to driver and VCU; SoC and limits to VCU
Solver	Fixed-step `ode4` at 10 ms, or variable-step `ode45`; `ode15s` if stiff
Initial conditions	$v(0) = 0$, $z(0) = z_0$, integrators reset
Assumptions	Point mass, averaged power electronics, no tyre slip
Simulation outputs	Speed trace, RMSE, torque-speed points, battery energy

4.3 Backward-Looking Model: Drive Cycle Perspective

4.3.1 Introduction to Backward-Looking Models

A **backward-looking model** reverses the logic. It assumes that the vehicle follows the prescribed speed exactly. Starting from that speed, it calculates the force required at the wheels, then the torque and speed at the motor, then the electrical power at the inverter and the battery.

Calculations run from the road back to the energy source, hence the name [2, 11]. The method is also called the *drive-cycle* or *quasi-static* approach, since each sample is treated as a steady operating point.

Its key features are:

- the speed profile is an *input*; there is no driver model;
- no differential equation for speed is solved, so there is no speed feedback;
- the only integrated quantities are energy and SoC;
- demand is assumed to be met, whether or not the hardware can deliver it.

Figure 4.14 shows the structure. Compare it with Figure 4.6: the arrows run from vehicle to battery, there is no driver and no feedback, and road conditions enter directly at the vehicle block.

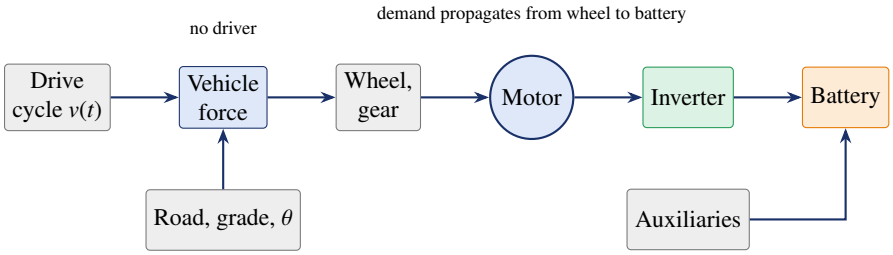


Figure 4.14: Complete backward-looking BEV model.

4.3.2 Calculation Sequence

The sequence is

Drive Cycle → Vehicle Speed → Acceleration → Road Load → Tractive Force → Wheel Torque → Motor Torque and Speed → Motor Power → Electrical Power → Battery Current and Energy.

Figure 4.15 lists each step. Each box needs only the previous box and a few parameters, so the calculation can be done row by row in a spreadsheet, which makes it an excellent teaching tool. The derivations below follow these boxes from top to bottom.

4.3.3 Drive-Cycle Input and Vehicle Acceleration

A drive cycle is a table of samples (t_k, v_k), usually at 1 Hz (Section 4.12). Speed must first be converted to m/s. Since speed is the input, acceleration comes from differentiating the data. The central difference,

$$a_k = \frac{v_{k+1} - v_{k-1}}{t_{k+1} - t_{k-1}}, \tag{4.18}$$

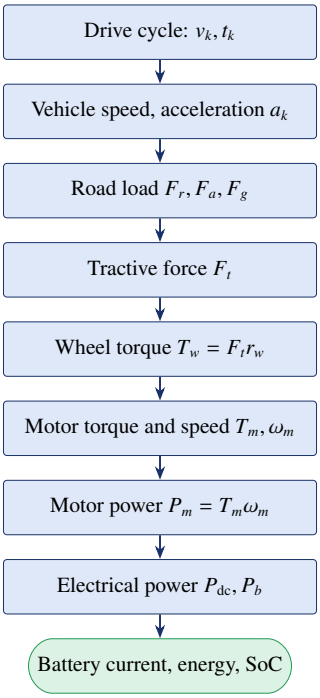


Figure 4.15: Backward-looking powertrain calculation flow.

is second-order accurate and does not shift the acceleration in time. A forward difference $a_k = (v_{k+1} - v_k)/\Delta t$ is simpler but represents the interval after sample k. If data are noisy, the speed must be filtered first, as differentiation amplifies noise.

Example 2: Vehicle acceleration from a speed-time profile

Problem. A speed trace reads 0, 7.2, 18, 28.8, 36, 36 km/h at $t = 0, 2, 4, 6, 8, 10$ s. Find the acceleration in each interval.

Given. Six samples, $\Delta t = 2$ s.

Required. Interval accelerations and the central-difference estimate at $t = 4$ s.

Governing equation. Forward difference and (4.18).

Solution. In m/s the samples are 0, 2, 5, 8, 10, 10.
$a = (2 - 0)/2 = 1.0$; $(5 - 2)/2 = 1.5$; $(8 - 5)/2 = 1.5$; $(10 - 8)/2 = 1.0$; $(10 - 10)/2 = 0$ m/s^2.
Central at $t = 4$ s: $a = (8 - 2)/(6 - 2) = 1.5$ m/s^2.

Result. Accelerations are 1.0, 1.5, 1.5, 1.0 and 0 m/s^2.

Physical interpretation. The car launches, reaches a steady acceleration and then eases into a cruise.

Engineering significance. A 1.5 m/s^2 urban acceleration uses far less than this car's capability. Forgetting to divide by 3.6 would make the force 3.6 times too large, and this is the most

common student error.

4.3.4 Road Load, Wheel Force and Wheel Torque

At each sample, the total tractive force is the sum of rolling resistance, drag, gradient and acceleration forces (Section 4.8 derives each term):

$$F_t = \underbrace{C_{rr} m g \cos\theta}_{F_r} + \underbrace{\tfrac{1}{2}\rho C_d A v^2}_{F_a} + \underbrace{m g \sin\theta}_{F_g} + \underbrace{\delta m a}_{F_{\text{acc}}}. \tag{4.19}$$

At standstill the rolling term is set to zero, since a static tyre has no rolling loss. F_t may be negative, which means braking is required. The wheel torque and power are

$$T_w = F_t r_w, \qquad \omega_w = \frac{v}{r_w}, \qquad P_w = F_t v = T_w \omega_w. \tag{4.20}$$

4.3.5 Motor Speed, Motor Torque and Motor Power

Going through the gear backward, with η_g applied according to the direction of power flow,

$$\omega_m = \frac{G v}{r_w}, \qquad T_m = \begin{cases} \dfrac{T_w}{G \eta_g}, & T_w \geq 0, \\[2mm] \dfrac{T_w \eta_g}{G}, & T_w < 0, \end{cases} \qquad P_m = T_m \omega_m. \tag{4.21}$$

So $P_m = P_w / \eta_g$ when motoring and $P_m = P_w \eta_g$ when braking. The magnitude of the mechanical power reaching the motor is reduced by gear loss in both directions of energy transfer from source to sink.

4.3.6 Component Efficiency and Transmission Losses

Efficiencies may be constants, as in this unit, or they may be two-dimensional maps $\eta_m(T_m, \omega_m)$ that vary from about 0.7 at very light load to above 0.95 at the sweet spot. Using a map greatly improves the prediction of energy consumption in urban driving, where a motor operates at low torque. Gear loss consists of tooth friction, roughly proportional to torque, and churning, roughly proportional to speed.

4.3.7 Electrical Power, Inverter Demand and Battery Current

Moving from the shaft to the DC side uses (4.10). Writing this explicitly for the backward calculation,

$$P_e = \frac{P_m}{\eta_m} \quad (P_m \geq 0), \qquad P_e = P_m \eta_m \quad (P_m < 0), \tag{4.22}$$

$$P_{\text{dc}} = \frac{P_e}{\eta_{\text{inv}}} \quad (P_e \geq 0), \qquad P_{\text{dc}} = P_e \eta_{\text{inv}} \quad (P_e < 0). \tag{4.23}$$

The auxiliary load is added at the battery as in (4.14). Battery current then follows from (4.15). Figure 4.16 shows the whole chain in the motoring direction. Each step applies one loss only once. If P_w/η_{dt} is already taken as DC power, η_g must not be applied again at the motor, otherwise the loss is double counted.

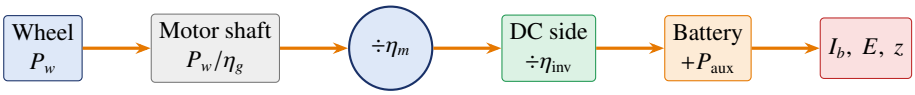


Figure 4.16: Wheel-to-battery energy calculation.

4.3.8 Drive-Cycle-Based Simulation Architecture

Figure 4.17 shows how the calculation is organised in software. The cycle database feeds an interpolation and differentiation stage. The road-load block uses vehicle parameters, the powertrain look-ups use efficiency maps, and the battery block returns the outputs. There are no loops anywhere in this diagram, so the whole model is a sequence of array operations that can run in milliseconds. That speed is what makes large parameter sweeps practical.

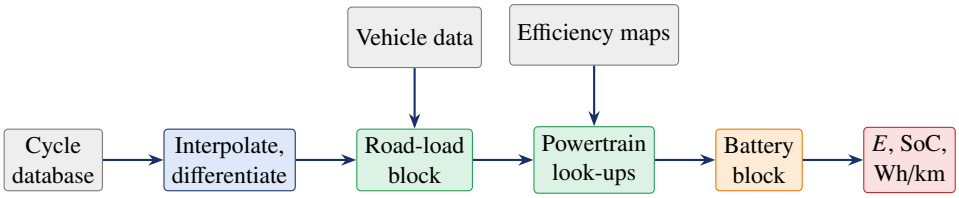


Figure 4.17: Drive-cycle-based simulation architecture.

4.3.9 Regenerative Braking in a Backward Model

When $P_w < 0$ the car needs braking and energy is available. Figure 4.18 shows the logic: braking power is checked against limits, the acceptable share passes through gear, motor and inverter (multiplying efficiencies), and the remainder goes to the friction brakes as heat. With $\kappa \in [0, 1]$ the fraction of braking power routed through the motor, the DC-side power becomes

$$P_{\mathrm{dc}} = \max(\kappa P_w \eta_g \eta_m \eta_{\mathrm{inv}},\ -P_{\mathrm{chg}}), \qquad P_w < 0. \tag{4.24}$$

Since both arguments are negative, the larger is the one of smaller magnitude, which enforces the charge limit. The friction brakes dissipate the remainder.

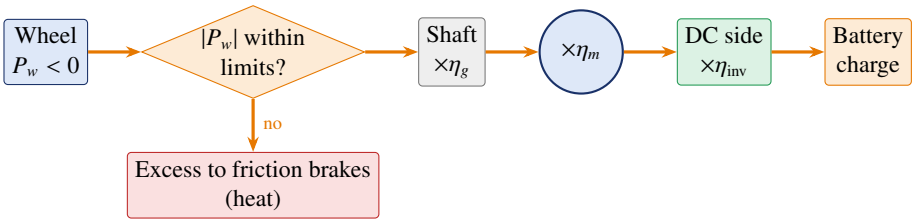


Figure 4.18: Backward-looking regenerative braking architecture.

4.3.10 Battery Energy, Regenerative Energy and SoC

Two energies must be tracked separately, and their difference is the net:

$$E_{\text{dis}} = \sum_{P_{b,k}>0} P_{b,k}\Delta t, \qquad E_{\text{reg}} = \sum_{P_{b,k}<0} \left|P_{b,k}\right| \Delta t, \qquad E_{\text{net}} = E_{\text{dis}} - E_{\text{reg}}. \tag{4.25}$$

In discrete form, the SoC update is

$$z_{k+1} = z_k - \frac{I_k \Delta t}{3600 Q}. \tag{4.26}$$

These quantities use the battery terminal power. The chemical energy withdrawn from the cell is slightly higher, because the internal resistance loss $I^2 R_0$ lies behind the terminals. The loss is captured by (4.15), since I is computed to deliver P_b at the terminals; it should not be added again.

Example 3: Backward-looking drive-cycle simulation

Problem. The reference BEV drives a short representative profile on a 1% uphill grade: 0 to 36 km/h in 10 s; cruise 20 s; 36 to 54 km/h in 10 s; 54 to 0 km/h in 12 s; idle 8 s. Calculate road loads, torque, speed, power, current and energy for each segment, using mid-segment values.

Given. Table 4.3; $P_{\text{aux}} = 600$ W; $V_{oc} = 350.4$ V held constant; full regeneration ($\kappa = 1$) with no active limit, as the power is under 50 kW; $\theta = \arctan 0.01$.

Required. a, F_r, F_a, F_g, F_t, T_w, T_m, ω_m, P_m, P_b, I, energy.

Governing equations. (4.19)–(4.21), (4.10), (4.15), (4.25).

Solution. For segment 1, $a = 10/10 = 1$ m/s^2 and $\bar{v} = 5$ m/s.
$F_r = 0.012(1500)(9.81)\cos\theta = 176.6$ N; $F_a = 0.396(5^2) = 9.9$ N; $F_g = 147.1$ N; $F_{\text{acc}} = 1.05(1500)(1) = 1575$ N.
$F_t = 1908.6$ N, $T_w = 591.7$ N m, $\omega_m = 9(5)/0.31 = 145.2$ rad/s, $T_m = 591.7/(9 \times 0.97) = 67.8$ N m.
$P_w = 9.54$ kW, $P_m = 9.84$ kW, $P_{\text{dc}} = 9.84/0.9021 = 10.91$ kW, $P_b = 11.51$ kW, $I = 33.0$ A, $E = 11.51(10)/3.6 = 32.0$ Wh.
Braking segment: $a = -1.25$ m/s^2, $\bar{v} = 7.5$ m/s, $F_t = -1622.8$ N, $P_w = -12.17$ kW, $T_m = -503.1(0.97)/9 = -54.2$ N m, $P_m = -11.81$ kW, $P_{\text{dc}} = -11.81(0.9021) = -10.65$ kW, $P_b = -10.05$ kW, $I = -28.6$ A. Results for all segments are in the table below.

Segment	Δt (s)	$\bar{v}$ (m/s)	F_t (N)	T_m (N m)	ω_m (rad/s)	P_m (kW)	P_b (kW)	I (A)	E (Wh)
Accelerate	10	5.0	1908.6	67.8	145.2	9.84	11.51	33.0	32.0
Cruise	20	10.0	363.3	12.9	290.3	3.75	4.75	13.6	26.4
Accelerate	10	12.5	1173.1	41.7	362.9	15.12	17.36	49.9	48.2
Brake	12	7.5	−1622.8	−54.2	217.7	−11.81	−10.05	−28.6	−33.5
Idle	8	0	0	0	0	0	0.60	1.7	1.3

Distance: $D = 50 + 200 + 125 + 90 + 0 = 465$ m.
$E_{\text{dis}} = 107.9$ Wh, $E_{\text{reg}} = 33.5$ Wh, $E_{\text{net}} = 74.4$ Wh.
Consumption: $e = 74.4/0.465 = 160$ Wh/km.
Charge balance: $\sum I\Delta t = 771$ A s $= 0.214$ Ah, so $\Delta z = 0.214/144 = 0.15\%$.

Result. Net energy is 74.4 Wh, 160 Wh/km on this grade, with 31% of discharge energy recovered.

Physical interpretation. Acceleration dominates peak force; the cruise force is about one-fifth of the launch force. Note that the 1% grade uses more force than drag at these speeds.

Engineering significance. Mid-segment averaging is a hand approximation; a real model uses every 1 s sample. The structure, however, is identical to the full simulations in Section 4.15.

4.3.11 Simulation Algorithm

Algorithm 2 gives the procedure and Figure 4.19 the flowchart. The decision on the sign of P_w selects the direction of the efficiency calculation. There is one pass per sample, and no iteration is needed.

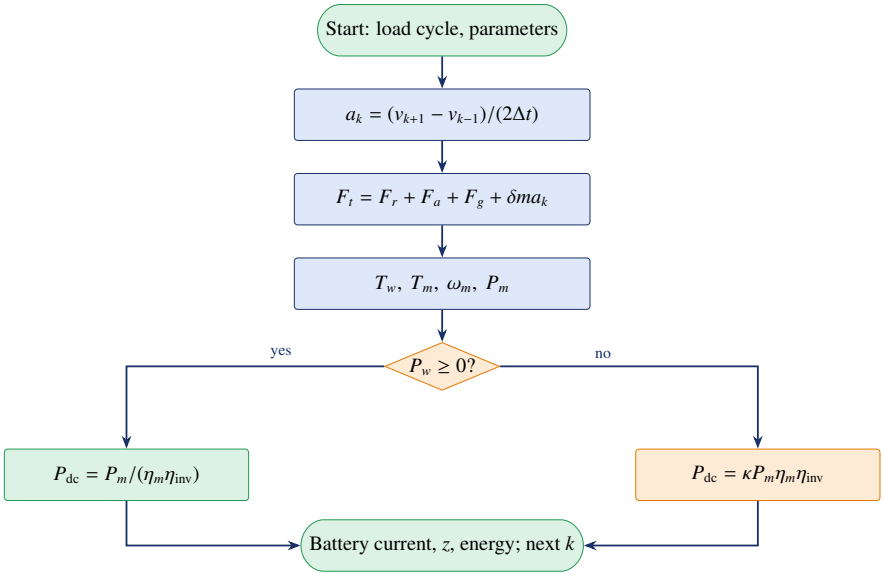


Figure 4.19: Backward-looking simulation flowchart.

Algorithm 2 Backward-looking BEV simulation

Input: Cycle (t_k, v_k) with v in m/s, grade θ_k, Table 4.3 parameters, z_0
Output: $F_{t,k}$, $T_{m,k}$, $P_{b,k}$, I_k, z_k, E_{net}, e

```
Initialise z_1 ← z_0, E_dis, E_reg, D ← 0
for k = 1 to N do
    a_k ← (4.18); F_{t,k} ← (4.19); P_{w,k} ← F_{t,k} v_k
    ω_{m,k} ← G v_k / r_w
    if P_{w,k} ≥ 0 then
        P_{dc,k} ← P_{w,k} / (η_g η_m η_inv)
    else
        P_{dc,k} ← (4.24), with low-speed fade
    end if
    P_{b,k} ← P_{dc,k} + P_aux; I_k ← (4.15) with V_oc(z_k)
    Accumulate E_dis or E_reg by the sign of P_{b,k}; D ← D + v_k Δt
    z_{k+1} ← z_k − I_k Δt / (3600 Q)
end for
E_net ← E_dis − E_reg; e ← E_net / D
```

4.3.12 Assumptions and Limitations

The method assumes that the vehicle follows the trace perfectly, that each sample is quasi-static, and that the road and wheels do not slip. Its limitations follow directly:

- It cannot show that a car *fails* to follow a cycle; torque above the motor limit is simply computed. An output check must flag such samples.
- Driver and control dynamics are absent, so it cannot be used for tuning controllers or for hardware-in-the-loop tests.
- Acceleration from numerical differences depends on sample rate; 1 Hz data hide short peaks.
- Thermal and other transient limits must be handled outside the main calculation.

4.3.13 MATLAB/Simulink Implementation of the Backward-Looking Model

A backward model is natural in MATLAB as vectorised code, but it can equally be built as open-loop blocks, as in Figure 4.20. Table 4.6 gives the implementation details. The MATLAB-style lines below show how compact the calculation is.

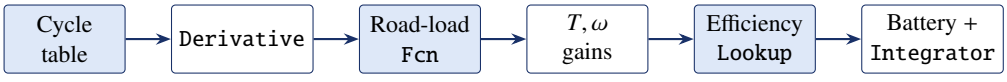


Figure 4.20: Conceptual Simulink diagram of the backward-looking BEV model.

Table 4.6: Backward-looking BEV model: conceptual implementation

Inputs	Cycle $v(t)$, grade, parameters, efficiency maps
Outputs	F_t, T_m, ω_m, P_b, I, z, Wh/km
States	z only (energy accumulators are bookkeeping)
Equations	(4.18)–(4.26)
Blocks	`Lookup Table`, `Derivative` or `gradient`, `Fcn`, `Gain`, `Integrator`
Feedback	None for speed; $z \rightarrow V_{oc}$ only
Solver	Fixed step equal to the cycle sample time
Initial conditions	$z(0) = z_0$
Assumptions	Perfect tracking, quasi-static operation
Simulation outputs	Energy, SoC, peak and rms torque, limit violations

```
v = cyc.kmh/3.6;  a = gradient(v, cyc.t);
Ft = m*g*Crr*(v>0) + 0.5*rho*Cd*A*v.^2 + delta*m*a;
Pw = Ft.*v;  Pdc = Pw/eta_dt;
Pdc(Pw<0) = max(Pw(Pw<0)*eta_dt, -Pchg);
Pb = Pdc + Paux;
I = (Voc - sqrt(Voc^2 - 4*R*Pb))/(2*R);
z = z0 - cumtrapz(cyc.t, I)/(3600*Q);
e = trapz(cyc.t, Pb)/3600 / (trapz(cyc.t, v)/1000);  % Wh/km
```

4.4 Comparison of Forward-Looking and Backward-Looking Models

4.4.1 Modelling Philosophy and Architecture

The two approaches ask different questions. A backward model asks: *if the vehicle drove this cycle, what would it cost?* A forward model asks: *given this driver and this hardware, what will the vehicle do?* Figure 4.21 places their structures one above the other. The forward model has a feedback line and passes through a driver; the backward model is a straight line, ending in energy. Everything in Table 4.7 follows from that difference.

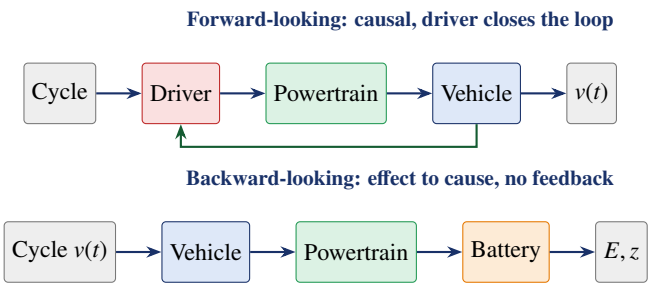


Figure 4.21: Forward versus backward modelling structures.

Table 4.7: Comparison of forward-looking and backward-looking models

Parameter	Forward-looking	Backward-looking
Modelling philosophy	Cause to effect, as in a real car	Effect to cause, wheel to battery
Input requirements	Reference speed, driver gains, full control data	Speed trace, parameters, efficiencies
Simulation architecture	Closed loop	Open-loop chain
Driver model	Required	Not required
Control system	Modelled explicitly	Rule-based post-processing only
Powertrain modelling	Torque is limited by physics	Torque computed; limits only checked
Vehicle dynamics	$\dot{v}$ integrated	a differentiated from data
Computational complexity	High: ODE with small step	Low: algebraic per sample
Simulation speed	Moderate	Very fast
Energy consumption prediction	Accurate; depends slightly on driver tuning	Accurate and repeatable for a given trace
Component sizing	Verifies a design	Screens many designs
Control strategy evaluation	Yes	Not possible
Dynamic performance analysis	Yes, acceleration and tracking	Not possible
Regenerative braking	Controller and limits interact dynamically	Static split with limits
Model accuracy	Better in transients	Good on average energy
Advantages	Realistic, HIL-ready	Simple, fast, repeatable
Disadvantages	Tuning effort, slower	Hardware limits are not enforced
Engineering applications	Control development, driveability	Early sizing, range studies, optimisation

The table shows that neither model dominates. Speed and repeatability are the backward model's strengths, while realism and causality are the forward model's.

4.4.2 Why Each Is Useful

Backward models for energy studies. Energy depends on the speed trace and the efficiency chain, not on the fine detail of a controller. Since the backward model follows the trace exactly, every candidate design is compared on the same work done at the wheels, and thousands of runs are affordable. For the reference BEV on MIDC, the backward net energy is 1.302 kWh and the forward PI-driver result is 1.289 kWh, a difference of 1%. Tracking the cycle to within about 0.3 km/h rms therefore has little effect on energy.

Forward models for control and actual response. Only a causal model has a vehicle that can fall behind its target. If a controller limits torque, a forward model shows the speed falling, the driver pressing harder and the integrator winding up. Pedal mapping, jerk limits, regenerative blending and driveability cannot be studied any other way.

4.4.3 Combined and Hybrid Modelling Approaches

ADVISOR combined the two ideas [11, 12]. A backward pass computes what is demanded, a check finds what the components can supply, and a forward pass returns what was actually delivered, as in Figure 4.22. If the limit is not reached, the vehicle follows the trace, retaining the backward model's speed. If it is reached, the delivered speed replaces the demand, and the trace miss is reported. Other hybrids use a fast backward model for design optimisation and a forward model for final verification, which is the common industrial practice. Gao *et al.* review these structures in [1].

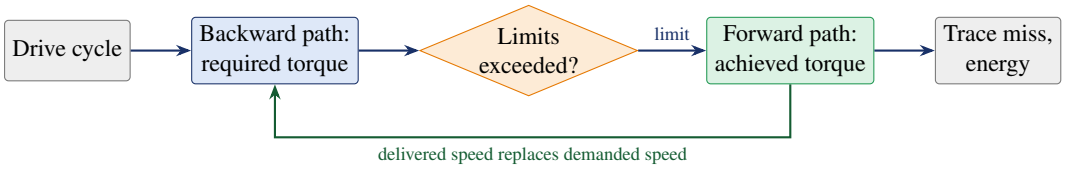


Figure 4.22: Combined backward/forward modelling approach.

4.5 Modelling of Driver

4.5.1 Introduction and Role of the Driver in EV Simulation

In a forward-looking simulation, the driver model is the block that makes the vehicle follow a speed target. It stands in for the human, or for the test technician who follows a trace on a dynamometer. Its job is not to imitate a person perfectly, but to generate realistic, repeatable pedal commands that keep the vehicle within an allowed band around the reference. The driver matters in three ways: it closes the speed loop, it sets how aggressively the powertrain is used, and it therefore influences energy consumption and component stress.

4.5.2 Driver Reference Speed, Actual Speed and Tracking Error

The reference speed $v^*(t)$ is the target, the actual speed $v(t)$ comes from the chassis model, and their difference is the speed tracking error e_v of (4.4). Figure 4.23 sketches typical behaviour. During a ramp the actual speed lags and e_v is positive. It overshoots slightly as the ramp ends, because the integrator has stored a force demand. In the brake phase, the lag has the opposite sign. Test standards permit only a small tolerance around the trace, so a good driver model should stay within roughly ±2 km/h and ±1 s of it.

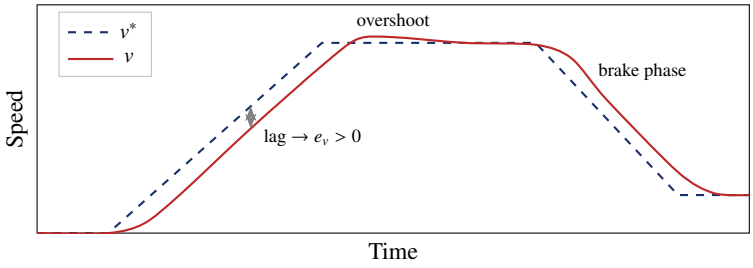


Figure 4.23: Reference speed and actual speed tracking (conceptual sketch).

4.5.3 Driver Model Architecture

Figure 4.24 shows the general model. The reference may first pass through a preview block. Together with measured speed it enters the speed controller. A response lag τ_d represents human reaction, and deadband and saturation logic shape the command before it is split into accelerator and brake. In a simple driver the preview and lag blocks are left out. The block order follows the human process: look ahead, compare, decide, act.

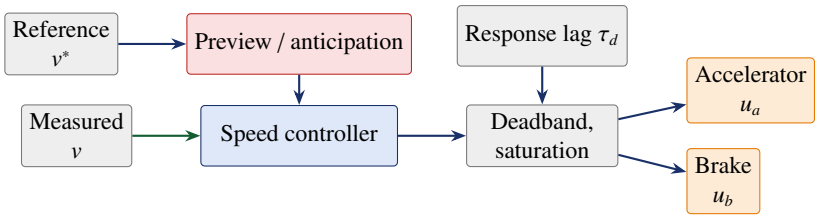


Figure 4.24: Driver model block diagram.

4.5.4 Driver Response Time, Behaviour and Aggressiveness

Human reaction times are in the order of 0.3–1 s. A first-order lag on the command represents this:

$$\tau_d \frac{du_d}{dt} + u_d = u, \tag{4.27}$$

where u_d is the delayed command and τ_d is in s. **Driver behaviour** is described by three linked parameters. **Aggressiveness** corresponds to high gains and hard use of the pedals, leading to high peak power and poor efficiency. **Smoothness** corresponds to low gains, a longer preview and gentle transitions. **Tracking accuracy** improves with gain, but only up to the point where overshoot and pedal chatter begin. Table 4.8 compares the common approaches.

Table 4.8: Driver modelling approaches

Approach	Equation	Tracking	Use
Proportional	$u = K_p e_v$	Steady error remains	Quick checks
PI	$+K_i \int e_v\, dt$	Zero error at steady speed	Standard cycle driver
PID	$+K_d \dot{e}_v$	Less overshoot, noise sensitive	Tight tracking
Preview	Feedforward + PI	Little lag	High-fidelity simulation
Human-like	Delay + stochastic	Realistic scatter	Driver-behaviour studies

The table shows that PI control provides the best balance for most work, and this is why it is the default driver in simulators.

4.5.5 Proportional, PI and PID Control

Figure 4.25 is the speed tracking feedback loop in transfer-function form, with controller $C(s)$, driver lag, a force gain k_T (N per unit demand) and the vehicle inertia. Road load enters as a disturbance. The three controllers are

$$u = K_p e_v \qquad \text{(P)}, \tag{4.28}$$

$$u = K_p e_v + K_i \int e_v\, dt \qquad \text{(PI)}, \tag{4.29}$$

$$u = K_p e_v + K_i \int e_v\, dt + K_d \frac{de_v}{dt} \qquad \text{(PID)}. \tag{4.30}$$

Proportional control needs a non-zero error to produce any force. At steady speed it must balance F_{res}, so the error settles at

$$e_{ss} = \frac{F_{\text{res}}}{K_p k_T}. \tag{4.31}$$

PI control removes this offset, as the integrator holds whatever output is needed while $e_v = 0$. Physically, the integral term learns the road load. **PID control** adds anticipation from the rate of error. Speed derived from a sensor is noisy, so K_d is used with a filter.

Ignoring lag and treating road load as a disturbance, the loop with PI control has the characteristic equation $\delta m s^2 + k_T K_p s + k_T K_i = 0$. Comparing it with $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$ gives the tuning rules

$$K_p = \frac{2\zeta\omega_n \delta m}{k_T}, \qquad K_i = \frac{\omega_n^2 \delta m}{k_T}. \tag{4.32}$$

As an illustration, with $\delta m = 1575$ kg, $k_T = 7885$ N, $\omega_n = 0.21$ rad/s and $\zeta = 1.4$, the result is $K_p \approx 0.12$ s/m. These formulas give a starting point only. The simulations here use a higher gain of $K_p = 0.6$ s/m for tighter tracking of the steep modal ramps, and any tuning should be checked against the actual cycle [28].

Figure 4.26 shows the driver with a PI controller. The proportional and integral paths add

to form u_r, which is clipped to $[-1, 1]$. If saturation is active, the integrator must not keep charging, or the vehicle will overshoot badly when the error reverses. The anti-windup path feeds back the difference between clipped and unclipped outputs:

$$\dot{\xi} = K_i e_v + K_{aw}(u - u_r), \tag{4.33}$$

with ξ the integrator state, K_{aw} in 1/s, and $u = \mathrm{sat}(u_r)$.

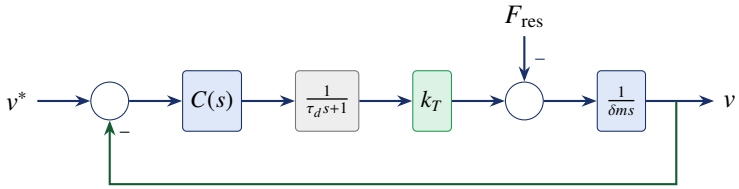


Figure 4.25: Speed tracking feedback loop.

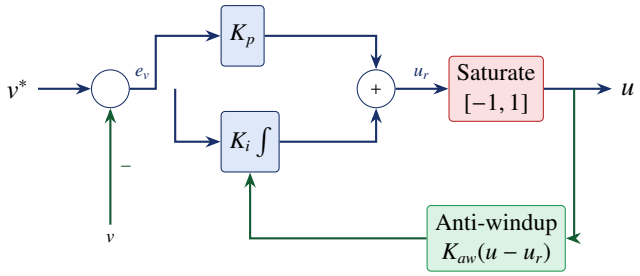


Figure 4.26: Driver model with PI controller and anti-windup.

4.5.6 Preview-Based Driver Model

Humans look ahead. A preview driver uses the future reference $v^*(t + t_p)$ over a look-ahead time t_p to predict the force that will be needed, as in Figure 4.27. The feedforward force is

$$\hat{F} = \delta m \, \frac{v^*(t + t_p) - v(t)}{t_p} + F_{\text{res}}(v), \tag{4.34}$$

and a PI loop trims the remaining error. Much of the demand is then supplied before an error appears, which shortens lag and reduces overshoot.

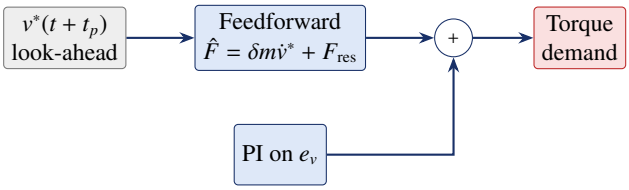


Figure 4.27: Preview-based driver model.

4.5.7 Accelerator and Brake Command Generation

Pedal commands need logic beyond (4.6), otherwise noise around $u = 0$ causes the pedals to chatter. Figure 4.28 shows a **deadband** scheme:

$$u_a = \mathrm{sat}_0^1(u - u_{\mathrm{db}}), \qquad u_b = \mathrm{sat}_0^1(-u - u_{\mathrm{db}}), \tag{4.35}$$

where sat_0^1 limits its argument to $[0, 1]$ and u_{db} is a small band, such as 0.02. Inside the band the vehicle coasts. Saturation represents a fully pressed pedal. **Acceleration and deceleration control** results from the same law: while $e_v > 0$ the accelerator is active, and a reference that falls below actual speed makes $e_v < 0$ and calls on the brake. A lift-off regenerative torque in the VCU means that light braking may need no brake pedal at all.

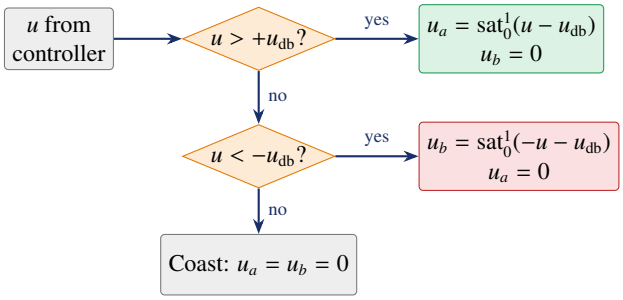


Figure 4.28: Accelerator and brake command generation.

4.5.8 Driver Demand and Torque Request Generation

The driver's command becomes a physical torque request. With demand u, the **torque demand** in N m is

$$T_{\mathrm{dem}} = \begin{cases} u_a \, T_{\max}(\omega_m), & u \geq 0, \\ -u_b \, F_{b,\max} \, r_w / G, & u < 0, \end{cases} \tag{4.36}$$

The negative branch expresses the requested brake force as an equivalent motor torque. Alternatively, a feedforward driver directly computes the force needed, $T_{\mathrm{dem}} = \hat{F} r_w / (G \eta_g)$.

Algorithm 3 Driver speed tracking

Input: $v^*(t)$, v, K_p, K_i, K_{aw}, Δt, u_{db}
Output: u_a, u_b, e_v

1: Initialise $\xi \leftarrow 0$
2: **for** each step **do**
3: $\quad e_v \leftarrow v^* - v$; $u_r \leftarrow K_p e_v + \xi$
4: $\quad u \leftarrow \min(1, \max(-1, u_r))$
5: $\quad \xi \leftarrow \xi + \Delta t \, [K_i e_v + K_{aw}(u - u_r)]$
6: $\quad$ Compute u_a, u_b with Algorithm 4
7: **end for**

Algorithm 3 updates the integrator after computing the output, so that anti-windup uses the present saturation. Algorithm 4 adds a brake hold at rest, which prevents the integrator from

Algorithm 4 Accelerator and brake demand generation

Input: u, u_{db}, speeds v, v^*, hold threshold $v_{\min}$
Output: u_a, u_b

```
if v* ≈ 0 and v < v_min then
    u_a ← 0; u_b ← u_hold                ▷ hold the car at rest
else if u > u_db then
    u_a ← min(1, u − u_db); u_b ← 0
else if u < −u_db then
    u_b ← min(1, −u − u_db); u_a ← 0
else
    u_a ← 0; u_b ← 0                     ▷ coast
end if
```

drifting while the car is stationary.

4.5.9 Simulated Waveforms of Driver Speed Tracking

The waveforms below are **simulated results** for the reference BEV, using the forward-looking model with $K_p = 0.6$, $K_i = 0.1$, and $\tau_T = 0.2$ s. The reference is the 195 s elementary urban part of the official MIDC speed table [16].

Figure 4.29 shows the reference alone. Three bursts reach 15, 32 and 50 km/h, each with idling between them. Figure 4.30 overlays actual speed: at this scale the two are almost coincident, so Figure 4.31 enlarges the third burst. The actual curve slightly lags each ramp, and the corners are rounded by the lag and limited gain.

Figure 4.32 shows the tracking error. Its peaks of about 1.5 km/h occur at the start of each acceleration, where the torque must first build up, and it decays through integral action during cruise. The RMS error is 0.36 km/h, which is well within test tolerance.

Figure 4.33 shows the accelerator command. Peaks near 27% coincide with the start of each ramp. Cruise needs only 2–3%, which illustrates how small the steady road load is compared with the torque available. The steps follow the gear-change plateaus built into the cycle. Figure 4.34 shows the brake command, with 15% hold at rest and lower values during decelerations. The pedals are never active together.

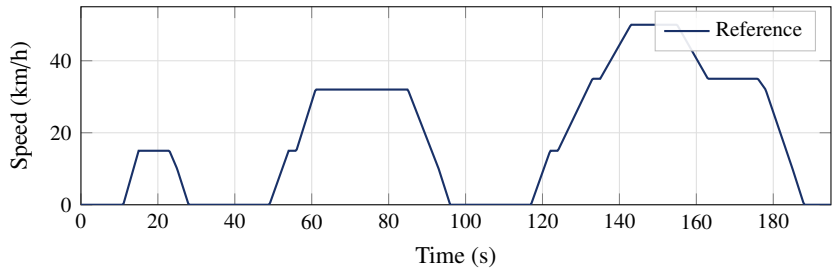


Figure 4.29: Reference speed versus time (official MIDC Part One trace).

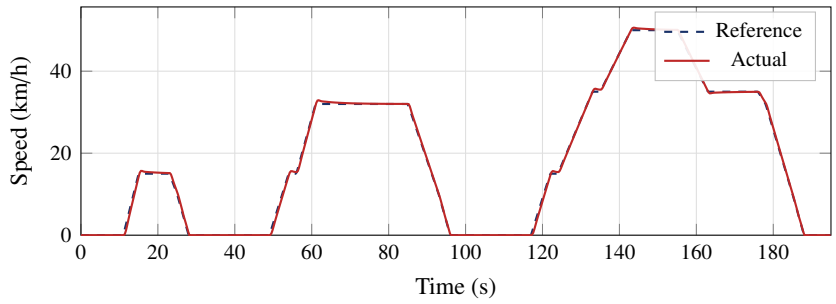


Figure 4.30: Reference and actual speed versus time (simulated, hypothetical BEV).

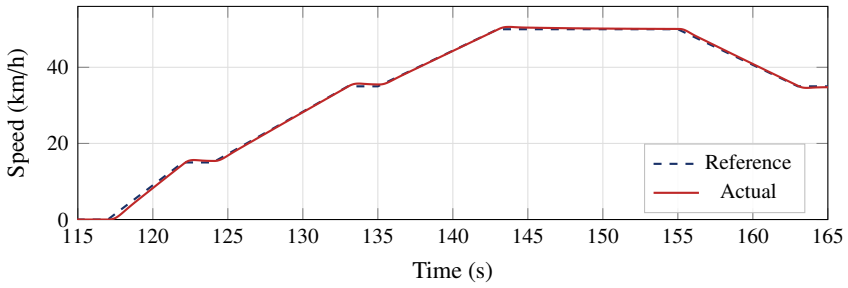


Figure 4.31: Detail of speed tracking during the third acceleration burst (simulated).

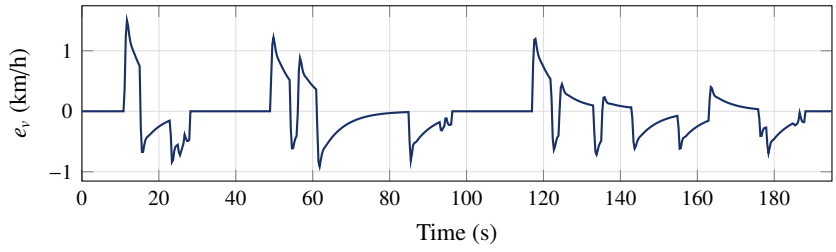


Figure 4.32: Speed tracking error (simulated).

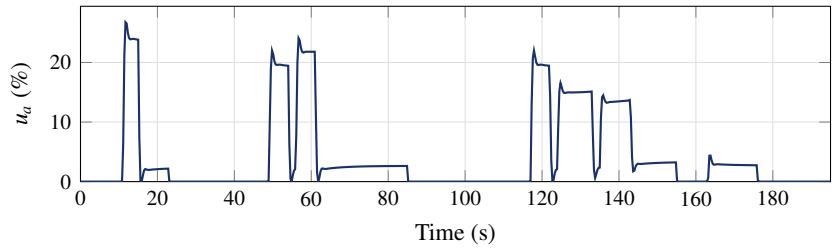


Figure 4.33: Accelerator pedal command (simulated).

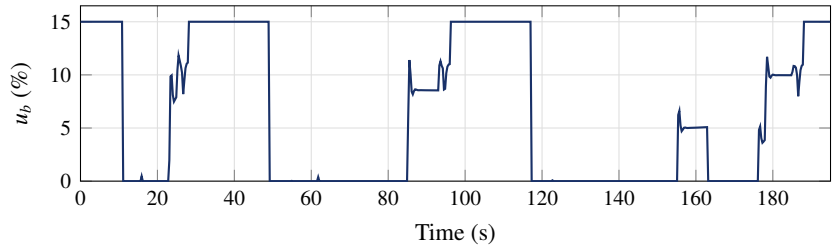


Figure 4.34: Brake pedal command, including hold at rest (simulated).

Example 4: Driver PI controller calculation

Problem. A driver with gains $K_p = 0.6$ s/m and $K_i = 0.1$ /m tracks $v^* = 12$ m/s. Sampled speeds at $\Delta t = 0.5$ s are 10.0, 10.5 and 10.9 m/s. Find the demand, assuming the integrator is updated before use and starts at zero.

Given. $\xi_0 = 0$, $u \in [-1, 1]$.

Required. e_v, ξ, u_r, u for three samples.

Governing equations. (4.29), with $\xi_{k+1} = \xi_k + K_i e_v \Delta t$.

Solution. Sample 1: $e_v = 2.0$, $\xi = 0 + 0.1(2)(0.5) = 0.100$, $u_r = 0.6(2) + 0.1 = 1.30$, clipped to $u = 1.00$.
Sample 2: $e_v = 1.5$, $\xi = 0.175$, $u_r = 0.9 + 0.175 = 1.075$, clipped to 1.00.
Sample 3: $e_v = 1.1$, $\xi = 0.230$, $u_r = 0.66 + 0.23 = 0.89$, so $u = 0.89$.

Result. $u = 1.00, 1.00, 0.89$. With $T_{\max} = 280$ N m the torque request is 280, 280 and 249 N m.

Physical interpretation. The driver floors the pedal while far from target and eases off as the gap closes, with the integral part slowly rising.

Engineering significance. During the first two samples the output was saturated, so a real implementation should hold or bleed the integrator using (4.33); otherwise overshoot would follow.

4.5.10 MATLAB/Simulink Implementation of the Driver Model

Figure 4.35 and Table 4.9 give the conceptual implementation. Simulink's `PID Controller` block has built-in saturation and anti-windup. The output is split by a `Switch` or simple `Max` blocks.

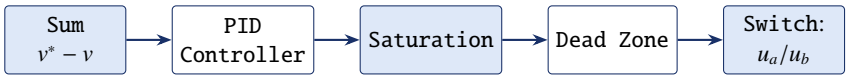


Figure 4.35: Conceptual Simulink diagram of the driver model.

4.6 Modelling of Brake Control Unit and Regenerative Braking

4.6.1 Introduction to EV Braking Systems

Braking has two jobs in an EV: to stop the car safely and to recover energy. In a conventional car, all kinetic energy becomes heat. In an EV the motor can act as a generator and return part of it to the battery [3, 5]. Friction brakes remain essential, as the motor alone cannot provide

Table 4.9: Driver model: conceptual Simulink implementation

Inputs	$v^*(t)$, v
Outputs	u_a, u_b, or a torque request
States	Integrator ξ, lag u_d
Equations	(4.4), (4.29), (4.33), (4.35)
Blocks	`Sum`, `PID Controller`, `Saturation`, `Dead Zone`, `Transfer Fcn`
Feedback	Actual speed from the chassis subsystem
Solver	Same as vehicle; 10–100 ms fixed step is adequate
Initial conditions	$\xi(0) = 0$
Assumptions	Perfect speed sensing, no route preview unless modelled
Simulation outputs	Pedal traces, RMSE, trace violations

full emergency braking at all speeds, states of charge and temperatures. Together with their control, the two systems make up the EV braking system.

4.6.2 Conventional Friction Braking

In a friction brake, hydraulic pressure clamps pads onto a disc. The brake torque at one wheel is

$$T_{\text{fr}} = 2\mu_p p_b A_p r_{\text{eff}}, \tag{4.37}$$

with μ_p the pad friction coefficient, p_b the line pressure in Pa, A_p the piston area in m^2, r_{eff} the effective disc radius in m, and 2 for the two faces of the disc. All braking energy becomes heat in the disc. The system is strong and works down to zero speed, but it recovers nothing.

4.6.3 Regenerative Braking and Motor Operation as a Generator

In regenerative braking, the inverter commands negative torque. Figure 4.36 shows the process. The wheels drive the rotor, and the inverter sets $i_q^* < 0$ so that stator current opposes the rotation. The torque on the rotor is therefore a braking torque, while the back EMF exceeds the inverter's fundamental voltage so that power flows toward the DC link. No switch changes state in the machine: the difference between motoring and generating lies solely in the sign of the current command.

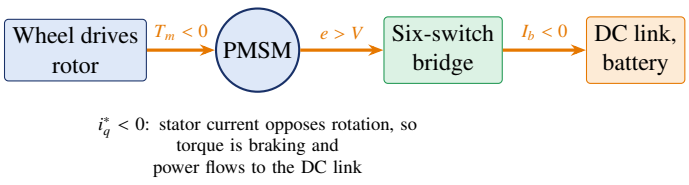


Figure 4.36: Motor operating in generator mode.

Figure 4.37 shows the battery side. Current enters the positive terminal, so $I < 0$ by the discharge-positive convention. The terminal voltage then rises above open-circuit voltage according to $V_t = V_{oc} - IR_0$. The rise matters because the BMS limits upper cell voltage: near full charge there is little room for IR_0, so charge acceptance falls.

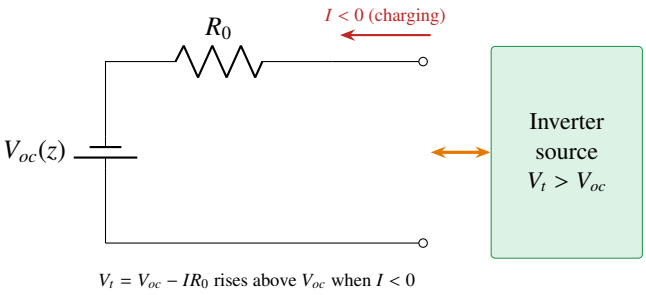


Figure 4.37: Battery charging during regenerative braking.

4.6.4 Regenerative Power Flow and Energy Recovery

Figure 4.38 shows where kinetic energy goes. Part is taken by road load, which does useful slowing for free. Part may go to friction heat. The remainder passes through the gear, the motor, the inverter and the battery's internal resistance, losing some energy at each stage, and is stored in the battery. The figure explains why recovery can never approach 100%: the energy must pass through the chain twice in a full motoring-braking round trip.

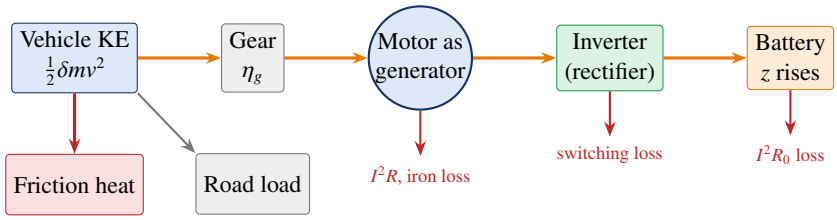


Figure 4.38: Regenerative braking energy flow.

4.6.5 Brake Control Unit and EV Braking Architecture

Figure 4.39 shows the architecture. The brake pedal signal enters the **brake control unit** (BCU), which splits demand into a regenerative torque request to the motor controller and a friction request to the hydraulic modulator. Feedback comes from the wheels (speed and slip) and from the BMS (charge limit). Figure 4.40 expands the unit. Pedal inputs become a force demand. A capability block computes what the motor can absorb at this moment, using battery, motor and inverter limits. The split block then produces $T_m^* < 0$ and pressures p_f, p_r. ABS and stability control can override the split at any time, because safety always outranks energy recovery.

4.6.6 Brake Pedal Input and Brake Torque Demand

The pedal travel or pressure sensor gives $u_b \in [0, 1]$. A map gives the total demand at the wheels,

$$F_b^* = u_b F_{b,\max}, \qquad T_b^* = F_b^* r_w. \tag{4.38}$$

$F_{b,\max}$ is set by tyre adhesion, about μmg, and the map is shaped so that pedal feel is consistent.

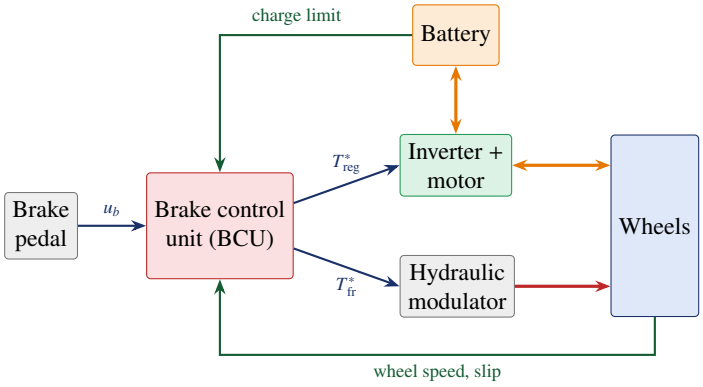


Figure 4.39: EV braking architecture.

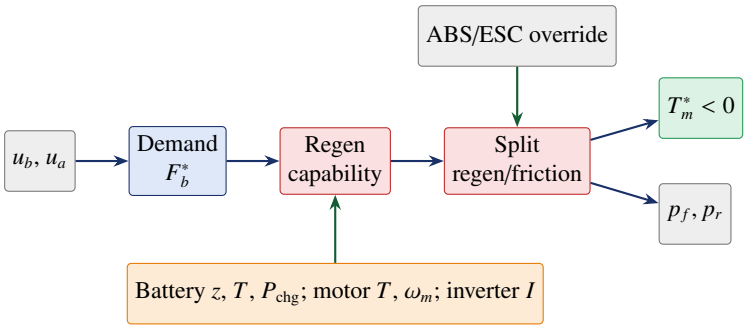


Figure 4.40: Brake control unit block diagram.

4.6.7 Total Braking Force, Deceleration and Torque

Treat the car as decelerating on a level road. All forces act backward, so Newton's law gives

$$\delta m\, a_d = F_b + F_r + F_a, \tag{4.39}$$

with deceleration $a_d > 0$ in m/s^2. Therefore the required brake force in N is

$$F_b = \delta m a_d - F_r - F_a, \tag{4.40}$$

and the total braking torque at the wheels is

$$T_b = F_b r_w = T_{b,\mathrm{reg}} + T_{\mathrm{fr}}. \tag{4.41}$$

For constant a_d, stopping time and distance from speed v_0 are $t_s = v_0/a_d$ and $d_s = v_0^2/(2a_d)$.

4.6.8 Regenerative and Mechanical Braking Power

Total brake power at the wheels is $P_{b,w} = F_b v = T_b \omega_w$ in W. With κ the fraction taken by the motor,

$$P_{\text{reg},w} = \kappa F_b v \qquad \text{(regenerative power at wheel)}, \tag{4.42}$$

$$P_{\text{mech}} = (1-\kappa) F_b v \qquad \text{(friction power, heat)}, \tag{4.43}$$

$$P_{\text{reg},b} = P_{\text{reg},w}\, \eta_g \eta_m \eta_{\text{inv}} \qquad \text{(power at the battery terminals)}. \tag{4.44}$$

The last equation applies the efficiencies once each, in the direction of flow. Inside the battery, a further loss $I^2 R_0$ is incurred, which appears when the stored energy is measured chemically.

4.6.9 Recoverable Energy, Efficiency and Energy Recovery Ratio

The kinetic energy at speed v_0 is

$$E_k = \tfrac{1}{2} \delta m v_0^2. \tag{4.45}$$

Part of this is spent on road load during the stop. Brake energy at the wheels is $E_{b,w} = \int F_b v \, dt < E_k$. Energy reaching the battery is $E_{\text{rec}} = \int P_{\text{reg},b} \, dt$. Three useful measures are

$$\eta_{\text{reg}} = \frac{E_{\text{rec}}}{E_{b,w}} \qquad \text{(regenerative braking efficiency)}, \tag{4.46}$$

$$\text{ERR} = \frac{E_{\text{rec}}}{E_k} \qquad \text{(energy recovery ratio)}, \tag{4.47}$$

$$\phi_{\text{fr}} = \frac{E_{\text{fr}}}{E_{b,w}} \qquad \text{(friction braking contribution)}. \tag{4.48}$$

4.6.10 Limits on Regenerative Braking

Regenerative torque is available only within several limits, each of which may be the active one.

- **Motor torque limit.** $|T_m| \le T_{\text{pk}}(\omega_m)$ on the generator side of the envelope.
- **Inverter current limit.** Phase current cannot exceed the device rating; this acts through the same torque limit.
- **Battery charge acceptance.** Charge power is limited to $P_{\text{chg}}(z, \Theta)$, which falls in a high SoC and at low temperature. Lithium plating risk makes cold charging particularly restricted.
- **Battery SoC limit.** Above about 95% SoC regeneration is largely unavailable, for example just after a full charge.
- **Low-speed limit.** Back EMF and power both vanish as $\omega_m \to 0$, so regeneration fades in below about 5–15 km/h. Friction takes over for the final stop, or the inverter briefly holds torque while drawing a small battery current.

- **Vehicle stability.** Regeneration on one axle can upset the car on slippery roads, so ABS or ESC activation cancels it.
- **Fault conditions.** Over-voltage on the DC link and sensor faults disable regeneration.

Combining the motor and battery limits gives the maximum regenerative torque at the motor shaft in N m:

$$T_{\text{reg,lim}} = f_v(v)\,\min\!\left(T_{\text{pk}}(\omega_m),\ \frac{P_{\text{chg}}}{\eta_m \eta_{\text{inv}} \omega_m}\right), \tag{4.49}$$

where f_v is a fade factor, 0 below v_{cut}, rising linearly to 1 at v_{full}. In this unit, $v_{\text{cut}} = 5$ km/h and $v_{\text{full}} = 15$ km/h.

4.6.11 Blended Braking and the Blending Model

Blended braking combines regeneration and friction so that the total always equals the driver's demand. Figure 4.41 shows the architecture. The demand passes through a minimum block limited by the regenerative capability, and the friction actuator receives whatever remains. Their outputs add at the wheel. With T_b^* the wheel-level demand, the mathematical model is

$$T_{\text{reg}} = \min(T_b^*,\ G\,T_{\text{reg,lim}}), \tag{4.50}$$

$$T_{\text{fr}} = T_b^* - T_{\text{reg}}. \tag{4.51}$$

Both are wheel torques in N m. Because friction is the slower actuator, the controller must coordinate the two so that the car's deceleration does not change while the split moves. Figure 4.42 shows the torque distribution versus speed for a constant 1120 N m wheel demand. Regeneration is absent at very low speed, climbs to a plateau set by the motor's torque limit, and falls at high speed where the power limit applies. Friction fills the gap above it, so friction is busy at both ends of the speed range. The figure explains why friction brakes in an EV must be full-size even though they are used much less.

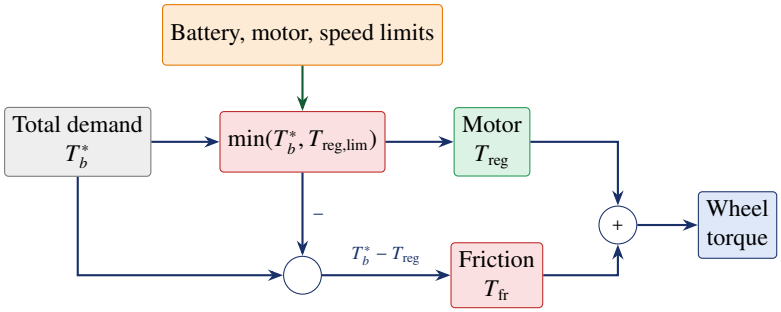


Figure 4.41: Blended braking architecture.

Tables 4.10 and 4.11 compare the alternatives.

The tables show that regeneration wins on energy and wear, friction wins on availability and robustness, and blending gives both at the price of software complexity.

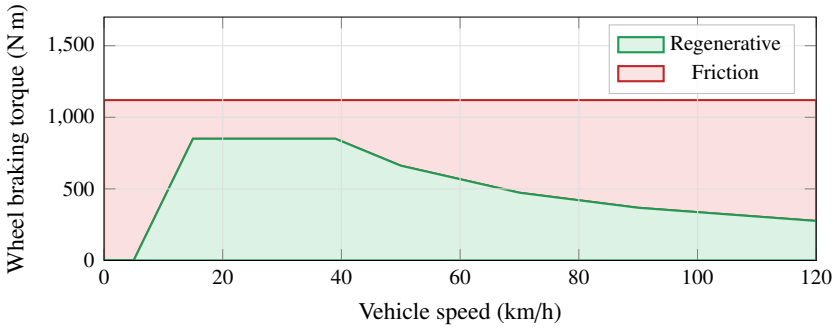


Figure 4.42: Regenerative and friction braking torque distribution versus speed (illustrative).

Table 4.10: Conventional braking versus regenerative braking

Aspect	Friction	Regenerative
Energy destination	Heat in disc	Battery, after losses
Available at standstill	Yes	No
Maximum torque	Set by tyre grip	Set by motor and battery
Response	Tens of ms to pressure	Few ms
Wear	Pads and discs	Almost none
Depends on SoC and temperature	No (fade when hot)	Strongly

Table 4.11: Friction braking versus blended braking

Aspect	Friction only	Blended
Energy recovered	None	Most of mild braking energy
Pedal linkage	Direct hydraulic	Usually decoupled (brake-by-wire)
Control complexity	Low	High, needs torque coordination
Pad and disc life	Normal	Much longer
Failure mode	Mechanical back-up inherent	Must fall back to friction
Emissions	Brake dust	Reduced brake dust

4.6.12 Brake Force Distribution, Front and Rear Axle Braking

Deceleration moves load to the front axle. With wheelbase L, CG height h_{cg} and distances l_f and l_r from the CG to the front and rear axles, axle loads in N for deceleration a_d are

$$F_{zf} = \frac{mg}{L}\left(l_r + \frac{a_d}{g}h_{cg}\right), \qquad F_{zr} = \frac{mg}{L}\left(l_f - \frac{a_d}{g}h_{cg}\right). \tag{4.52}$$

The ideal distribution divides the brake force in proportion to these loads, so that both axles reach the adhesion limit together. In practice the distribution is kept front-biased, since locking the rear wheels first makes the car unstable. A single front motor can regenerate only through the front axle, and a rear motor only through the rear, so the rear share must stay within stability limits. For a drive axle share β, the regenerative torque cannot exceed βT_b^* without deviating from the intended distribution unless the system permits a safe bias.

4.6.13 ABS Interaction and Brake-by-Wire Systems

Under hard braking on a slippery road, ABS modulates pressure at each wheel to hold longitudinal slip near the peak of the friction curve. Regenerative torque acts through a gearbox and has a different dynamic response, so it is usually faded out quickly when ABS activates, leaving control to the hydraulic system. In some designs, the fast motor torque assists the wheel slip control.

Brake-by-wire decouples pedal from hydraulics: a sensor reads the pedal, a simulator provides feel, and an electric pump or booster generates pressure. This is what allows the invisible exchange of friction for regenerative torque. In fault conditions, a hydraulic fallback connects the pedal directly to the wheels.

4.6.14 Regenerative Braking under Different Operating Conditions

- **Urban stop-and-go.** Frequent mild stops suit regeneration very well; a large share of traction energy can be recovered.
- **Highway.** Few stops mean drag does the braking, so little energy is available.
- **Long downhill.** Power is available for a long time, so thermal limits and a full battery may end regeneration. Friction brakes must be able to take over.
- **Cold battery.** Charge acceptance is low until the pack warms.
- **Full battery.** Regeneration is reduced or absent just after charging.
- **Emergency braking.** Friction dominates; the tyres set the limit.

4.6.15 Control Algorithms

Algorithms 5 to 7 implement the logic above. The first computes capability, the second completes the split so that the demand is met, and the third shares the friction part across the axles.

Algorithm 5 Regenerative braking control

Input: T_b^*, ω_m, v, z, Θ_b, ABS flag
Output: Regenerative wheel torque T_{reg}

if ABS active **or** fault **or** $z > z_{\max}$ **then**
    $T_{\text{reg}} \leftarrow 0$
else
    $P_{\text{chg}} \leftarrow \text{BMS limit}(z, \Theta_b)$
    $f_v \leftarrow \min(1, \max(0, (v - v_{\text{cut}})/(v_{\text{full}} - v_{\text{cut}})))$
    $T_{\text{reg,lim}} \leftarrow$ (4.49)
    $T_{\text{reg}} \leftarrow \min(T_b^*, G T_{\text{reg,lim}})$
end if

Algorithm 6 Blended braking control

Input: u_b, $F_{b,\max}$, r_w, G, η_g
Output: T_m^*, T_{fr}^*

$T_b^* \leftarrow u_b F_{b,\max} r_w$
$T_{\text{reg}} \leftarrow$ Algorithm 5
$T_{\text{fr}}^* \leftarrow T_b^* - T_{\text{reg}}$
$T_m^* \leftarrow -T_{\text{reg}} \eta_g / G$ ▷ gear loss helps braking
if T_{reg} is decreasing fast **then**
    Rate-limit T_{reg} so friction pressure can follow
end if

Algorithm 7 Brake torque distribution

Input: T_{fr}^*, T_{reg}, a_d, L, l_f, l_r, h_{cg}, drive axle
Output: $T_{\text{fr},f}$, $T_{\text{fr},r}$

$T_b \leftarrow T_{\text{fr}}^* + T_{\text{reg}}$
$F_{zf} \leftarrow$ (4.52); $\beta_f \leftarrow F_{zf}/(mg)$
$T_f \leftarrow \beta_f T_b$; $T_r \leftarrow (1 - \beta_f) T_b$
if drive axle is front **then**
    $T_{\text{fr},f} \leftarrow \max(0, T_f - T_{\text{reg}})$; $T_{\text{fr},r} \leftarrow T_r$
else
    $T_{\text{fr},r} \leftarrow \max(0, T_r - T_{\text{reg}})$; $T_{\text{fr},f} \leftarrow T_f$
end if

4.6.16 Numerical Examples

Example 5: Calculation of braking force and braking torque

Problem. The reference BEV brakes from 72 km/h to rest at a constant $a_d = 2.5$ m/s^2 on a level road. Find the brake force and torque at the start, and the stopping time and distance.

Given. Table 4.3.

Required. F_b, T_b, t_s, d_s.

Governing equations. (4.39)–(4.41).

Solution. $v_0 = 20$ m/s. Total retarding force $= \delta m a_d = 1.05(1500)(2.5) = 3937.5$ N.
Road load at 20 m/s: 176.6 + 158.4 = 335.0 N, so $F_b = 3602.5$ N.
$T_b = 3602.5(0.31) = 1116.8$ N m in total at the wheels.
$t_s = 20/2.5 = 8$ s, $d_s = 400/5 = 80$ m.

Result. $F_b = 3.60$ kN, $T_b = 1117$ N m, $t_s = 8$ s, $d_s = 80$ m.

Physical interpretation. Road load does 8.5% of the braking at the start, and less as speed falls.

Engineering significance. Referred to the motor, T_b is only 124 N m, which is within its capability. Motor torque is not the limit for this stop; power is.

Example 6: Recoverable energy, regenerative efficiency and comparison with friction braking

Problem. For the stop in Example 5, find kinetic energy, recoverable energy with $\kappa = 0.8$ and regenerative braking efficiency, and compare with friction braking alone.

Given. $\kappa = 0.8$. Regeneration power limit 50 kW at the DC side, with fade between 15 and 5 km/h.

Required. E_k, $E_{b,w}$, E_{rec}, η_{reg}, ERR.

Governing equations. (4.45)–(4.47).

Solution. $E_k = \frac{1}{2}(1575)(20^2) = 315$ kJ $= 87.5$ Wh.
Integrating $F_b v$ over 8 s, with F_a falling as speed falls: $E_{b,w} = 81.9$ Wh. Road load took the other 5.6 Wh.
Ideal chain value: 0.8(0.875)(81.9) = 57.3 Wh.
Peak wheel power is 72.1 kW, so $0.8 \times 0.875 \times 72.1 = 50.4$ kW at the DC side, marginally above the 50 kW limit. Numerical integration including the limit and low-speed fade gives $E_{\text{rec}} = 56.1$ Wh.

$\eta_{\text{reg}} = 56.1/81.9 = 68.5\%$; ERR $= 56.1/87.5 = 64.1\%$.
Comparison:

Quantity	Friction only	Blended ($\kappa = 0.8$)
Energy returned to battery (Wh)	0	56.1
Energy lost as heat or conversion loss (Wh)	81.9	25.8
Regenerative efficiency η_{reg}	0	68.5%

Result. $E_{\text{rec}} = 56.1$ Wh, $\eta_{\text{reg}} = 68.5\%$, against zero with friction only.

Physical interpretation. One stop returns the energy needed to drive about 390 m at 144 Wh/km. This looks small, but a city trip contains many stops.

Engineering significance. The limits cost only 2% of recovered energy here. They matter far more at higher deceleration or a low charge limit, which is why a higher battery charge rating directly yields extra recovery.

Figure 4.43 shows this event simulated. Speed falls linearly over 8 s from the two-second mark. The total demand stays almost flat, and regenerative torque tracks 80% of it until low speed, where the fade begins at about 15 km/h and friction takes over. Charging power peaks at 50 kW when the limit is active and falls with speed, since torque is nearly constant. Recovered energy is the integral of this curve; it flattens toward the end of the stop as speed vanishes.

Example 7: Blended braking torque calculation

Problem. At 50 km/h, the driver requests $a_d = 2$ m/s^2. Using a simplified demand $F = ma_d$, find the split between regenerative and friction torque when the shaft power limit is 40 kW. Repeat at high SoC, with a shaft limit of 10 kW. Neglect gear losses.

Given. $m = 1500$ kg, $r_w = 0.31$ m, $G = 9$, $T_{\text{pk}} = 280$ N m.

Required. T_b^*, T_{reg}, T_{fr}.

Governing equations. (4.49), (4.50), (4.51).

Solution. $F = 3000$ N, $T_b^* = 930$ N m. $\omega_m = 9(13.89)/0.31 = 403.2$ rad/s.
Case 1: $T_{m,\text{lim}} = \min(280,\ 40\,000/403.2) = 99.2$ N m, or 892.8 N m at the wheel.
$T_{\text{reg}} = 892.8$ N m, $T_{\text{fr}} = 37.2$ N m. Regenerative share: 96%.
Case 2: $T_{m,\text{lim}} = 24.8$ N m, or 223.2 N m at the wheel, so $T_{\text{fr}} = 706.8$ N m. Regenerative share: 24%.

Result. $T_{\text{reg}}/T_{\text{fr}} = 892.8/37.2$ at normal SoC and $223.2/706.8$ at high SoC.

Physical interpretation. The driver feels the same deceleration, but the source of the braking changes completely.

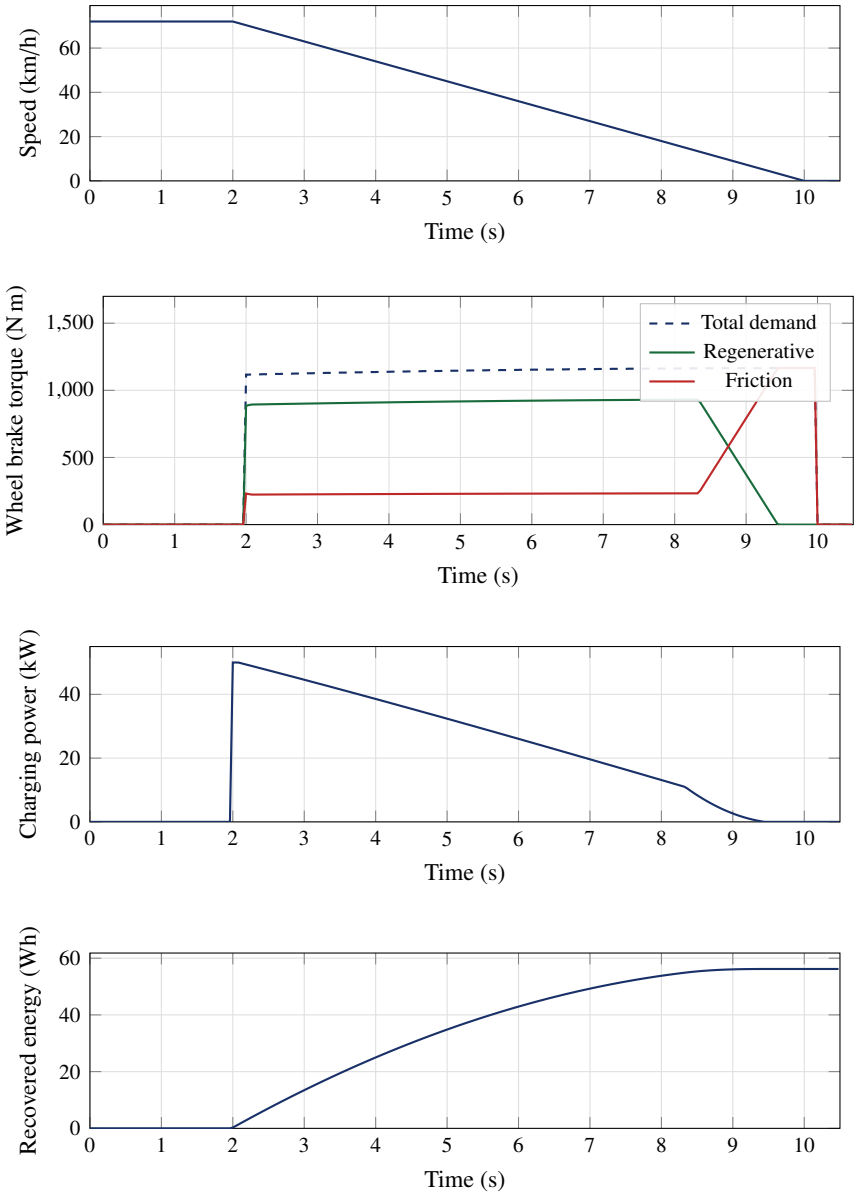


Figure 4.43: Simulated stop of the hypothetical BEV from 72 km/h: speed, wheel braking torque, battery charging power and cumulative recovered energy.

Engineering significance. The friction system must absorb the entire difference within milliseconds as limits change, which is the essential job of blending control.

4.6.17 MATLAB/Simulink Implementation of Brake Control and Regeneration

Figure 4.44 and Figure 4.45 show the conceptual structure of the BCU and of the regenerative capability calculation. Table 4.12 gives the details for both.

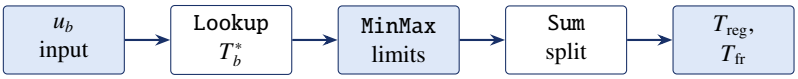


Figure 4.44: Conceptual Simulink diagram of the brake control unit.

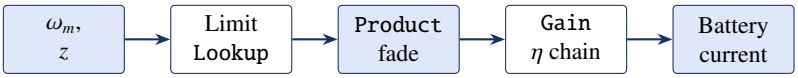


Figure 4.45: Conceptual Simulink diagram of the regenerative braking path.

Table 4.12: Brake control unit and regeneration: conceptual Simulink implementation

Inputs	u_b, ω_m, v, z, BMS P_{chg}, ABS flag
Outputs	T_{reg}, T_{fr}, T_m^*, charging power, recovered energy
States	Friction actuator lag; recovered energy accumulator
Equations	(4.38), (4.49)–(4.51), (4.44)
Blocks	`Lookup Table`, `MinMax`, `Sum`, `Rate Limiter`, `Switch`, `Integrator`
Feedback	Wheel speed, SoC and charge limit
Solver	Fixed step 1–10 ms if pressure dynamics are modelled
Initial conditions	Zero brake pressure
Assumptions	No wheel lock unless a slip model is included
Simulation outputs	Torque split, κ, η_{reg}, ERR

4.7 Modelling of Vehicle Control Strategy

4.7.1 Introduction to EV Vehicle Control

The vehicle control strategy is the logic that decides, at each moment, what torque the powertrain should deliver. It sits between human intention and the machines. A good strategy gives a car that responds predictably, protects the hardware, uses energy well and behaves safely when something fails.

4.7.2 Vehicle Control Unit and Supervisory Control

The **vehicle control unit** (VCU) is the ECU that hosts this logic. Figure 4.46 shows its connections. Inputs on the left are pedal positions, gear selector, ignition and vehicle signals. On the right, the VCU exchanges limits and SoC with the BMS, sends a torque command to the motor controller, receives speed, and commands thermal systems and the DC-DC converter. The links are normally CAN or Ethernet. The figure shows that the VCU is the hub through which every torque request passes.

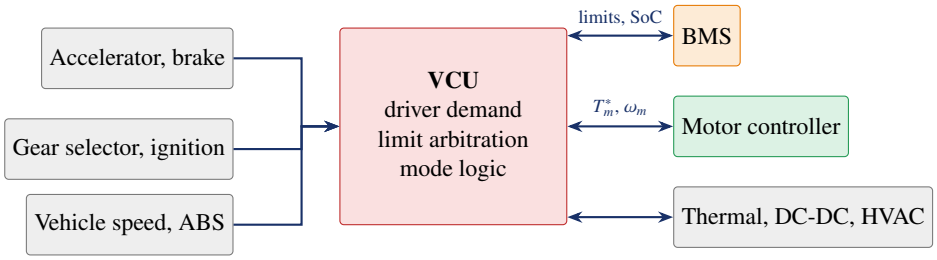


Figure 4.46: Vehicle control unit architecture.

Supervisory control refers to the upper decision layer. Figure 4.47 shows the usual three-layer hierarchy. The supervisory layer, running about every 100 ms, chooses modes and manages energy. The vehicle dynamics layer, about 10 ms, computes torque demand. The component layer, about 100 μs, runs the current loops and cell protection. Commands go down and measurements return up. This separation of time scales simplifies design: each layer can treat the layer below as an ideal actuator.

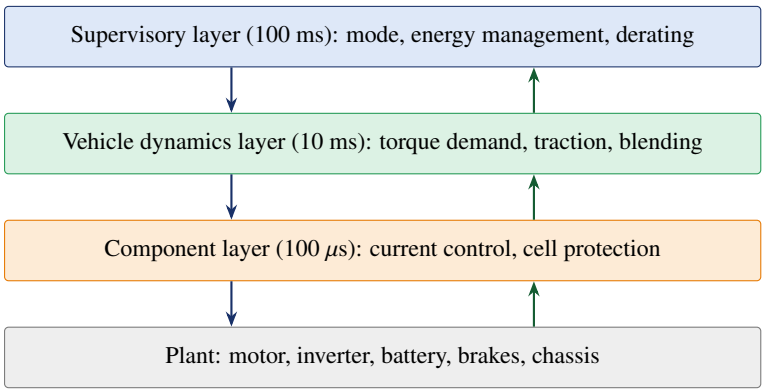


Figure 4.47: Supervisory vehicle control hierarchy.

4.7.3 Driver Demand Processing and Pedal Interpretation

The accelerator pedal does not command power directly. The VCU interprets it through a map from pedal position and speed to torque. A **linear** map uses (4.7). A **progressive** map uses a smaller slope at low pedal for fine control:

$$T_{\text{dem}} = T_{\max}(\omega_m)\, u_a^{\gamma}, \qquad \gamma \geq 1. \tag{4.53}$$

With $\gamma = 1.5$, half pedal gives 35% of available torque. Many EVs also add a **lift-off** regeneration term, and at very low speed a creep torque that imitates an automatic car:

$$T_{\text{lift}} = -k_{\text{lift}}\, f_v(v)\,(1 - u_a/u_0), \quad u_a < u_0; \qquad T_{\text{creep}} = k_c \max(0, v_c - v), \tag{4.54}$$

where u_0 is the zero-torque pedal position, k_{lift} the lift-off torque in N m, k_c the creep gain in N m s/m and v_c the creep speed in m/s. **One-pedal driving** is simply a large k_{lift}. The brake pedal, as already described, is interpreted by the BCU, and when both pedals are pressed a brake-override rule removes the traction torque.

4.7.4 Torque Command Generation and Motor Torque Control

Figure 4.48 shows the path from pedal to torque command. The map and additional terms are summed, a rate limiter restricts jerk, and a saturation stage applies limits. The result T_m^* goes to the motor controller, where it becomes current references (Unit 3). The rate limit is important for comfort, because a motor can step torque in milliseconds, and an unrestricted step would jerk the driveline. The total demand is

$$T_{\text{dem}} = f(u_a, \omega_m) + T_{\text{lift}} + T_{\text{creep}} + T_{\text{brk}}, \tag{4.55}$$

and the rate limit in discrete form is

$$T_k = T_{k-1} + \text{sat}_{-r\Delta t}^{r\Delta t}(T_{\text{dem},k} - T_{k-1}), \tag{4.56}$$

with r in N m/s.

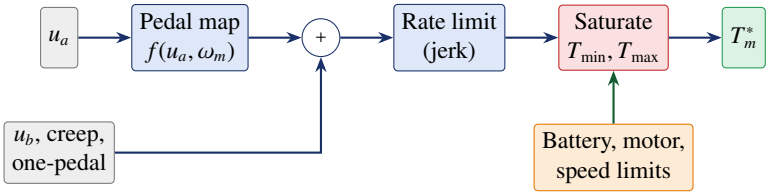


Figure 4.48: Torque demand generation.

4.7.5 Motor, Battery, Speed and Thermal Limits

Figure 4.49 shows how limits converge on a single block. Each limit gives a maximum and a minimum, and the tightest wins. The **motor limit** is the envelope,

$$T_{\text{mot}}(\omega_m) = \min\left(T_{\text{pk}}, \frac{P_{\text{pk}}}{|\omega_m|}\right). \tag{4.57}$$

The **battery power limit** gives speed-dependent torque limits. With P_{dis} and P_{chg} the BMS limits,

$$T_{\text{bat}}^{+} = \frac{(P_{\text{dis}} - P_{\text{aux}})\,\eta_m \eta_{\text{inv}}}{\omega_m}, \qquad T_{\text{bat}}^{-} = -\frac{P_{\text{chg}} + P_{\text{aux}}}{\eta_m \eta_{\text{inv}}\,\omega_m}. \tag{4.58}$$

In the charge direction the auxiliaries absorb some regenerated power, hence the plus sign. The **thermal limit** scales torque by a derating factor k_Θ, going from 1 at temperature Θ_1 to a low value at Θ_2,

$$k_\Theta = \mathrm{sat}^{1}_{k_{\min}}\left(1 - \frac{\Theta - \Theta_1}{\Theta_2 - \Theta_1}\right). \tag{4.59}$$

A **speed limit** tapers torque to zero as v approaches $v_{\max}$. Finally the **torque command saturation** is

$$T_m^* = \min(\max(T_{\mathrm{dem}}, T_{\min}), T_{\max}), \tag{4.60}$$

$$T_{\max} = k_\Theta \min(T_{\mathrm{mot}}, T_{\mathrm{bat}}^{+}), \qquad T_{\min} = -k_\Theta \min(T_{\mathrm{mot}}, \left|T_{\mathrm{bat}}^{-}\right|). \tag{4.61}$$

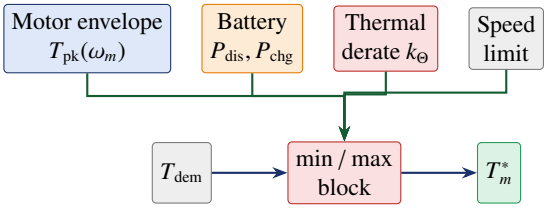


Figure 4.49: Battery and motor constraint management.

4.7.6 Powertrain Operating Modes

Figure 4.50 shows a typical state machine, which defines the following modes:

- **Standby.** Contactors are open. Key-on and successful self-checks lead to Ready.
- **Standstill (Ready).** High voltage is live with $T_m^* = 0$, or creep torque in D.
- **Motoring.** Forward traction with $T_m^* > 0$.
- **Regenerative braking.** Forward motion with $T_m^* < 0$, entered by a brake press or lift-off.
- **Reverse.** Selected at $v \approx 0$. Torque sign reversed and speed limited to a low value.
- **Fault/limp.** Torque is reduced or removed, according to severity.

Each arc in the figure carries its guard condition. Transitions between Forward and Reverse require $v \approx 0$, because a direct change of sign at speed would shock the driveline. This makes the state machine a safety mechanism as well as a sequencer.

4.7.7 Traction Control Overview

Traction control limits wheel slip when torque exceeds what the tyres can transmit. Slip during traction is $\lambda = (\omega_w r_w - v)/(\omega_w r_w)$. If λ exceeds about 0.1–0.15, past the peak of the tyre curve, then the VCU reduces torque. An electric motor is much better at this than an engine, because torque can be reduced within a few milliseconds.

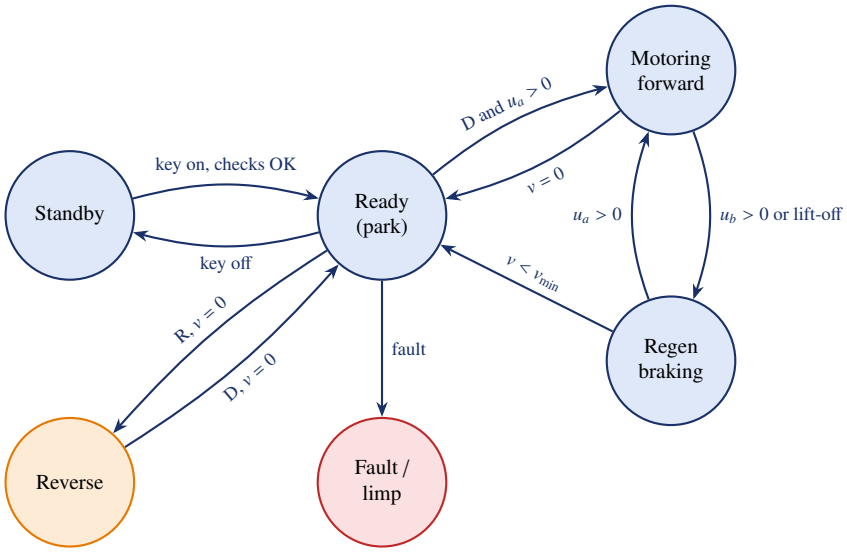


Figure 4.50: Operating mode state machine.

4.7.8 Protection and Fault Conditions

The VCU reacts to over-voltage, under-voltage, over-current, over-temperature, insulation faults, sensor disagreement and CAN timeouts. Reactions are graded: a warning, a derate, limp mode or an immediate torque cut with contactors opening. The accelerator signal typically uses two redundant sensors, and a plausibility check forces $T_m^* = 0$ if they disagree.

4.7.9 Energy Management and Strategy Selection

Energy management in a BEV has fewer degrees of freedom than in a hybrid, because there is only one energy source, but several choices remain: torque split in a dual-motor car, regeneration strength, HVAC power, thermal conditioning of the pack and eco-mode power limits. The choice of control strategy depends on available computing power, safety requirements and the benefit gained. Table 4.13 summarises the options, and Section 4.13 treats each in depth.

Table 4.13: Control approaches in vehicle control

Approach	Typical VCU role
Rule-based	Mode selection and derating
Threshold-based	Temperature and SoC limits with hysteresis
State-machine-based	Start-up, shut-down and fault handling
PI/PID-based	Cruise control, speed limit, current loops
Optimisation-based	Offline tuning of maps, torque split
Model predictive control	Eco-driving, advanced torque vectoring
Fuzzy logic	Regeneration share, driver-style adaptation
Energy-efficient torque control	Efficiency-map-based dual-motor split

As the table suggests, production controllers remain mostly rule-based and state-machine-

based, as these are simple to certify, with optimisation used offline to generate their maps.

4.7.10 Rule-Based Control Strategy

Figure 4.51 gives a rule-based priority chain. Fault checks come first, the brake next and the accelerator last, so that a safety-related condition can never be masked by a traction request. Any branch leads directly to a defined torque action.

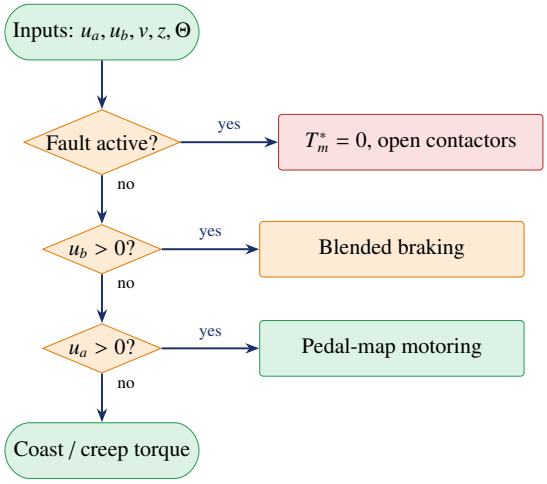


Figure 4.51: Rule-based vehicle control strategy.

4.7.11 Model Predictive Control Architecture

Figure 4.52 shows MPC in vehicle control. An optimiser uses a prediction model to choose a sequence of future commands that minimise a cost while honouring limits. Only the first command is applied, and the process repeats at the next step using the newly measured state. Section 4.13 gives the mathematical form.

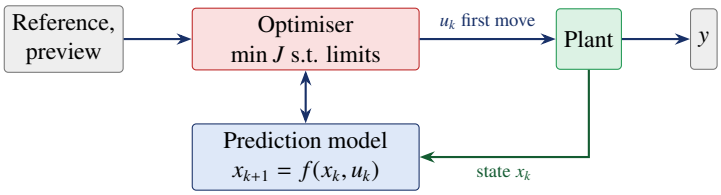


Figure 4.52: Model predictive control architecture in vehicle control.

4.7.12 Acceleration and Braking Control

Cruise control and adaptive cruise control wrap a PI loop around the same torque path. An acceleration controller computes

$$T_{\text{dem}} = \frac{r_w}{G\eta_g}\left(\delta m a^* + \hat{F}_{\text{res}}(v)\right) + K_{p,a}(a^* - a), \tag{4.62}$$

which is feedforward of inertia and road load plus feedback on the acceleration error. A negative result passes to the blended brake model.

4.7.13 Supervisory Control Logic

Algorithms 8 to 11 set out the logic. Algorithm 8 is the top-level task, and calls the other three: demand generation, mode selection and limit application. Every feedback signal in the earlier figures (speed, SoC, temperature, fault flags) is read here.

Algorithm 8 EV supervisory control

Input: u_a, u_b, gear g_s, v, ω_m, z, Θ, fault flags, BMS limits
Output: T_m^*, T_{fr}^*, mode M, contactor and auxiliary commands

$M \leftarrow$ Algorithm 10
if $M =$ Fault **then**
    Reduce T_m^* according to fault class; set warning
else
    $T_{\text{dem}} \leftarrow$ Algorithm 9
    $T_m^* \leftarrow$ Algorithm 11
    if $T_{\text{dem}} < 0$ **then** call Algorithm 6
    end if
end if
Schedule HVAC, cooling and DC-DC from Θ and z

Algorithm 9 Torque demand generation

Input: u_a, u_b, ω_m, v, previous T_{k-1}, r, Δt, drive mode
Output: T_{dem}

$T \leftarrow T_{\max}(\omega_m)\, u_a^{\gamma}$
if $u_a < u_0$ **then** $T \leftarrow T + T_{\text{lift}}$
end if
if $v < v_c$ **and** $u_b = 0$ **then** $T \leftarrow T + T_{\text{creep}}$
end if
if $u_b > 0$ **then** $T \leftarrow -T_b^*/G$ ▷ brake override
end if
$T_{\text{dem}} \leftarrow T_{k-1} + \text{sat}_{-r\Delta t}^{r\Delta t}(T - T_{k-1})$

Algorithm 10 Operating mode selection

Input: Current mode M, key, gear g_s, u_a, u_b, v, fault flag
Output: New mode M

if fault **then** $M \leftarrow$ Fault
else if key off **then** $M \leftarrow$ Standby
else if $g_s =$ R **and** $|v| < v_{\min}$ **then** $M \leftarrow$ Reverse
else if $g_s =$ D **and** $u_b > 0$ **and** $v > v_{\min}$ **then** $M \leftarrow$ Regen
else if $g_s =$ D **and** $u_a > 0$ **then** $M \leftarrow$ Motoring
else if $v < v_{\min}$ **then** $M \leftarrow$ Ready
end if

Rule order in Algorithm 10 is significant: fault is checked first, so no other condition can override it.

Algorithm 11 Motor and battery power limitation

Input: T_{dem}, ω_m, P_{dis}, P_{chg}, P_{aux}, Θ
Output: T_m^*
1: $T_{\text{mot}} \leftarrow$ (4.57); $(T_{\text{bat}}^+, T_{\text{bat}}^-) \leftarrow$ (4.58)
2: $k_\Theta \leftarrow$ (4.59)
3: $T_{\max} \leftarrow k_\Theta \min(T_{\text{mot}}, T_{\text{bat}}^+)$
4: $T_{\min} \leftarrow -k_\Theta \min(T_{\text{mot}}, |T_{\text{bat}}^-|)$
5: $T_m^* \leftarrow \min(T_{\max}, \max(T_{\min}, T_{\text{dem}}))$

4.7.14 MATLAB/Simulink Implementation of the Vehicle Control Strategy

Stateflow models the mode logic naturally, while map and limit blocks handle continuous torque calculation, as in Figure 4.53. Table 4.14 gives details.

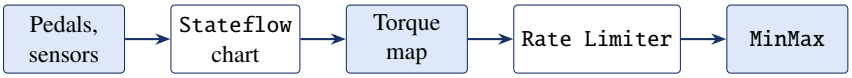


Figure 4.53: Conceptual Simulink diagram of the vehicle control strategy.

Table 4.14: Vehicle control strategy: conceptual Simulink implementation

Inputs	u_a, u_b, gear, v, ω_m, z, temperatures, limits
Outputs	T_m^*, T_{fr}^*, mode, derate flags
States	Stateflow state, rate limiter memory
Equations	(4.53)–(4.61)
Blocks	`Stateflow Chart`, `2-D Lookup Table`, `Rate Limiter`, `MinMax`, `Saturation Dynamic`
Feedback	Speed, battery limits, temperatures, fault flags
Solver	Discrete sample time, 10 ms, as in the real ECU
Initial conditions	Standby mode, zero torque
Assumptions	Ideal communication unless CAN delay is modelled
Simulation outputs	Mode history, active limit, delivered torque

4.8 Modelling of Vehicle Chassis

4.8.1 Introduction to Vehicle Chassis Modelling

The chassis model turns forces into motion. For energy and powertrain work a **longitudinal** model is sufficient. It treats the vehicle as a single mass moving along the road, ignores steering and suspension, and focuses on the forces in the direction of travel [7–9]. Figure 4.54 shows the picture: the tractive force at the driven wheels pushes forward, while aerodynamic drag and rolling resistance oppose motion. Weight acts at the centre of gravity. This single diagram is the origin of most of the equations in this unit.

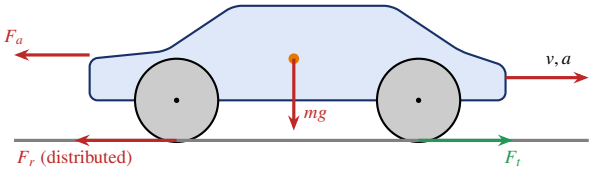


Figure 4.54: Vehicle chassis longitudinal dynamics.

4.8.2 Vehicle Parameters

Figure 4.55 defines the geometry and Table 4.15 lists the parameters and how each influences the model.

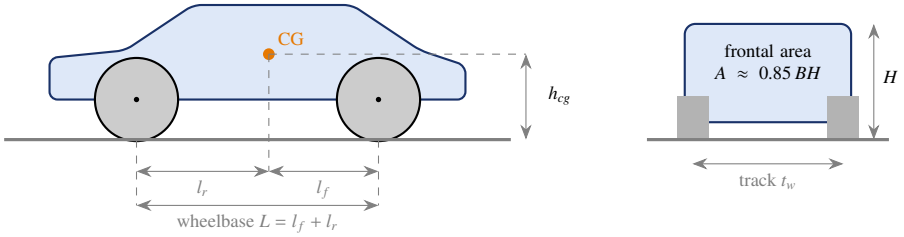


Figure 4.55: Centre of gravity and vehicle geometry.

Table 4.15: Vehicle chassis parameters

Parameter	Symbol	Typical car	Main effect
Vehicle mass	m	1200–2500 kg	Rolling, grade and inertial forces
Wheelbase	L	2.4–3.1 m	Load transfer, ride
Track width	t_w	1.4–1.7 m	Roll stability, cornering
Centre of gravity	h_{cg}, l_f, l_r	h_{cg} 0.45–0.6 m	Load transfer, traction limit
Frontal area	A	2.0–2.8 m^2	Aerodynamic drag
Drag coefficient	C_d	0.20–0.35	Aerodynamic drag
Wheel radius	r_w	0.28–0.37 m	Torque-to-force ratio
Rolling coefficient	C_{rr}	0.007–0.013	Rolling loss

A battery pack mounted low in the floor is responsible for a low h_{cg} compared with an engine car, which reduces load transfer and improves stability. Pack mass also raises m, and that affects every term except aerodynamic drag.

4.8.3 Forces Acting on a Moving Vehicle

Figure 4.56 shows a vehicle on a slope of angle θ. The body weight mg resolves into a component perpendicular to the road, which sets the normal loads F_{zf} and F_{zr}, and a component $F_g = mg\sin\theta$ along the road. Five forces act along the road: tractive force, rolling resistance, aerodynamic drag, gradient force and inertia. The figure directly corresponds to (4.71).

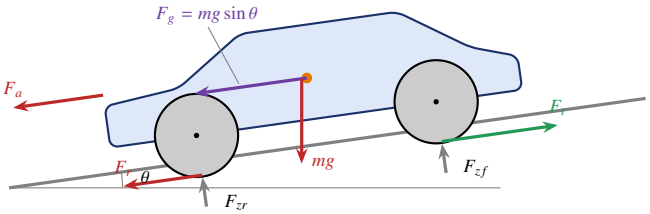


Figure 4.56: Forces acting on a moving vehicle on a gradient.

4.8.4 Newton's Second Law for Vehicle Motion

Applying Newton's second law along the road gives

$$m_{\text{eq}}\frac{dv}{dt} = F_{\text{tractive}} - F_{\text{rolling}} - F_{\text{aero}} - F_{\text{gradient}}, \tag{4.63}$$

where $m_{\text{eq}} = \delta m$ includes rotating inertia, as derived below. Forces are in N, speed in m/s and dv/dt in m/s^2.

4.8.5 Rolling Resistance Force

A rolling tyre deforms at the contact patch. Material in the tyre wall is compressed and then rebounds, and the rubber returns less energy than it absorbed. This hysteresis shifts the normal force slightly ahead of the wheel centre by a distance e, as seen in Figure 4.57, creating a moment $F_z e$ that opposes rotation. Dividing that moment by the wheel radius gives an equivalent force at the road,

$$F_{\text{rolling}} = F_r = \frac{F_z e}{r_w} = C_{rr} F_z = C_{rr}\, mg\cos\theta, \tag{4.64}$$

with $C_{rr} = e/r_w$ the dimensionless rolling resistance coefficient. On a small grade $\cos\theta \approx 1$. C_{rr} rises gently with speed and depends on tyre pressure, temperature and road surface. A low-rolling-resistance EV tyre can have a value below 0.008.

4.8.6 Aerodynamic Drag Force

Moving air separates from the body, and the wake leaves a pressure deficit behind the car. The drag force is

$$F_{\text{aero}} = F_a = \tfrac{1}{2}\rho\, C_d A\,(v + v_w)^2, \tag{4.65}$$

where ρ is air density in kg/m^3 (about 1.2 at 20°C at sea level), C_d is the drag coefficient, A the frontal area in m^2, and v_w the head-wind speed in m/s. The product $\frac{1}{2}\rho v^2$ is the dynamic pressure in Pa. Dimensionally, kg/m^3×m^2×m^2/s^2 = kg m/s^2 = N. With zero wind, the reference car has $F_a = 0.396v^2$.

4.8.7 Gradient Resistance

The weight component along a slope is

$$F_{\text{gradient}} = F_g = mg\sin\theta, \qquad \tan\theta = \frac{\%\ \text{grade}}{100}. \tag{4.66}$$

For a grade below about 10%, $\sin\theta \approx \tan\theta$, so a 1% grade costs about $0.01mg$. Going downhill makes F_g negative, and gravity helps.

4.8.8 Acceleration Resistance and Rotational Mass Correction

Accelerating the car requires linear inertial force ma plus torque to spin up the rotating parts. The kinetic energy of the vehicle is

$$E_k = \tfrac{1}{2}mv^2 + \tfrac{1}{2}J_w\omega_w^2 + \tfrac{1}{2}J_m\omega_m^2, \tag{4.67}$$

with J_w the total wheel inertia, J_m the motor rotor and gear inertia at motor speed, in $\text{kg}\,\text{m}^2$. Substituting $\omega_w = v/r_w$ and $\omega_m = Gv/r_w$,

$$E_k = \tfrac{1}{2}\left(m + \frac{J_w}{r_w^2} + \frac{G^2 J_m}{r_w^2}\right)v^2 = \tfrac{1}{2}\delta m v^2. \tag{4.68}$$

This defines the **rotational mass factor**

$$\delta = 1 + \frac{J_w + G^2 J_m}{m r_w^2}, \tag{4.69}$$

which gives the **acceleration resistance**

$$F_{\text{acceleration}} = F_{\text{acc}} = \delta m\frac{dv}{dt}. \tag{4.70}$$

For example, with $J_w = 4\ \text{kg}\,\text{m}^2$, $J_m = 0.05\ \text{kg}\,\text{m}^2$, $G = 9$, $m = 1500$ kg and $r_w = 0.31$ m, the value is $\delta = 1 + (4 + 4.05)/144.2 = 1.056$. Note that the motor term, multiplied by G^2, matches the four wheels despite its small rotor. The reference car uses $\delta = 1.05$. This is the **drivetrain inertia** effect.

4.8.9 Total Tractive Force

Rearranging (4.63) gives the complete force balance:

$$F_{\text{tractive}} = F_{\text{rolling}} + F_{\text{aero}} + F_{\text{gradient}} + F_{\text{acceleration}}. \tag{4.71}$$

Written out with the symbols used throughout,

$$F_t = C_{rr}mg\cos\theta + \tfrac{1}{2}\rho C_d A v^2 + mg\sin\theta + \delta m\frac{dv}{dt}. \tag{4.72}$$

- F_t: force the powertrain must deliver at the tyre contact, N.

- $C_{rr} mg \cos\theta$: roughly constant, it dominates at low speed.
- $\frac{1}{2}\rho C_d A v^2$: grows with the square of speed, it dominates at highway speed.
- $mg \sin\theta$: recoverable, because it stores potential energy.
- $\delta m\, dv/dt$: recoverable, because it stores kinetic energy.

Only the rolling and drag terms represent energy permanently lost to the environment.

4.8.10 Wheel Torque, Vehicle Power, Acceleration and Speed Integration

Wheel torque in N m and vehicle power in W are

$$T_w = F_t r_w, \qquad P_w = F_t v. \tag{4.73}$$

Given a known tractive force, vehicle acceleration is

$$a = \frac{dv}{dt} = \frac{F_t - F_r - F_a - F_g}{\delta m}, \tag{4.74}$$

which is (4.12) without a friction brake. Discretising over step Δt gives **vehicle speed integration**:

$$v_{k+1} = v_k + a_k \Delta t, \qquad x_{k+1} = x_k + v_k \Delta t. \tag{4.75}$$

Algorithms 12 and 13 implement these steps in the forward and backward directions respectively.

Algorithm 12 Vehicle longitudinal dynamics (forward step)

Input: T_w, F_b, v_k, θ, Δt, chassis parameters
Output: v_{k+1}, x_{k+1}, a_k

$F_t \leftarrow \min(T_w/r_w,\ \mu F_{z,\text{drive}})$ ▷ tyre limit
$F_r \leftarrow C_{rr} mg \cos\theta$ **if** $v_k > 0$ **else** 0
$F_a \leftarrow \frac{1}{2}\rho C_d A v_k^2$; $F_g \leftarrow mg \sin\theta$
$a_k \leftarrow (F_t - F_b - F_r - F_a - F_g)/(\delta m)$
if $v_k = 0$ **and** $a_k < 0$ **and** $\theta = 0$ **then** $a_k \leftarrow 0$
end if
$v_{k+1} \leftarrow \max(0, v_k + a_k \Delta t)$; $x_{k+1} \leftarrow x_k + v_k \Delta t$

Algorithm 13 Tractive force calculation (backward step)

Input: v_k in m/s, a_k, grade s in %, parameters
Output: $F_r, F_a, F_g, F_{\text{acc}}, F_t, T_w, P_w$

$\theta \leftarrow \arctan(s/100)$
$F_r \leftarrow C_{rr} mg \cos\theta$ **if** $v_k > 0$ **else** 0
$F_a \leftarrow \frac{1}{2}\rho C_d A v_k^2$; $F_g \leftarrow mg \sin\theta$; $F_{\text{acc}} \leftarrow \delta m a_k$
$F_t \leftarrow F_r + F_a + F_g + F_{\text{acc}}$
$T_w \leftarrow F_t r_w$; $P_w \leftarrow F_t v_k$

4.8.11 Tyre Characteristics and Tyre-Road Interaction

Figure 4.57 shows the wheel on the left and the tyre's force characteristic on the right. A tyre transmits force only by slipping slightly. Traction slip is

$$\lambda = \frac{\omega_w r_w - v}{\omega_w r_w}. \tag{4.76}$$

The ratio $\mu = F_x/F_z$ rises steeply at small slip, peaks near $\lambda \approx 0.1$–0.2 and then falls [10]. The peak is about 0.9–1.0 on dry roads and much less on wet or icy roads. The maximum transmissible force is therefore

$$F_{x,\max} = \mu F_{z,\text{drive}}, \tag{4.77}$$

which caps the useful torque at the wheels regardless of the motor. At normal levels of force, slip is small enough that simple models ignore it.

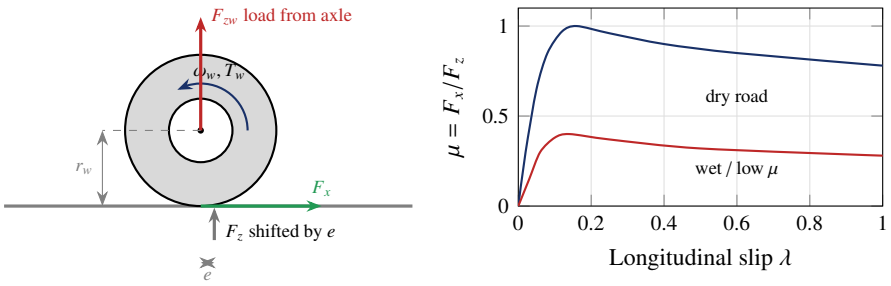


Figure 4.57: Wheel and tyre force representation, with illustrative slip curves.

4.8.12 Wheel Rotational Dynamics

When slip matters, each wheel needs its own rotational equation. Summing torques about the axle,

$$J_w \frac{d\omega_w}{dt} = T_w - T_{\text{fr}} - F_x r_w - F_z e. \tag{4.78}$$

Figure 4.58 shows how this couples with the chassis. Drive torque minus the tyre reaction accelerates the wheel. Wheel and vehicle speed together set slip, slip sets tyre force, and the tyre force accelerates the chassis. Since a wheel has small inertia, and F_x is a steep function of slip, this model is stiff and needs a small step or implicit solver. Traction control and ABS studies need it, while range studies do not.

4.8.13 Load Transfer

Inertia acts at the CG, which is above the road. During acceleration this creates a moment that shifts load from front to rear; during braking it shifts it from rear to front. Taking moments about the rear contact point on a level road gives the front axle load, and similarly for the rear:

$$F_{zf} = \frac{mg\, l_r - \delta m a\, h_{cg}}{L}, \qquad F_{zr} = \frac{mg\, l_f + \delta m a\, h_{cg}}{L}. \tag{4.79}$$

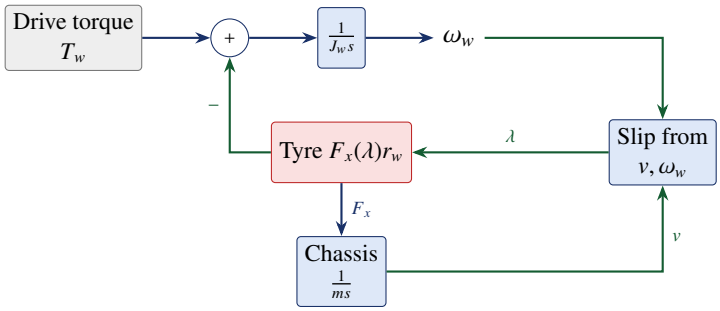


Figure 4.58: Wheel rotational dynamics coupled to the chassis.

Figure 4.59 illustrates the effect. The load transfer $\Delta F_z = \delta m a\, h_{cg}/L$ loads the rear tyres under acceleration, so a rear-driven car has more available grip than a front-driven one. For instance, at $a = 3$ m/s^2, $h_{cg} = 0.55$ m and $L = 2.7$ m, the transfer is 963 N, which is 6.5% of the total weight.

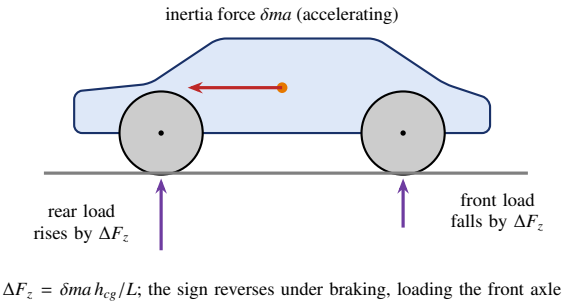


Figure 4.59: Load transfer during acceleration; the direction reverses in braking.

4.8.14 Vehicle Dynamics Assumptions

The longitudinal model in this unit assumes that the car is a rigid point mass, that the road and tyre conditions are uniform, that there is no lateral motion and that suspension pitch can be neglected. Rolling and drag coefficients are taken as constant, and tyre slip is neglected. These assumptions are valid for energy prediction, and are poor for handling or limit braking.

4.8.15 Numerical Examples

The following examples form one chain. The reference BEV of Table 4.3 travels at 60 km/h ($v = 16.67$ m/s) up a 4% grade ($\theta = \arctan 0.04 = 2.29°$) while accelerating at 0.5 m/s^2, with no wind. These operating values are hypothetical.

Example 8: Calculation of rolling resistance

Problem. Find the rolling resistance force of the car in the stated condition.

Given. $m = 1500$ kg, $C_{rr} = 0.012$, $g = 9.81$ m/s^2, $\theta = 2.29°$.

Required. F_r.

Governing equations. (4.64).

Solution. $F_r = 0.012(1500)(9.81)\cos 2.29^\circ = 176.4$ N.

Result. $F_r = 176.4$ N.

Physical interpretation. The small grade barely changes the normal load, as $\cos\theta = 0.9992$.

Engineering significance. With $C_{rr} = 0.008$ the force falls to 117.6 N, which shows the value of low-rolling-resistance tyres.

Example 9: Calculation of aerodynamic drag

Problem. Find the drag force at 60 km/h in still air, and at 120 km/h.

Given. $\rho = 1.2$ kg/m^3, $C_d = 0.30$, $A = 2.2$ m^2.

Required. F_a at both speeds.

Governing equations. (4.65).

Solution. $F_a = 0.5(1.2)(0.30)(2.2)(16.67)^2 = 110.0$ N. At 120 km/h (33.33 m/s), $F_a = 440.0$ N.

Result. $F_a = 110$ N at 60 km/h and 440 N at 120 km/h.

Physical interpretation. Doubling speed quadruples the force.

Engineering significance. Drag power rises eightfold, from 1.83 kW to 14.67 kW, so motorway speed dominates highway consumption.

Example 10: Calculation of gradient resistance

Problem. Find the gradient force on the 4% grade.

Given. $m = 1500$ kg, $\theta = 2.29^\circ$.

Required. F_g.

Governing equations. (4.66).

Solution. $F_g = 1500(9.81)\sin 2.29^\circ = 588.1$ N. The approximation $mg\tan\theta = 588.6$ N differs by only 0.1%.

Result. $F_g = 588.1$ N.

Physical interpretation. Even a mild slope exceeds rolling and drag forces combined at this speed.

Engineering significance. This force stores potential energy, part of which regeneration can recover on the way down.

Example 11: Calculation of total tractive force

Problem. Find the total tractive force, and also the acceleration if 1800 N acts at the wheels on a level road at the same speed.

Given. Results of Examples 8 to 10; $\delta = 1.05$, $a = 0.5$ m/s^2.

Required. F_{acc}, F_t, and a for the level case.

Governing equations. (4.70), (4.72), (4.74).

Solution. $F_{\text{acc}} = 1.05(1500)(0.5) = 787.5$ N.
$F_t = 176.4 + 110.0 + 588.1 + 787.5 = 1662.1$ N.
Level road: $F_r = 176.6$ N, $F_a = 110.0$ N, so $a = (1800 - 286.6)/1575 = 0.96$ m/s^2.

Result. $F_t = 1662$ N; $a = 0.96$ m/s^2.

Physical interpretation. Acceleration and gradient forces, which store energy, make up 83% of the total.

Engineering significance. Rolling and drag are comparatively small here, but they are the only terms that are permanently lost.

Example 12: Calculation of wheel torque

Problem. Find the wheel torque and wheel speed for the tractive force of Example 11.

Given. $F_t = 1662.1$ N, $r_w = 0.31$ m.

Required. T_w, ω_w.

Governing equations. (4.20).

Solution. $T_w = 1662.1(0.31) = 515.2$ N m; $\omega_w = 16.67/0.31 = 53.76$ rad/s.

Result. $T_w = 515.2$ N m at 53.8 rad/s.

Physical interpretation. This is the total for the driven axle, shared equally by its two wheels.

Engineering significance. Half-shafts and wheel bearings are designed for such values, and for much greater peaks.

Example 13: Calculation of motor torque and speed

Problem. Find the motor torque and speed for this wheel condition.

Given. $T_w = 515.2$ N m, $G = 9$, $\eta_g = 0.97$.

Required. T_m, ω_m.

Governing equations. (4.21).

Solution. $T_m = 515.2/(9 \times 0.97) = 59.0$ N m.
$\omega_m = 9(53.76) = 483.9$ rad/s, or 4621 r/min.

Result. $T_m = 59.0$ N m at 4621 r/min.

Physical interpretation. The gear multiplies torque and speed by opposite factors, and the loss falls on the torque side.

Engineering significance. The motor is at about a fifth of its peak torque and just above its peak-power base speed.

Example 14: Calculation of motor mechanical power

Problem. Find the wheel and motor mechanical powers, and verify them.

Given. $F_t = 1662.1$ N, $v = 16.67$ m/s, $T_m = 59.02$ N m, $\omega_m = 483.9$ rad/s.

Required. P_w, P_m.

Governing equations. (4.73), (4.21).

Solution. $P_w = 1662.1(16.67) = 27.70$ kW.
$P_m = 59.02(483.9) = 28.56$ kW.
Check: $P_w/\eta_g = 27.70/0.97 = 28.56$ kW.

Result. $P_w = 27.7$ kW, $P_m = 28.6$ kW.

Physical interpretation. The difference of 0.86 kW is heat in the gears.

Engineering significance. Agreement between the force route and the torque route confirms that the conversions are consistent.

4.8.16 Influence of Vehicle Parameters on EV Performance

- **Vehicle mass.** Raises rolling, grade and inertial forces in proportion. Regeneration returns part of the inertial energy, but rolling loss is lost. A 10% mass increase raises urban consumption substantially and has less effect on the highway.

- **Frontal area and drag coefficient.** The product C_dA sets highway consumption; power rises with v^3. A 10% reduction in C_dA gives a 10% lower drag power at all speeds.
- **Wheel radius.** A larger radius lowers force for a given wheel torque and lowers motor speed for a given road speed. It is therefore equivalent to a change in G/r_w, and must be considered together with the gear ratio.
- **Rolling resistance.** Important at low and moderate speeds, and P_r rises only linearly with speed. Tyre choice is one of the cheapest ways to gain range.

Table 4.16 collects these and related factors, using the simulation results of Section 4.15. It groups each factor by whether it affects mainly urban or highway driving.

Table 4.16: Major factors affecting EV energy consumption

Factor	Urban	Highway	Mechanism
Mass	Strong	Moderate	Inertia, rolling force
C_dA	Weak	Very strong	v^2 force, v^3 power
C_{rr}	Strong	Moderate	Tyre hysteresis
Speed	Weak	Very strong	Drag
Stop frequency	Strong	Negligible	Braking loss, regeneration limits
Auxiliary load	Very strong	Moderate	Fixed power, long trip time
Regeneration	Strong	Weak	Recovers inertial energy
Ambient temperature	Strong	Moderate	HVAC, battery resistance
Road gradient	Route dependent	Route dependent	Potential energy, part recovered

4.8.17 MATLAB/Simulink Implementation of Vehicle Longitudinal Dynamics

Figure 4.60 shows the block structure, with the integrators forming the system states and road load fed back as a function of speed. Table 4.17 summarises the implementation.

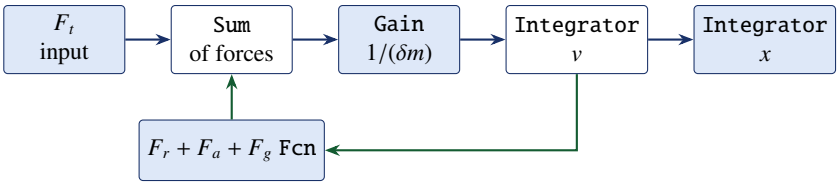


Figure 4.60: Conceptual Simulink diagram of vehicle longitudinal dynamics.

4.9 Sizing of Components: Steady-State Energy Balance Equation

Table 4.17: Vehicle longitudinal dynamics: conceptual Simulink implementation

Inputs	F_t or T_w, F_b, grade θ, wind v_w
Outputs	v, a, x, ω_w
States	v, x, and ω_w if a slip model is used
Equations	(4.72), (4.74), (4.75)
Blocks	`Sum`, `Gain`, `Fcn`, `Integrator` (limited at zero)
Feedback	Road load depends on v
Solver	`ode45` or fixed `ode4`; stiff solver if slip is modelled
Initial conditions	$v(0) = 0$, $x(0) = 0$
Assumptions	Rigid point mass, no slip, uniform road
Simulation outputs	Speed and distance traces, force breakdown

4.9.1 Introduction and Purpose of Component Sizing

Component sizing selects battery, inverter, motor and gear ratings so that a vehicle meets its targets at the lowest cost and mass. The sizing is iterative, since a bigger battery adds mass, which raises road load, which demands more power and energy. A **steady-state energy balance** is the natural first step: it shows how much power each component must handle at a defined operating point, with no integration required. The peak values are then treated in Section 4.11.

4.9.2 Steady-State Operation and the Vehicle Power Balance

In **steady-state operation** speed is constant, so $dv/dt = 0$ and all states are stationary. Stored kinetic energy does not change, and all input power goes to road load and losses. Figure 4.61 shows the steady power flow. Battery power divides between auxiliaries and the traction path. Traction power passes through inverter, motor and gear, and emerges at the wheel, where it meets road load.

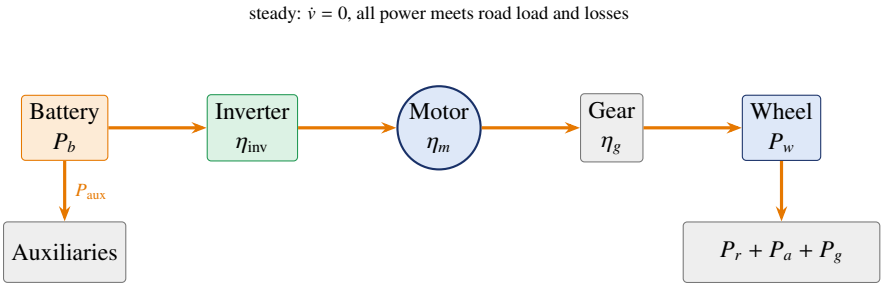


Figure 4.61: BEV steady-state power flow.

The starting relation is that wheel power equals tractive force times speed:

$$P_{\text{wheel}} = F_{\text{tractive}}\, v. \tag{4.80}$$

Substituting the force components of (4.72) gives the **vehicle power demand** in W:

$$P_w = \underbrace{C_{rr} m g \cos\theta\, v}_{P_r} + \underbrace{\tfrac{1}{2}\rho C_d A v^3}_{P_a} + \underbrace{m g \sin\theta\, v}_{P_g} + \underbrace{\delta m\, a\, v}_{P_{\text{acc}}}. \tag{4.81}$$

Each term has a different dependence on speed. Rolling and gradient power are proportional to v, while aerodynamic power rises with v^3. Figure 4.62 plots these terms for the reference car. On a level road, rolling power leads below about 75 km/h and drag leads above it. A modest 2% grade adds almost as much power as the entire level road load at that crossover, which shows why range falls so strongly in hilly country.

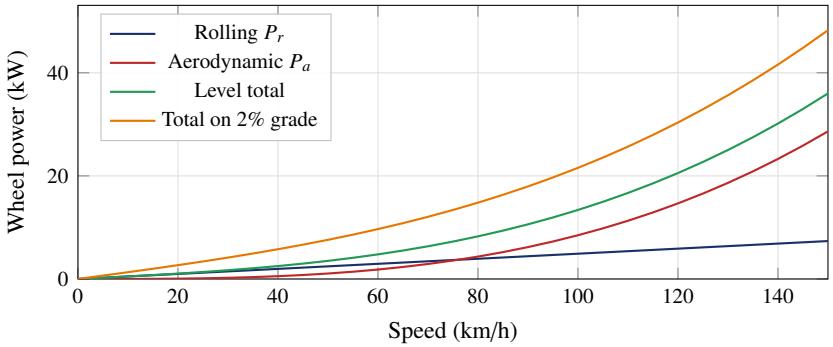


Figure 4.62: Vehicle power demand versus speed (calculated, hypothetical BEV).

4.9.3 Mechanical, Electrical and Battery Power

Follow the power from wheel to battery. In the motoring direction each efficiency divides:

$$P_m = \frac{P_w}{\eta_g} \qquad \text{mechanical power at the motor shaft,} \tag{4.82}$$

$$P_e = \frac{P_m}{\eta_m} \qquad \text{AC electrical power at the motor terminals,} \tag{4.83}$$

$$P_{\text{dc}} = \frac{P_e}{\eta_{\text{inv}}} \qquad \text{DC power at the inverter input,} \tag{4.84}$$

$$P_b = P_{\text{dc}} + P_{\text{aux}} \qquad \text{power at the battery terminals,} \tag{4.85}$$

$$P_{\text{oc}} = V_{oc} I = P_b + I^2 R_0 \qquad \text{chemical power released in the cell.} \tag{4.86}$$

Combining (4.82)–(4.85), the **battery-side power balance** for motoring is

$$P_b = \frac{P_w}{\eta_g \eta_m \eta_{\text{inv}}} + P_{\text{aux}}, \qquad P_w \geq 0. \tag{4.87}$$

With power flow reversed, for a share κ of negative wheel power,

$$P_b = \kappa P_w\, \eta_g \eta_m \eta_{\text{inv}} + P_{\text{aux}}, \qquad P_w < 0. \tag{4.88}$$

Auxiliary load always adds to battery power, whichever direction traction power flows, and it may absorb some regenerated power directly.

Figure 4.63 marks every interface along the chain. The distinction matters, because the literature often simply says "power" without naming the interface. Measured at the battery terminals, $P_b = V_t I$ already contains drivetrain losses but not the internal resistance loss, which sits to the left of the terminal. Only one efficiency applies across each block; applying them twice is the most frequent source of error in energy calculations. Figure 4.64 shows the efficiency chain itself, in which the drivetrain efficiency is $\eta_{\text{dt}} = \eta_{\text{inv}}\eta_m\eta_g = 0.875$ for the reference car, and the battery efficiency, $\eta_{\text{bat}} = V_t/V_{oc}$ during discharge, is treated separately.

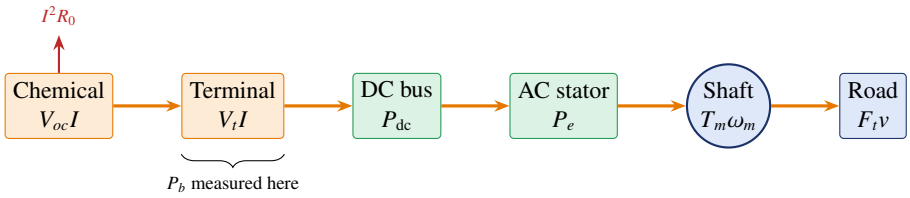


Figure 4.63: Battery-to-wheel energy conversion, showing each power interface.

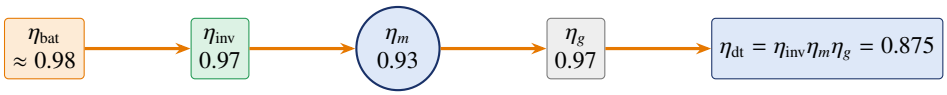


Figure 4.64: Component efficiency chain (reference BEV).

4.9.4 Motor, Inverter and Transmission Efficiency, Auxiliary Power

Component efficiency depends on the operating point. A motor efficiency map has its highest values at moderate torque and mid speeds. A gear pair is highly efficient, typically 0.95–0.98. An inverter normally exceeds 0.96 at moderate load, with efficiency falling at very low load since switching and gate-drive losses remain. Auxiliaries include the DC-DC converter, cooling pumps, and particularly HVAC, which can draw from about 0.5 kW to several kW. Unlike road load, auxiliary *power* is independent of speed, so its *energy per km* rises as speed falls.

4.9.5 Steady-State Energy Balance Equation

For a trip of duration T at constant speed v, energy equals power times time. Integrating (4.87) with $D = vT$ gives

$$E_b = \frac{(F_r + F_a + F_g)D}{\eta_g\eta_m\eta_{\text{inv}}} + P_{\text{aux}}\frac{D}{v}. \tag{4.89}$$

Dividing by distance gives the energy per metre in J/m, which is numerically a force in N. This is useful: the consumption of a steadily driven car in Wh/km is simply the road-load force divided by the efficiency and by 3.6, plus the auxiliary term,

$$e = \frac{1}{3.6}\left(\frac{F_{\text{road}}}{\eta_{\text{dt}}} + \frac{P_{\text{aux}}}{v}\right) \ \text{Wh/km}. \tag{4.90}$$

4.9.6 Regenerative Energy, Energy Losses and Their Allocation

Over a trip the **dynamic energy balance** must include storage terms. The wheel energy delivered by the powertrain equals dissipated road work plus any change in stored energy:

$$\int P_w \, dt = \int (F_r + F_a) v \, dt + \Delta\left(\tfrac{1}{2}\delta m v^2\right) + mg \, \Delta h. \tag{4.91}$$

Table 4.18 compares the two balances.

Table 4.18: Steady-state and dynamic energy balances

Aspect	Steady-state	Dynamic
Speed	Constant	Varies with time
Kinetic energy term	Zero	Exchanged and partly recovered
Calculation	Algebraic	Integrated over the cycle
Regeneration	Only downhill	Every deceleration
Result	Base load, continuous rating	Cycle energy, peak power

The steady balance is thus a special case of the dynamic balance with no storage change, which is why it can give continuous power but not peak power.

4.9.7 Operating Cases

- **Level-road cruising.** $\theta = 0$, $a = 0$: $P_w = P_r + P_a$. For the reference car at 90 km/h, $P_w = 10.6$ kW.
- **Hill climbing.** $P_g = mgv \sin\theta$ adds to demand, and dominates on steep roads.
- **Constant-speed driving.** This is the general steady case with $a = 0$ and any grade.
- **Acceleration.** $P_{\text{acc}} = \delta m a v$ rises with speed for a constant a, so peak power occurs at the end of the acceleration. This case is not steady, but the same power chain applies.
- **Deceleration.** $P_{\text{acc}} < 0$. If $|P_{\text{acc}}| < P_r + P_a$, the car simply coasts down and no braking is needed.
- **Regenerative braking.** When $P_w < 0$, (4.88) applies. A long descent is the steady version, in which $P_g < 0$ can power the battery.

4.9.8 Powertrain Component Selection, Constraints and Safety Margins

Component selection must satisfy limits on current, voltage, temperature and mechanical speed. A **design margin** is added because efficiency estimates are uncertain, batteries age, ambient conditions vary and parts vary between batches. Typical margins are 10–20% on power and 10–15% on energy, and larger on current where semiconductor junction temperature is critical.

Example 15: Steady-state BEV energy balance

Problem. The reference BEV cruises at 90 km/h up a 2% grade with HVAC on, giving $P_{\text{aux}} = 1$ kW. Find the rolling, aerodynamic and gradient powers, wheel power, motor shaft and input power, inverter input power, battery power and current, and overall efficiency.

Given. Table 4.3; $V_{oc} = 350.4$ V; $R_0 = 0.048\ \Omega$.

Required. $P_r, P_a, P_g, P_w, P_m, P_e, P_{\text{dc}}, P_b, I, \eta_{\text{dt}}, \eta_{\text{ov}}$.

Governing equations. (4.81)–(4.86), (4.15).

Solution. $v = 25$ m/s; $\theta = 1.146°$.
$F_r = 176.5$ N, so $P_r = 4.41$ kW.
$F_a = 0.396(625) = 247.5$ N, so $P_a = 6.19$ kW.
$F_g = 294.2$ N, so $P_g = 7.36$ kW.
$P_w = 4.41 + 6.19 + 7.36 = 17.96$ kW.
Shaft: $P_m = 17.96/0.97 = 18.51$ kW. At $\omega_m = 725.8$ rad/s this means $T_m = 25.5$ N m.
Motor input: $P_e = 18.51/0.93 = 19.91$ kW.
Inverter input: $P_{\text{dc}} = 19.91/0.97 = 20.52$ kW.
Battery: $P_b = 20.52 + 1.00 = 21.52$ kW.
Current from (4.15): $I = 61.9$ A, so $V_t = 350.4 - 61.9(0.048) = 347.4$ V.
Internal loss: $I^2 R_0 = 0.18$ kW.
$\eta_{\text{dt}} = 17.96/20.52 = 0.875$; overall $\eta_{\text{ov}} = P_w/P_b = 17.96/21.52 = 0.834$.

Result. $P_w = 17.96$ kW, $P_b = 21.52$ kW, $I = 61.9$ A, $\eta_{\text{ov}} = 83.4\%$.

Physical interpretation. Stage losses are 0.56 kW (gear), 1.39 kW (motor) and 0.62 kW (inverter). Check: $17.96 + 0.56 + 1.39 + 0.62 + 1.00 = 21.53$ kW.

Engineering significance. Since only 25.5 N m is needed, the motor runs at light load, where a real efficiency map would be below 0.93. This is a limit of constant-efficiency analysis.

Figures 4.65 and 4.66 show this result graphically. Figure 4.65 shows power remaining after each stage, from 21.52 kW at the battery to 17.96 kW at the road, with the red caps marking stage losses. Figure 4.66 ranks the destinations of the power. Gradient, drag and rolling take most of it, and the motor is the largest of the conversion losses.

Algorithms 14 and 15 give the calculation in reusable form. Note that Algorithm 15 handles both signs of power, which is the only difference from (4.87).

4.10 Modelling of Battery and Battery Cell

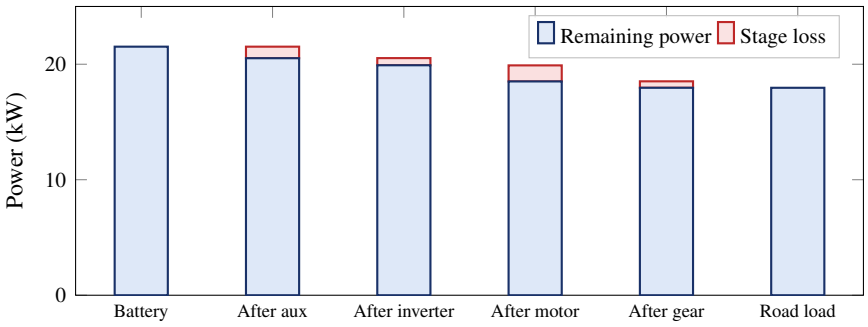


Figure 4.65: Energy balance across the drivetrain for Example 15.

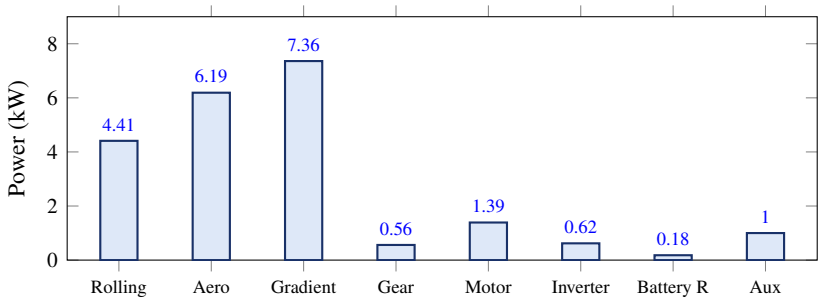


Figure 4.66: Power consumed in each destination for Example 15.

Algorithm 14 Steady-state energy balance calculation

Input: v, grade, P_{aux}, vehicle and efficiency data, V_{oc}, R_0
Output: $P_r, P_a, P_g, P_w, P_b, I, \eta_{\text{ov}}$, loss list

1: Compute F_r, F_a, F_g with $a = 0$
2: $P_w \leftarrow (F_r + F_a + F_g)v$
3: $P_m \leftarrow P_w/\eta_g$; $P_e \leftarrow P_m/\eta_m$; $P_{\text{dc}} \leftarrow P_e/\eta_{\text{inv}}$
4: $P_b \leftarrow P_{\text{dc}} + P_{\text{aux}}$; $I \leftarrow$ (4.15)
5: Losses $\leftarrow \{P_m - P_w,\ P_e - P_m,\ P_{\text{dc}} - P_e,\ I^2R_0\}$
6: Check $|P_b - P_w - \sum(\text{drivetrain losses}) - P_{\text{aux}}| < \varepsilon$

Algorithm 15 Motor and inverter power estimation

Input: T_m, ω_m, maps $\eta_m(T, \omega)$, $\eta_{\text{inv}}(P)$
Output: P_m, P_e, P_{dc}

1: $P_m \leftarrow T_m\omega_m$
2: **if** $P_m \geq 0$ **then**
3: $\quad P_e \leftarrow P_m/\eta_m$; $P_{\text{dc}} \leftarrow P_e/\eta_{\text{inv}}$
4: **else**
5: $\quad P_e \leftarrow P_m\eta_m$; $P_{\text{dc}} \leftarrow P_e\eta_{\text{inv}}$
6: **end if**

4.10.1 Introduction to Battery Modelling in EV Simulation

Unit 2 developed equivalent circuit models in detail. Here the concern is how a battery model is used *within* a vehicle simulation. The vehicle model hands the battery a power demand, and the battery model must return voltage, current, SoC and losses. The role of the battery model in a BEV simulation appears in Figure 4.67: the demand enters the model, states are updated, V_t and I pass to the powertrain, and the BMS converts states into power limits that restrict the demand. A thermal model may be coupled in both directions.

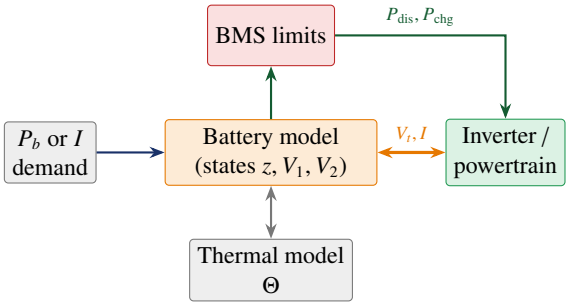


Figure 4.67: Battery model in a BEV simulation.

4.10.2 Cell-Level, Module-Level and Pack-Level Modelling

A cell model describes one electrochemical unit, usually identified from laboratory tests. Cells form modules, and modules form a pack, as in Figure 4.68, where three strings of 96 series cells are drawn, for 96s3p. Assuming identical cells, scaling is simple:

$$V_{\text{pack}} = N_s V_{\text{cell}}, \quad Q_{\text{pack}} = N_p Q_{\text{cell}}, \quad R_{\text{pack}} = \frac{N_s}{N_p} R_{\text{cell}}, \quad E_{\text{pack}} = V_{\text{pack}} Q_{\text{pack}}. \tag{4.92}$$

In this equation, N_s and N_p are the numbers of series and parallel cells. For the reference pack, $V = 96(3.65) = 350.4$ V, $Q = 3(48) = 144$ Ah, $R = (96/3)(1.5\ \text{m}\Omega) = 48\ \text{m}\Omega$, $E = 50.46$ kWh. Each cell carries I/N_p. Note that the drawing shows parallel-connected strings, while many packs connect cells in parallel first (3p96s); the scaling is the same. A real pack also includes busbar and contactor resistance, and cell-to-cell variation is handled by cell balancing (Unit 2).

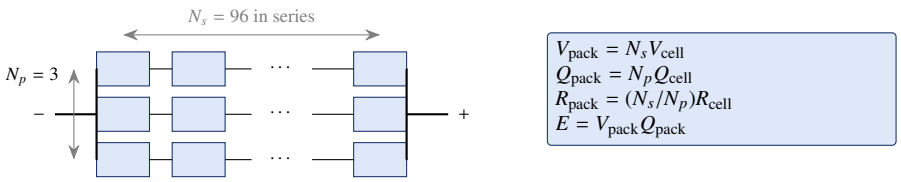


Figure 4.68: Battery pack series-parallel architecture.

4.10.3 Open-Circuit Voltage, Terminal Voltage and Ideal Source Model

The **open-circuit voltage** (OCV) $V_{oc}(z)$ is the rest voltage at SoC z. The **terminal voltage** V_t is what the load sees. In the **ideal voltage source model** of Figure 4.69, $V_t = V_{oc}$ for any current. This neglects all losses, so it can only be used for the crudest energy estimate.

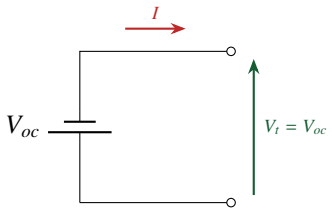


Figure 4.69: Battery cell equivalent circuit: ideal voltage source.

4.10.4 Internal Resistance and the Rint Model

The **Rint model** of Figure 4.70 puts a resistance in series with the source:

$$V_t = V_{oc}(z) - IR_0, \tag{4.93}$$

with $I > 0$ for discharge. The **battery power** at the terminals and the **internal resistance loss** are

$$P_b = V_t I = V_{oc} I - I^2 R_0, \qquad P_{\text{loss}} = I^2 R_0. \tag{4.94}$$

The discharge efficiency of the battery at any instant is $\eta_{\text{dis}} = V_t / V_{oc}$ and the charge efficiency is $\eta_{\text{chg}} = V_{oc}/V_t$. At 250 A the reference pack loses 3 kW. This model is simple and fast, and it serves well for energy studies, but it responds instantly to current steps and so lacks the slow voltage recovery of real cells.

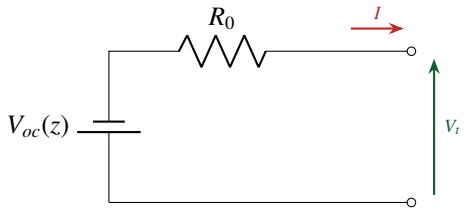


Figure 4.70: Rint battery model.

4.10.5 First-Order RC Equivalent Circuit Model

Adding one RC pair, as in Figure 4.71, represents charge-transfer and diffusion polarisation. The **polarisation voltage** V_1 obeys

$$\frac{dV_1}{dt} = -\frac{V_1}{R_1 C_1} + \frac{I}{C_1}, \qquad V_t = V_{oc}(z) - IR_0 - V_1, \tag{4.95}$$

with time constant $\tau_1 = R_1 C_1$ in s. The model captures the fast voltage drop and subsequent slow sag under load, followed by recovery at rest.

4.10.6 Second-Order RC Equivalent Circuit Model

Figure 4.72 uses two RC pairs, one with a fast and the other with a slow time constant:

$$\frac{dV_1}{dt} = -\frac{V_1}{\tau_1} + \frac{I}{C_1}, \quad \frac{dV_2}{dt} = -\frac{V_2}{\tau_2} + \frac{I}{C_2}, \quad V_t = V_{oc}(z) - IR_0 - V_1 - V_2. \tag{4.96}$$

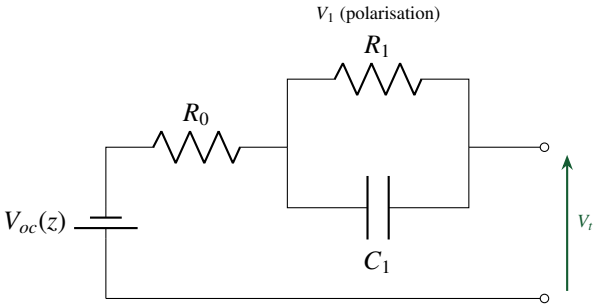


Figure 4.71: First-order (1-RC) Thevenin model.

Typical pairs are of the order of ten seconds and a hundred seconds or longer. Comparative studies suggest that low-order models are a good compromise for several lithium-ion chemistries, and that added RC pairs bring diminishing returns [14, 15].

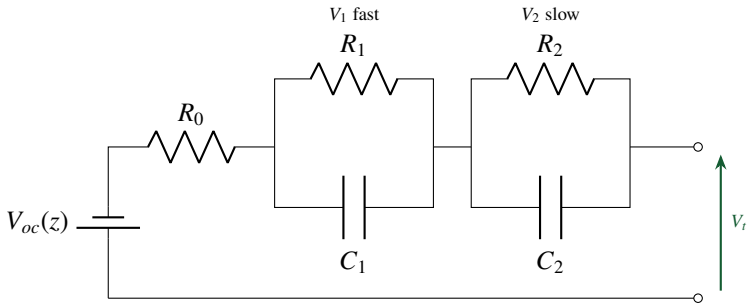


Figure 4.72: Second-order (2-RC) Thevenin model.

4.10.7 Battery Capacity, SoC Dynamics and Coulomb Counting

The SoC differential equation (4.16) expresses charge conservation. Its integral is the **Coulomb counting equation**:

$$z(t) = z(0) - \frac{1}{3600\,Q}\int_0^t \eta_c(\tau) I(\tau)\,d\tau, \tag{4.97}$$

where η_c is the Coulombic efficiency, 1 for discharge and about 0.99 or better for charge. Capacity Q is in Ah. Since integrated errors grow, a real BMS corrects Coulomb counting with voltage-based estimation, as shown in Unit 2. In simulation the current is exactly known, so the equation is perfect within the model. Figure 4.73 shows SoC feedback: the SoC sets V_{oc} and is compared with limits in the BMS, which in turn restricts the current drawn. SoC is therefore part of a feedback loop in a forward-looking model and is merely an output in a pure backward model, except for its effect on V_{oc}.

4.10.8 Battery Energy and Losses

The electrical energy delivered at the terminals and the chemical energy withdrawn are

$$E_t = \int V_t I\,dt, \qquad E_{\text{chem}} = \int V_{oc} I\,dt = E_t + \int I^2 R_0\,dt. \tag{4.98}$$

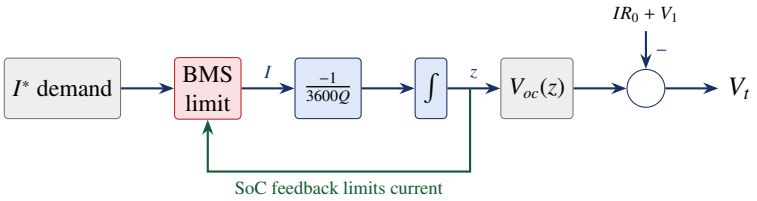


Figure 4.73: Battery SoC feedback architecture.

In SI these are in J. Dividing by 3600 gives Wh. With RC branches, the instantaneous loss adds V_1^2/R_1 for each pair. The vehicle energy consumption in Section 4.15 is defined at the terminals.

4.10.9 Discrete-Time Battery Model

For a piecewise-constant current over step Δt, the exact discrete solution of (4.96) is

$$z_{k+1} = z_k - \frac{\Delta t}{3600\,Q} I_k, \tag{4.99}$$

$$V_{1,k+1} = e^{-\Delta t/\tau_1} V_{1,k} + R_1(1 - e^{-\Delta t/\tau_1}) I_k, \tag{4.100}$$

$$V_{2,k+1} = e^{-\Delta t/\tau_2} V_{2,k} + R_2(1 - e^{-\Delta t/\tau_2}) I_k, \tag{4.101}$$

$$V_{t,k} = V_{oc}(z_k) - I_k R_0 - V_{1,k} - V_{2,k}. \tag{4.102}$$

These remain stable for any Δt, unlike forward Euler.

Algorithm 16 Battery model simulation

Input: $P_{b,k}$ or I_k, $Q, R_0, R_1, C_1, R_2, C_2$, $V_{oc}(z)$ table, z_0
Output: $V_{t,k}$, I_k, z_k, $P_{\text{loss},k}$

$z_1 \leftarrow z_0$; $V_1, V_2 \leftarrow 0$
for each step k **do**
  $V_e \leftarrow V_{oc}(z_k) - V_1 - V_2$
  if power is the input **then**
    $I_k \leftarrow (V_e - \sqrt{V_e^2 - 4R_0 P_{b,k}})/(2R_0)$
  end if
  $V_{t,k} \leftarrow V_e - I_k R_0$
  $V_1, V_2 \leftarrow$ (4.100), (4.101)
  Update SoC with Algorithm 17
end for

Algorithm 16 shows the key detail when power is the input. The current depends on the voltage and the voltage on the current, so the quadratic is solved first using the effective source V_e behind R_0. Algorithm 17 performs the Coulomb count and flags window violations.

4.10.10 Temperature Effects and Battery Degradation Overview

Low temperature raises internal resistance, slows diffusion and lowers usable capacity, which reduces both power and range. High temperature lowers resistance, but accelerates ageing. In

Algorithm 17 Battery SoC calculation

Input: z_k, I_k, Δt, Q, η_c, $z_{\min}$, $z_{\max}$
Output: z_{k+1}, limit flags
1: $\eta \leftarrow 1$ **if** $I_k > 0$ **else** η_c
2: $z_{k+1} \leftarrow z_k - \eta I_k \Delta t/(3600Q)$
3: **if** $z_{k+1} < z_{\min}$ **then** flag discharge limit
4: **end if**
5: **if** $z_{k+1} > z_{\max}$ **then** flag charge limit
6: **end if**

a detailed model, $R_0(z, \Theta)$ and $Q(\Theta)$ come from look-up tables, and a lumped thermal state is added:

$$C_{\text{th}} \frac{d\Theta}{dt} = I^2 R_0 - hA_s(\Theta - \Theta_a), \tag{4.103}$$

with C_{th} the heat capacity in J/K, h the heat transfer coefficient in W/(m^{2}K), A_s the surface area in m^2 and Θ_a the ambient temperature. **Degradation** appears as capacity fade and resistance growth, with calendar and cycle components, as described in Unit 2. In a vehicle simulation it is usually represented by scaling Q and R_0 with the state of health.

4.10.11 Parameter Identification and Validation

Parameters are identified from pulse tests: the instantaneous step gives R_0, and the following exponential gives R_1 and τ_1. Fitting is done by least squares on (4.102), and OCV is measured with low-current or rest-based tests. Validation uses an independent dynamic profile derived from a drive cycle, comparing simulated and measured V_t with (4.3). A cell model with an RMSE of a few tens of millivolts is generally acceptable for a vehicle simulation.

4.10.12 Comparison of Battery Models

Tables 4.19 and 4.20 compare the models. They show the general trade-off: each step up in complexity adds states and identification effort, for a gain that matters mainly in voltage-sensitive tasks such as power-limit prediction.

Table 4.19: Battery model types

Model	States	Effort	Best use
Ideal source	z or none	Minimal	Rough energy limit
Rint	z	Low	Energy and range
1-RC	z, V_1	Moderate	Transients, BMS design
2-RC	z, V_1, V_2	Moderate	Voltage accuracy, power limit
Electrochemical	Many	Very high	Cell design, ageing

Table 4.20: Rint versus 1-RC versus 2-RC models

Feature	Rint	1-RC	2-RC
Instantaneous drop	Yes	Yes	Yes
Sag under sustained load	No	Yes	Yes
Rest relaxation	No	One time constant	Fast and slow
Parameters (excluding OCV)	1	3	5
Time step needed	Any	$\lesssim \tau_1/5$ (Euler)	$\lesssim \tau_1/5$ (Euler)
Energy accuracy on a cycle	Good	Good	Good
Voltage accuracy	Modest	Good	Best

4.10.13 Numerical Examples

Example 16: Battery terminal voltage and power

Problem. A 96s3p pack uses 3.65 V, 48 Ah, 1.5 mΩ cells. At mid-SoC find pack voltage, resistance, terminal voltage, power and loss at 250 A discharge, the current in each cell, and the same quantities at 120 A charge.

Given. $N_s = 96$, $N_p = 3$.

Required. V_{oc}, R_0, V_t, P_b, P_{loss}, I_{cell}.

Governing equations. (4.92), (4.93), (4.94).

Solution. $V_{oc} = 350.4$ V, $R_0 = 48$ mΩ.
Discharge: $V_t = 350.4 - 250(0.048) = 338.4$ V; $P_b = 338.4(250) = 84.6$ kW; $P_{\text{loss}} = 250^2(0.048) = 3.0$ kW. Check: $V_{oc}I = 87.6$ kW $= 84.6 + 3.0$.
Cell: $I_{\text{cell}} = 83.3$ A and $V_{\text{cell}} = 3.65 - 83.3(0.0015) = 3.525$ V.
Charge ($I = -120$ A): $V_t = 350.4 + 5.76 = 356.2$ V; $P_b = -42.7$ kW; $P_{\text{loss}} = 0.69$ kW.

Result. At 250 A: 338.4 V, 84.6 kW, loss 3.0 kW (efficiency 96.6%). At −120 A: 356.2 V, −42.7 kW.

Physical interpretation. Terminal voltage falls on discharge and rises on charge, and loss grows with the square of the current.

Engineering significance. The cell runs at about 1.7C at 250 A. Upper voltage limits apply to the raised charge voltage, not to OCV.

Example 17: Cell and pack simulation: SoC variation under dynamic load

Problem. The pack of Example 16 starts at $z = 0.80$ and runs a stepped current profile: 250 A

for 10 s (acceleration), 60 A for 40 s (cruise), −120 A for 8 s (regeneration), rest for 20 s, then the first three steps again. Using the Rint model with $V_{oc} = 96(3.30 + 0.70z)$, find SoC, terminal voltage, power, loss, energy consumed and energy recovered, evaluating the voltage with the SoC at the end of each step.

Given. $Q = 144$ Ah, $R_0 = 48$ mΩ.

Required. A table of results and energy totals.

Governing equations. (4.93), (4.99), (4.98).

Solution. First step: $\Delta z = 250(10)/(3600 \times 144) = 0.482\%$, $z = 0.7952$, $V_{oc} = 370.2$ V, $V_t = 358.2$ V, $P_b = 89.6$ kW, $E = 248.8$ Wh.

Δt (s)	I (A)	z	V_{oc} (V)	V_t (V)	P_b (kW)	E (Wh)	Loss (Wh)
10	250	0.7952	370.2	358.2	89.6	248.8	8.3
40	60	0.7905	369.9	367.0	22.0	244.7	1.9
8	−120	0.7924	370.0	375.8	−45.1	−100.2	1.5
20	0	0.7924	370.0	370.0	0	0	0
10	250	0.7876	369.7	357.7	89.4	248.4	8.3
40	60	0.7829	369.4	366.5	22.0	244.4	1.9
8	−120	0.7848	369.5	375.3	−45.0	−100.1	1.5

Energy consumed = 986.2 Wh; energy recovered = 200.3 Wh; net terminal energy = 786.0 Wh. Internal loss total = 23.5 Wh. Chemical energy = 786.0 + 23.5 = 809.5 Wh. Check: $\Delta z\, Q\, \bar{V}_{oc} = 0.0152(144)(369.8) = 809$ Wh.

Result. SoC falls from 80.0% to 78.5%; 20.3% of discharge energy is recovered.

Physical interpretation. Voltage moves mainly with current, not with SoC, over this short window.

Engineering significance. Internal loss is 3% of the net energy here, even at a high current. The check against Δz confirms that nothing has been double counted.

4.10.14 Simulated Battery Waveforms

Figures 4.74 to 4.77 show the same profile simulated with a 2-RC model, extended by a second repeat and a short regeneration pulse. Parameters are hypothetical: $R_0 = 33.6$ mΩ, $R_1 = 9.6$ mΩ, $\tau_1 = 10$ s, $R_2 = 4.8$ mΩ, $\tau_2 = 100$ s, so the total resistance matches 48 mΩ.

Figure 4.74 is the imposed current. Figure 4.75 shows voltage: it steps at each current change, sags gently under sustained load and rises above OCV during charge. Figure 4.76 shows power, with the sign following current. Figure 4.77 shows SoC: its slope is proportional to current, so it falls fastest during the 250 A pulses, and it climbs during regeneration.

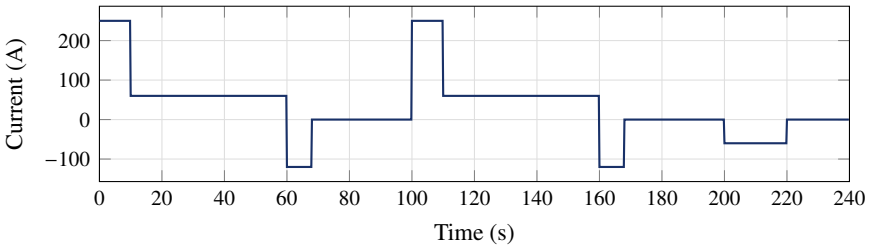


Figure 4.74: Battery current versus time (simulated profile).

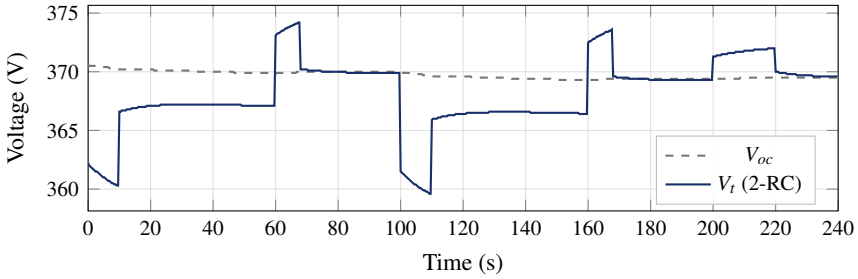


Figure 4.75: Battery voltage versus time (simulated, 2-RC model).

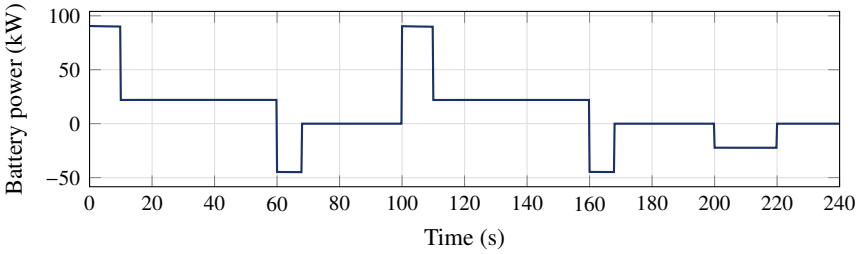


Figure 4.76: Battery power versus time (simulated).

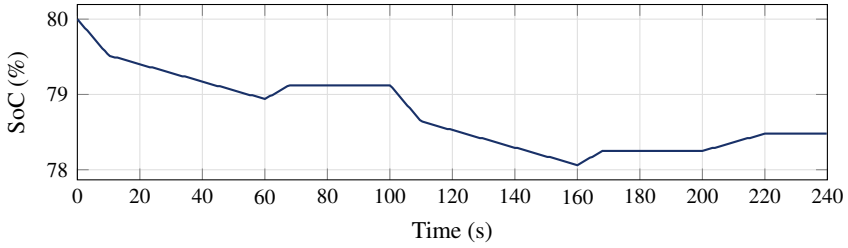
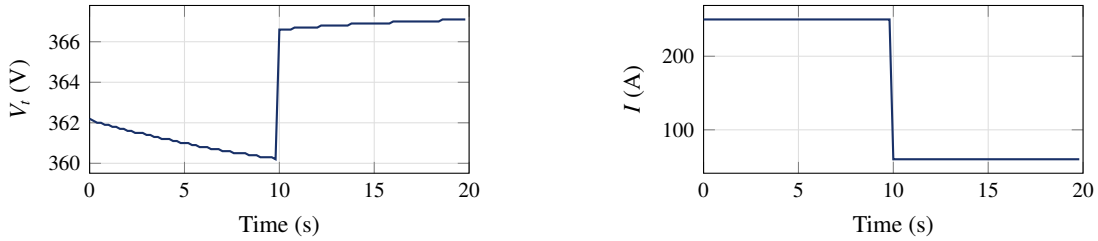


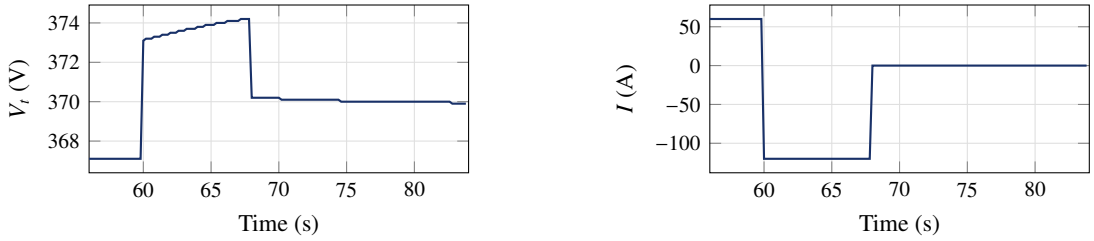
Figure 4.77: SoC versus time (simulated).

Figure 4.78 zooms on the battery response during acceleration: terminal voltage sits about 8 V below OCV and droops during the pulse, then rises at the end of the pulse by far more than V_1 changes, because the IR_0 term vanishes at once. Figure 4.79 shows the response during regenerative braking: voltage steps up, stays above OCV while charge flows, and steps back when current stops.



(a) Terminal voltage. **(b)** Current.

Figure 4.78: Battery response during acceleration (simulated).



(a) Terminal voltage. **(b)** Current.

Figure 4.79: Battery response during regenerative braking (simulated).

Figure 4.80 compares models under the same dynamic load. All share the same total resistance, so they settle to similar values. The Rint trace jumps straight to the final drop and is pessimistic immediately after a step, by about 3 V at the end of the first pulse here, whereas the RC models start from a smaller drop and approach the final value gradually.

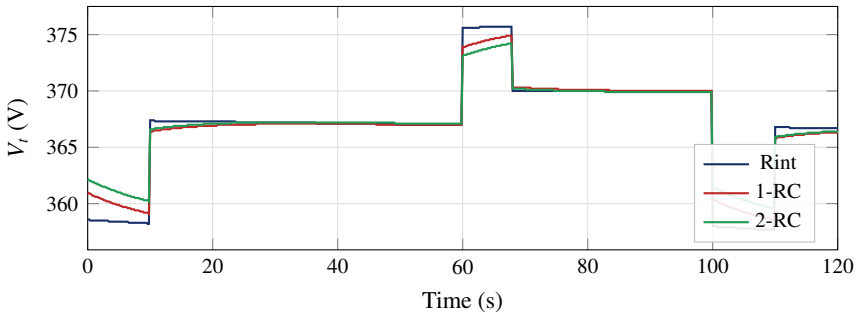


Figure 4.80: Battery terminal voltage under dynamic load for three models (simulated).

4.10.15 Effect of Model Complexity on Accuracy and Computation

Energy depends on $\int I^2 R\, dt$, and all three models use the same steady resistance, so they predict almost identical cycle energy. Voltage-sensitive questions are different: predicting the instantaneous power limit requires knowledge of the polarisation state, so RC models are needed.

Each additional RC pair adds one state, and at most two exponential updates per step, which is a negligible cost in a modern computer. The main cost is in the laboratory, because every parameter must be identified across SoC and temperature. A sensible rule is to use Rint for energy and sizing, 1-RC or 2-RC for transients and BMS work, and electrochemical models only for cell-level design.

4.10.16 MATLAB/Simulink Implementation of the Battery Model

Figure 4.81 gives the conceptual block chain, and Table 4.21 gives its details.

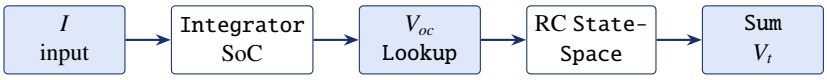


Figure 4.81: Conceptual Simulink diagram of the battery equivalent circuit model.

Table 4.21: Battery equivalent circuit model: conceptual Simulink implementation

Inputs	Current I or power demand P_b, ambient temperature
Outputs	V_t, z, P_{loss}, limits
States	z, V_1, V_2, optionally Θ
Equations	(4.93), (4.96), (4.97)
Blocks	`Integrator`, `1-D Lookup Table`, `State-Space`, `Sum`; or Simscape Battery
Feedback	$z \rightarrow V_{oc}$; V_t into current calculation
Solver	Fixed 0.1–1 s, or variable-step; `ode23t` for Simscape
Initial conditions	$z(0) = z_0$, $V_1(0) = V_2(0) = 0$ (rested)
Assumptions	Identical cells, lumped temperature, constant parameters over a step
Simulation outputs	Voltage, SoC and loss traces

4.11 Powertrain Dimensioning: Peak versus Continuous Performance

4.11.1 Introduction to Powertrain Sizing

Powertrain dimensioning converts vehicle targets into ratings for motor, gear, inverter and battery. Two conditions must be separated. **Peak** performance is what the car can do for seconds, such as overtaking. **Continuous** performance is what it can do indefinitely, such as climbing a long hill. Heat decides the difference. Peak ratings are limited by current and voltage, and continuous ratings by temperature. Figure 4.82 shows the sizing architecture: targets become forces, forces become motor and ratio choices, which then flow to inverter and battery ratings, and a drive cycle supplies the thermal duty.

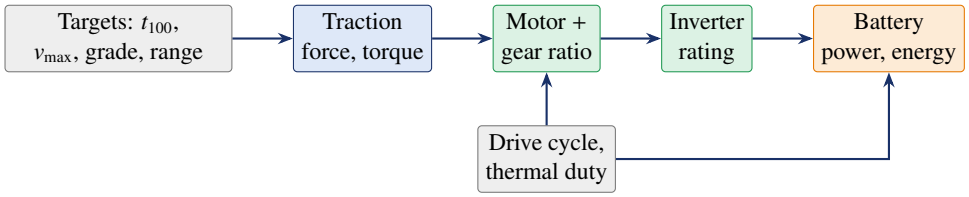


Figure 4.82: EV powertrain sizing architecture.

4.11.2 Motor Characteristics: Base Speed, Maximum Speed and Operating Regions

Figures 4.83 and 4.84 show the torque-speed and power-speed characteristics of the reference motor. Figure 4.85 labels the operating regions.

- **Constant torque region**, $0 \le \omega_m \le \omega_b$. Current is at its limit and flux is at full value. Torque is constant and voltage rises with speed, so power rises linearly.
- **Base speed** ω_b. The inverter reaches its voltage limit, and $P_{\text{pk}} = T_{\text{pk}}\omega_b$. The reference motor has $\omega_b = 110\,000/280 = 393$ rad/s, or 3750 r/min.
- **Constant power region**, $\omega_b < \omega_m \le \omega_c$. **Field weakening** reduces the net flux so that voltage stays constant, as derived in Unit 3. Torque falls as $1/\omega_m$.
- **Power fade region.** Above ω_c the available power also falls.
- **Maximum speed** ω_{max}. Set by rotor mechanical strength and bearings. The reference motor is taken as 12 000 r/min.

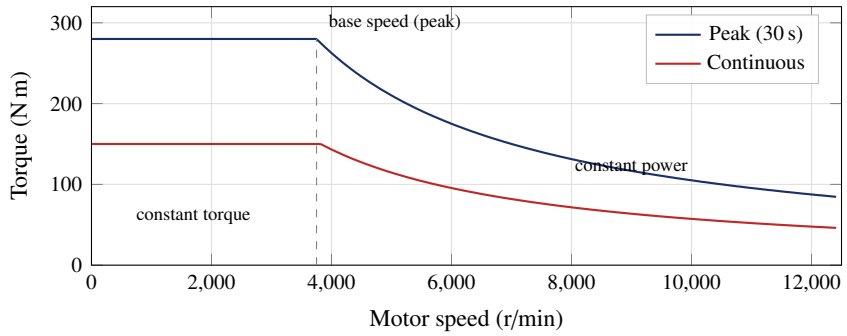


Figure 4.83: Motor torque-speed characteristic (hypothetical motor).

4.11.3 Peak Power and Continuous Power: Definitions

Peak power is the highest mechanical power available for a specified short time, commonly 10–60 s, starting from a defined temperature. **Continuous power** is the power that can be held indefinitely at the thermal limit, with the cooling system working normally; this is the **thermal steady state**. Peak and continuous torque are defined in the same way. Figure 4.86 compares ratings of the reference machine, and Table 4.22 summarises differences.

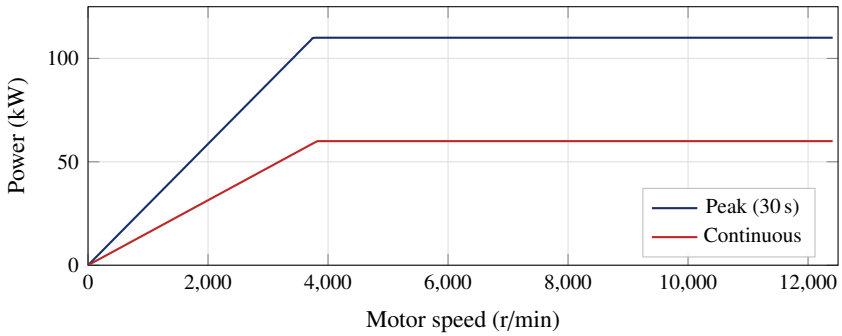


Figure 4.84: Motor power-speed characteristic (hypothetical motor).

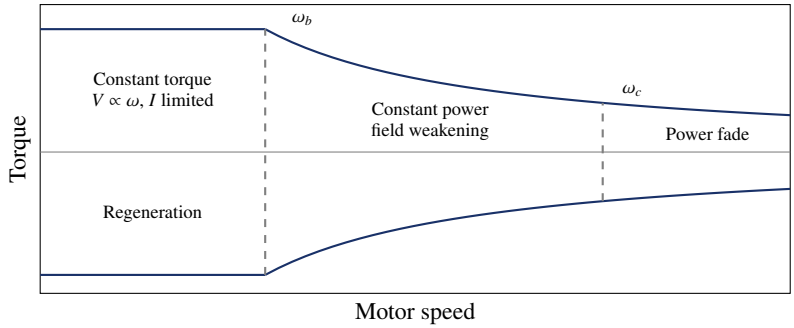


Figure 4.85: Motor operating regions; the mirror image at negative torque is regeneration (conceptual).

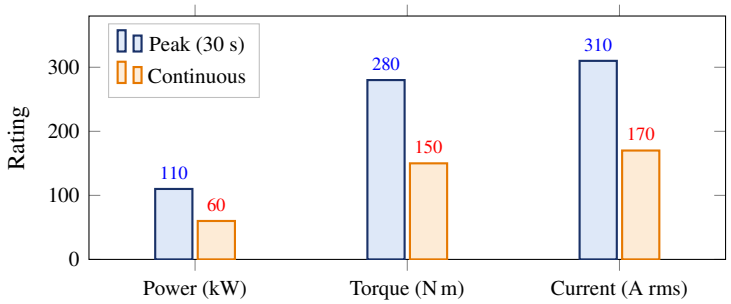


Figure 4.86: Peak versus continuous power, torque and current ratings (hypothetical motor).

Table 4.22: Peak versus continuous motor performance

Aspect	Peak	Continuous
Duration	Seconds to about a minute	Indefinite
Limited by	Current, voltage, magnet demagnetisation	Winding and magnet temperature
Governing quantity	Instantaneous current	Rms current over duty cycle
Sets	Acceleration, launch gradeability	Cruise, sustained hill climb
Typical ratio	1.0	0.4–0.7 of peak

The table shows that peak and continuous limits are set by different physics, and that peak ratings cannot simply be divided by a constant to find continuous ability; cooling design has a strong effect.

4.11.4 Thermal Limits, Motor Thermal Capacity and Duty Cycle

Winding losses scale with I^2, so torque has a squared effect on heat. A simple lumped thermal model of a motor is

$$C_{\text{th}}\frac{d\Delta\Theta}{dt} = P_{\text{loss}} - \frac{\Delta\Theta}{R_{\text{th}}}, \qquad \tau_{\text{th}} = R_{\text{th}}C_{\text{th}}, \tag{4.104}$$

where $\Delta\Theta$ is the temperature rise in K, C_{th} the **thermal capacity** in J/K, and R_{th} the thermal resistance in K/W. At steady state, $\Delta\Theta_{\text{ss}} = P_{\text{loss}}R_{\text{th}}$, which defines the continuous rating. The thermal capacity is what permits short overloads: a short burst heats the metal before temperature reaches the limit, and for an overload loss P_{ov} starting from a rise $\Delta\Theta_0$, the allowed time is

$$t_{\text{ov}} = \tau_{\text{th}} \ln \frac{P_{\text{ov}}R_{\text{th}} - \Delta\Theta_0}{P_{\text{ov}}R_{\text{th}} - \Delta\Theta_{\max}}. \tag{4.105}$$

A **duty cycle** is a repeating load pattern, and in **repeated acceleration and deceleration** the machine stays within thermal limits if its rms torque stays below the continuous torque (for a loss dominated by copper):

$$T_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T T_m^2\,dt} \le T_{\text{c}}. \tag{4.106}$$

For the reference car, T_{rms} is 30 N m on WLTC and 53 N m on US06, compared with T_c = 150 N m, so the motor is thermally comfortable on any standard cycle. Regeneration counts too, as braking torque heats the windings in the same way. A racetrack, however, can overheat the same machine.

4.11.5 Motor Current, Inverter Current and Battery Discharge Limits

Motor torque in the constant-torque region is proportional to current, so the **motor current limit** sets peak torque. The **inverter current limit** comes from junction temperature, and a semiconductor has a thermal time constant of only milliseconds to a few seconds, so its overload capability is far shorter than a motor's. DC and phase currents are

$$I_{\text{dc}} = \frac{P_{\text{dc}}}{V_{\text{dc}}}, \qquad I_{\text{ph,rms}} = \frac{P_e}{\sqrt{3}\,V_{LL}\cos\varphi}, \qquad V_{LL,\max} \approx \frac{V_{\text{dc}}}{\sqrt{2}}. \tag{4.107}$$

Here V_{LL} is the rms line voltage for space vector modulation and $\cos\varphi$ is the power factor. The **battery discharge limit** is set by the minimum cell voltage, temperature and SoC, and falls at low SoC and low temperature.

4.11.6 Gear Ratio, Wheel Torque and Motor Speed

The gear ratio connects motor capability to wheel needs. The top-speed requirement sets the upper limit of the ratio:

$$G \le \frac{\omega_{m,\max}\, r_w}{v_{\max}}. \tag{4.108}$$

Peak wheel torque and tractive force then follow:

$$T_{w,\max} = G\eta_g T_{\mathrm{pk}}, \qquad F_{t,\max} = \frac{G\eta_g T_{\mathrm{pk}}}{r_w}. \tag{4.109}$$

A higher ratio gives more wheel torque for a smaller motor, but more gear loss, higher motor speed and more iron loss at cruise. Motor speed follows (4.11).

4.11.7 Acceleration Requirements and Peak Wheel Power

The time to accelerate from 0 to v_f along the powertrain's limit curve follows from separating variables in (4.74),

$$t_a = \int_0^{v_f} \frac{\delta m}{F_t(v) - F_r - F_a(v)}\, dv. \tag{4.110}$$

Ignoring road load, an ideal drive with force $F = P/v_b$ below base speed v_b, and constant power P above it, gives the classic result $t_a = \delta m(v_b^2 + v_f^2)/(2P)$ [3]. Rearranging, then adding the mean road-load power gives the **peak wheel power**:

$$P_{w,\mathrm{pk}} = \frac{\delta m}{2t_a}(v_b^2 + v_f^2) + \frac{2}{3} mgC_{rr}v_f + \frac{1}{5}\rho C_d A v_f^3. \tag{4.111}$$

The coefficients $\frac{2}{3}$ and $\frac{1}{5}$ are approximations of the average resistive power during the run. Motor peak power, with the gear loss included, is $P_{m,\mathrm{pk}} = P_{w,\mathrm{pk}}/\eta_g$, and peak torque is $T_{\mathrm{pk}} = P_{m,\mathrm{pk}}/\omega_b$. Since the v_b^2 term is small for low v_b, a wide constant-power range reduces the required power for a given acceleration time.

4.11.8 Maximum Speed and Continuous Wheel Power

At top speed $a = 0$, so the wheel power is

$$P_{w,\max v} = \left(C_{rr}mg + \tfrac{1}{2}\rho C_d A v_{\max}^2\right) v_{\max}. \tag{4.112}$$

The **continuous wheel power** is taken from the most demanding sustained condition, such as a motorway cruise or a long climb,

$$P_{w,\mathrm{c}} = \max_{\text{cases}} \left[(C_{rr}mg\cos\theta + \tfrac{1}{2}\rho C_d A v^2 + mg\sin\theta)v\right]. \tag{4.113}$$

4.11.9 Gradeability and Hill Climbing

Gradeability is the steepest grade that the car can hold at a given speed. Setting $a = 0$ in (4.72), the required force is

$$F_t(v) - \tfrac{1}{2}\rho C_d A v^2 = mg(\sin\theta + C_{rr}\cos\theta). \tag{4.114}$$

At low speed drag is small. Writing $\beta = (F_t - F_a)/(mg)$, the exact solution is

$$\sin\theta = \frac{\beta - C_{rr}\sqrt{1 + C_{rr}^2 - \beta^2}}{1 + C_{rr}^2}, \tag{4.115}$$

For $\theta \ll 1$, this gives the often-used approximation grade $\approx 100(\beta - C_{rr})$ in %. Force is capped by adhesion on the driven axle, so a steep hill start can be limited by tyre grip.

Figure 4.87 plots the available tractive force and the road-load curves against speed. The force curve is flat up to about 49 km/h and then falls as a hyperbola. Each intersection with a road-load curve gives a steady speed on that grade. The difference between the curves is the force available for acceleration.

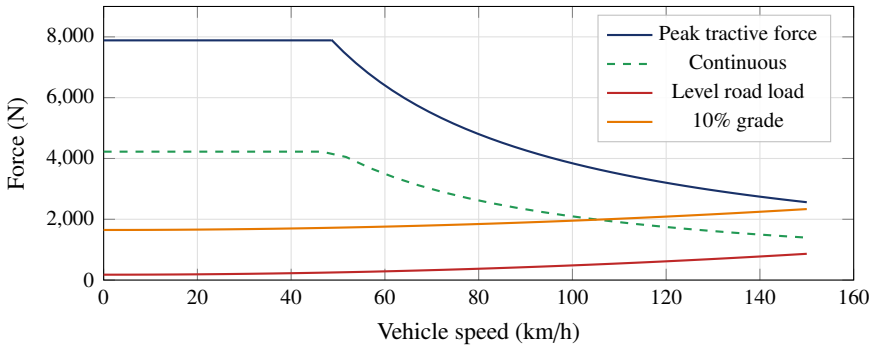


Figure 4.87: Tractive force envelope and road load (calculated, hypothetical BEV).

4.11.10 Battery and Inverter Power Rating, Peak and Continuous Current

The battery must supply peak motor power through both conversion stages, plus auxiliaries:

$$P_{b,\mathrm{pk}} = \frac{P_{m,\mathrm{pk}}}{\eta_m \eta_{\mathrm{inv}}} + P_{\mathrm{aux}}, \qquad P_{b,\mathrm{c}} = \frac{P_{m,\mathrm{c}}}{\eta_m \eta_{\mathrm{inv}}} + P_{\mathrm{aux}}. \tag{4.116}$$

Current then follows from (4.15), and the worst case is the lowest V_{oc} at low SoC. The C-rate is $P_{b,\mathrm{pk}}/E_{\mathrm{nom}}$, which is critical in small packs. At a high power-to-energy ratio, the battery can be the limiting component for a whole vehicle. Figure 4.88 shows that limits form a chain, in which available wheel torque equals the minimum over all the component limits. A design is balanced when these limits nearly coincide, since an oversized part in the chain is wasted cost.

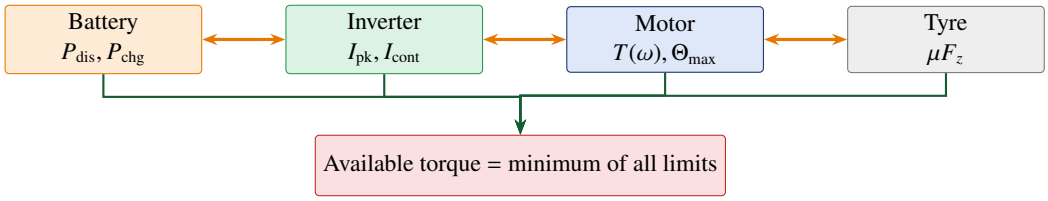


Figure 4.88: Battery and inverter power constraints on available torque.

4.11.11 Short-Duration Overload and Component Safety Margin

Overload capability differs among components. A motor can tolerate 30 s or more due to its large thermal mass. An inverter's capability is limited to seconds. A battery's peak is commonly specified as a 10 s or 30 s pulse. The overall peak rating is set by the shortest of these, and the BMS or inverter counts down remaining overload time. Safety margins should be added to the final selected rating, not to each intermediate step, or the design will be needlessly oversized.

4.11.12 Powertrain Efficiency

Over a cycle, the average efficiency is the ratio of energies, not the product of constant values. Efficiency also depends on sizing: an oversized motor runs at a smaller fraction of rated torque in normal driving, where efficiency tends to be lower. This is a further reason not to oversize.

4.11.13 Sizing Workflow and Algorithm

Figure 4.89 gives the sizing workflow. It runs from grade and speed requirements to gear ratio, acceleration requirement to peak power, cycle needs to continuous power, and finally to electrical ratings. If any limit fails, the loop returns to the ratio and motor selection. Algorithm 18 expresses the same procedure in steps.

Algorithm 18 Powertrain component sizing

Input: m, r_w, C_{rr}, C_dA, ρ, targets t_a, v_f, $v_{\max}$, grade, range D, cycle
Output: G, T_{pk}, P_{pk}, P_{c}, inverter current, $P_{b,\text{pk}}$, E_{nom}

repeat
  Select $\omega_{m,\max}$; $G \leftarrow$ (4.108)
  Choose base speed ω_b, so $v_b = \omega_b r_w / G$
  $P_{\text{pk}} \leftarrow$ margin $\times$ (4.111)$/\eta_g$; $T_{\text{pk}} \leftarrow P_{\text{pk}}/\omega_b$
  Check gradeability (4.114) and traction limit (4.77)
  $P_{\text{c}} \leftarrow$ margin $\times$ (4.113)$/\eta_g$; check T_{rms} (4.106)
  Inverter currents $\leftarrow$ (4.107)
  $P_{b,\text{pk}}, P_{b,\text{c}} \leftarrow$ (4.116); $I_{\max} \leftarrow$ (4.15) at lowest V_{oc}
  $E_u \leftarrow eD$ from cycle simulation; $E_{\text{nom}} \leftarrow E_u/$(window)
  Update m with battery mass
until m has converged and all limits are met

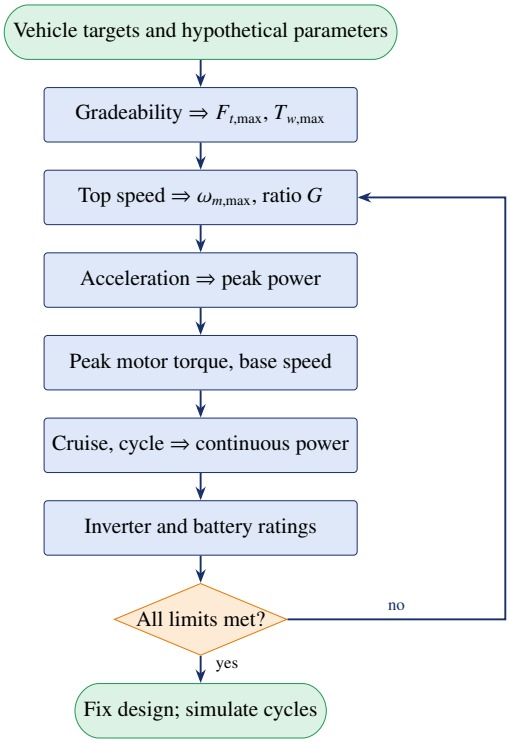


Figure 4.89: EV powertrain component sizing workflow.

4.11.14 Mandatory Case Study

Example 18: Peak versus continuous motor power sizing

Problem. Size a BEV powertrain for: 0–100 km/h in 9 s or less; top speed 150 km/h; 30% gradeability at 20 km/h; and sustained 100 km/h up an 8% grade at 1700 kg (full load).

Given (assumed). $m = 1500$ kg (test), $\delta = 1.05$, $C_{rr} = 0.012$, $\rho C_d A/2 = 0.396$ kg/m, $r_w = 0.31$ m, $\eta_g = 0.97$, motor maximum speed 12 000 r/min. WLTC is used as the drive cycle.

Required. Peak traction force, peak wheel torque, gear ratio, motor speed range, peak torque and power, continuous power.

Governing equations. (4.108), (4.111), (4.112), (4.113), (4.114), (4.106).

Solution. *Gradeability:* $\theta = \arctan 0.3 = 16.70°$ and $v = 5.56$ m/s.
$F_t = 0.012(14\,715)(0.958) + 0.396(30.9) + 14\,715(0.287) = 4410$ N, so $T_w = 1367$ N m. This sets a minimum force requirement.
Top speed: $v_{\max} = 41.67$ m/s, $F = 176.6 + 687.5 = 864$ N and $P_w = 36.0$ kW.
Gear ratio: $\omega_{m,\max} = 1257$ rad/s, so $G \le 1257(0.31)/41.67 = 9.35$. Choose $G = 9$. At $v_{\max}$, $\omega_m = 1210$ rad/s (11 550 r/min).
Grade torque at the motor: $1367/(9 \times 0.97) = 156.6$ N m.
Peak power: choose base speed near 3750 r/min, so $v_b = 13.5$ m/s. With $v_f = 27.78$ m/s and $t_a = 9$ s,

$\frac{\delta m}{2t_a}(v_b^2 + v_f^2) = 87.5(182 + 772) = 83.5$ kW; resistive terms: $\frac{2}{3}(14\,715)(0.012)(27.78) = 3.3$ kW and $\frac{1}{5}(0.792)(27.78)^3 = 3.4$ kW.
$P_{w,\mathrm{pk}} = 90.2$ kW and $P_m = 90.2/0.97 = 93.0$ kW. A margin of 18% gives $P_{\mathrm{pk}} = 110$ kW.
$T_{\mathrm{pk}} = 110\,000/392.7 = 280$ N m.
Peak wheel torque: $T_{w,\max} = 280(9)(0.97) = 2444$ N m, and peak traction force $F_{t,\max} = 7885$ N, far above the grade need.
Traction check: with $\mu = 0.9$ and 52% of weight on a driven axle, the tyre limit is 6887 N, so launch is limited by the tyres.
Continuous power: the cases are:

Condition	Force (N)	P_w (kW)	P_m (kW)
130 km/h, level, 1500 kg	693	25.0	25.8
100 km/h, 3%, 1500 kg	923	25.6	26.4
80 km/h, 6%, 1500 kg	1253	27.8	28.7
100 km/h, 8%, 1700 kg	1835	51.0	52.5

The last is the sizing case. A 14% margin gives $P_{\mathrm{c}} = 60$ kW and $T_c = 150$ N m. Cycle check: WLTC $T_{\mathrm{rms}} = 30$ N m, and $P_{m,\mathrm{rms}} = 12$ kW, far below that limit.

Result. $G = 9$, speed range to 11 550 r/min, $F_{t,\max} = 7885$ N, $T_{w,\max} = 2444$ N m, $T_{\mathrm{pk}} = 280$ N m, $P_{\mathrm{pk}} = 110$ kW, $P_{\mathrm{c}} = 60$ kW.

Physical interpretation. Peak power is set by acceleration, and continuous power by the worst sustained hill. Top speed needs only 36 kW at the wheels.

Engineering significance. Sizing for peak alone sets acceleration performance, and says nothing about whether the motor overheats on a mountain road. Sizing for continuous power alone would give a 60 kW machine with a 0–100 km/h time of about 14.6 s. Both must be examined. A full simulation of this design gives 7.5 s if traction limits are ignored, so the simple formula (4.111) is conservative.

Example 19: Inverter and battery power rating

Problem. For the design of Example 18, determine inverter power and current, and battery peak and continuous power and current. Assume nominal $V_{\mathrm{dc}} = 350.4$ V and $\cos\varphi = 0.9$.

Given. $\eta_m = 0.93$, $\eta_{\mathrm{inv}} = 0.97$, $P_{\mathrm{aux}} = 0.6$ kW, $R_0 = 0.048\ \Omega$, V_{oc} at $z = 0.1$: 323.5 V.

Required. P_e, P_{dc}, I_{ph}, I_{dc}, P_b, I, C-rate.

Governing equations. (4.107), (4.116), (4.15).

Solution. Motor input: $P_{e,\mathrm{pk}} = 110/0.93 = 118.3$ kW and $P_{e,\mathrm{c}} = 60/0.93 = 64.5$ kW.
Inverter input: $P_{\mathrm{dc,pk}} = 118.3/0.97 = 121.9$ kW, $P_{\mathrm{dc,c}} = 66.5$ kW.

$V_{LL} = 350.4/\sqrt{2} = 247.8$ V rms.
$I_{\text{ph,pk}} = 118\,300/(1.732 \times 247.8 \times 0.9) = 306$ A rms, and $I_{\text{ph,c}} = 167$ A rms.
Battery power: $P_{b,\text{pk}} = 122.5$ kW, $P_{b,\text{c}} = 67.1$ kW.
Nominal voltage: $I_{\text{pk}} = 368$ A (terminal 332.7 V), $I_{\text{c}} = 197$ A (terminal 341.0 V).
Low SoC ($V_{oc} = 323.5$ V): $I_{\text{pk}} = 403$ A (terminal 304 V), $I_{\text{c}} = 214$ A.
C-rate: $122.5/50.46 = 2.4$C peak; 1.3C continuous.

Result. The inverter should be rated at about 122 kW with at least 310 A rms peak and 170 A rms continuous. The battery needs 122.5 kW and about 400 A peak.

Physical interpretation. Low SoC raises current for the same power, because the voltage is lower.

Engineering significance. Selecting by continuous rating only would leave peak acceleration unavailable, and selecting by peak only says nothing about cooling. Device current and busbars are chosen from these two values, with the rms value governing heat.

4.11.15 Selection Based on Peak Power versus Continuous Power

Selecting a motor from **peak power** ensures acceleration, launch and overtaking, and it leads to a machine and inverter with a high current rating. Selecting from **continuous power** ensures sustained speed, hill climbing and repeatable performance, and it mainly determines the cooling system and machine volume. A correct design meets both: pick the machine whose peak curve covers the acceleration points and whose continuous curve covers the sustained points. Passenger cars tend to be peak-driven, since their duty is mild, while buses and trucks tend to be continuous-driven, because they spend long periods at high load.

4.11.16 MATLAB/Simulink Implementation of Powertrain Sizing

Sizing is mostly a script that sweeps design variables, as Figure 4.90 indicates. Each candidate's pass or fail is recorded, and successful ones are run through the cycle simulation. Table 4.23 lists details.

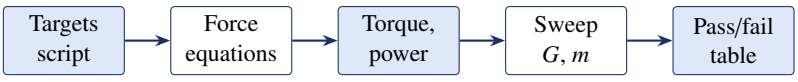


Figure 4.90: Conceptual workflow for powertrain sizing in MATLAB.

4.12 Types of Drive Cycles

4.12.1 Introduction and Definition

A **drive cycle** is a standardised speed-time profile, sampled typically once per second, that a vehicle follows on a chassis dynamometer or in simulation. Its purpose is comparability: every vehicle runs the same trace, so that consumption, emissions and range can be compared. Some

Table 4.23: Powertrain sizing: conceptual MATLAB implementation

Inputs	Targets, vehicle data, candidate G, P_{pk}, cycle
Outputs	Ratings, feasibility table, cost and mass estimates
States	None in static sizing; v in an acceleration check; $\Delta\Theta$ in a thermal check
Equations	(4.108)–(4.116), (4.110)
Blocks	Script loops, `ode45` for the acceleration run; Simulink thermal model
Feedback	Mass update from pack size, iteration to convergence
Solver	`ode45` on (4.110); `fmincon` if optimising
Initial conditions	First-guess mass and motor size
Assumptions	Constant efficiency in first pass, then maps
Simulation outputs	t_{100}, v_{max}, gradeability, T_{rms}, temperature rise

cycles are *modal*, built from straight-line accelerations, cruises and decelerations. Others are *transient*, derived from real traffic recordings. The **speed-time profile** is all the cycle specifies; road load settings, ambient temperature and test preparation are defined in the accompanying regulation.

4.12.2 Importance of Drive Cycles

Cycles support regulatory approval, published range figures, component sizing and control tuning. In simulation they provide the common input for backward and forward models. A cycle is only a sample of real driving, so a vehicle tuned to a single cycle may perform differently on the road. Regulators therefore moved from NEDC to WLTC, which is more dynamic.

4.12.3 Urban, Highway, Combined and Stop-and-Go Driving

Urban driving has low average speed, frequent stops and long idling, so inertial energy dominates. **Highway driving** has high steady speed and few stops, so aerodynamic drag dominates. **Combined driving** mixes the two. **Stop-and-go driving** is the extreme urban case. Acceleration and deceleration levels set the peak power and the regenerative opportunity. Table 4.24 compares urban and highway driving, and the simulated results from the reference car are in Section 4.15.

Table 4.24: Urban versus highway driving

Feature	Urban	Highway
Average speed	Below about 40 km/h	Above about 70 km/h
Idle share	10–30%	Near 0%
Dominant load	Inertia and rolling	Aerodynamic drag
Regeneration share	High	Low
Motor operating region	Low speed, varied torque	High speed, low torque
Auxiliary energy per km	High	Low
EV relative efficiency	Best	Lower

The table explains why EVs, unlike engine cars, are usually more efficient in the city. An engine car suffers at low load, whereas an EV's efficiency chain stays high and braking energy is recovered.

4.12.4 Official Data and Illustrative Profiles: How Figures in This Section Were Prepared

The following distinction applies to every speed-time figure below.

- **Official standardised data.** The MIDC figure was reconstructed from the operation tables in the Indian type-approval procedure [16]. WLTC Class 3b and NEDC use the regulation data as carried by the European Commission JRC's open reference code [20], with the WLTC samples confirmed against that code's published checksum. UDDS, FTP-75, HWFET and US06 use the EPA schedules [21] as packaged in NREL's FASTSim distribution [13]. Statistics for these cycles in Table 4.25 were computed directly from these data, and agree with the widely published figures.
- **Illustrative representative profiles.** The IDC and Artemis figures are *illustrative shapes only*, because the official second-by-second traces were not available from a verifiable public source at the time of writing. They must not be used as official data. The parameters quoted for those cycles come from the cited references.

4.12.5 Indian Driving Cycle (IDC)

The IDC is a short modal cycle developed for two- and three-wheelers in Indian urban conditions. Its elementary cycle lasts 108 s, peaks at 42 km/h, and it is repeated six times, giving 648 s, 3.94 km and an average of 21.9 km/h, with maximum acceleration of 0.64 m/s^2 [17]. Duration and distance figures in commercial brochures that quote "IDC range" relate to this gentle low-speed cycle, and therefore overstate real range. In EV simulation it is used for electric two- and three-wheelers. Figure 4.91 gives an illustrative shape of a single 108 s cycle, and it is not the official trace.

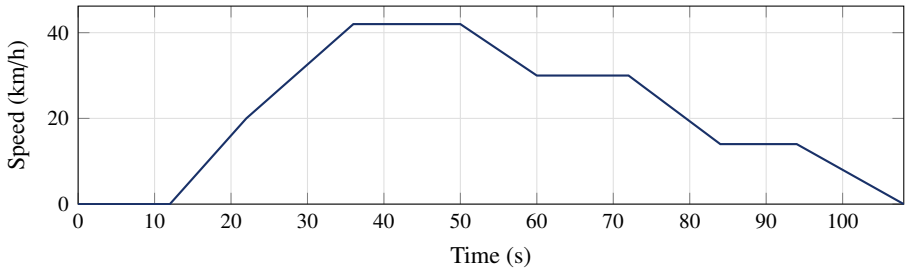


Figure 4.91: IDC-like speed profile: *illustrative only, not official data.*

4.12.6 Modified Indian Driving Cycle (MIDC)

The MIDC is used for type approval of light vehicles in India. It adapts the NEDC structure to Indian conditions: four elementary urban cycles of 195 s each, followed by a 400 s extra-

urban cycle whose maximum speed is reduced to 90 km/h [16]. The theoretical distance is 4(1.013) + 6.594 = 10.65 km, and integrating the speed table gives 10.67 km. Figure 4.92 shows the modal shape, in which long constant-speed plateaus are visible. Mean acceleration is low, so MIDC range values are optimistic relative to real traffic. It is the standard cycle quoted in Indian EV range claims.

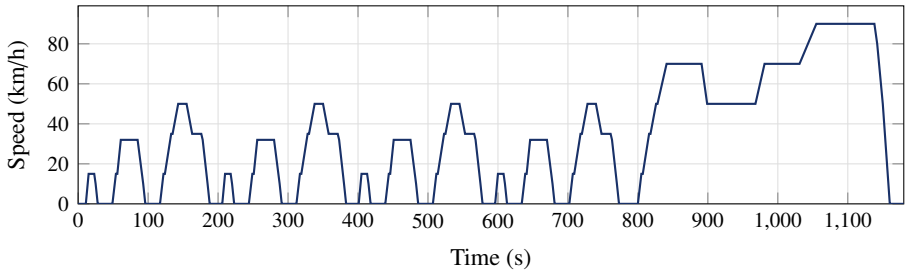


Figure 4.92: MIDC speed-time profile, reconstructed from the official operation tables [16].

4.12.7 Worldwide Harmonized Light Vehicles Test Cycle (WLTC)

The WLTC was developed under the United Nations from real driving data [18,19]. Its Class 3b version applies to most cars and has four phases: Low, Medium, High and Extra High, totalling 1800 s and 23.27 km, with a peak of 131.3 km/h. Figure 4.93 shows the highly transient shape. It is a combined cycle with more acceleration and shorter idling than NEDC, and it gives a more realistic energy figure. It is the basis of WLTP range values.

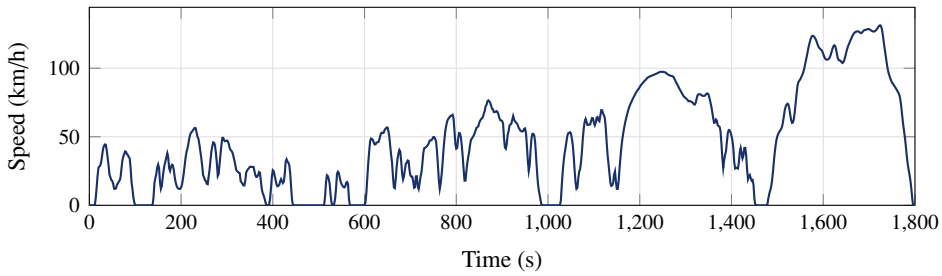


Figure 4.93: WLTC Class 3b speed-time profile (official regulation data [19,20]).

4.12.8 New European Driving Cycle (NEDC)

The NEDC was formerly the European approval cycle. It comprises four ECE urban cycles and one Extra-Urban Driving Cycle, 1180 s and 11.0 km, peaking at 120 km/h. As Figure 4.94 shows, it is purely modal with gentle accelerations. Its low dynamics led to its replacement by WLTC, but it remains valuable as a simple teaching cycle, and its structure lives on in MIDC.

4.12.9 EPA FTP-75

FTP-75 is the US city test. It consists of a cold-start phase and stabilised phase of the UDDS, a 10-minute soak, then a hot-start repeat of the first 505 s [21, 22]. Driven duration is 1874 s and distance 17.8 km, shown with the soak removed in Figure 4.95. For EVs the temperature

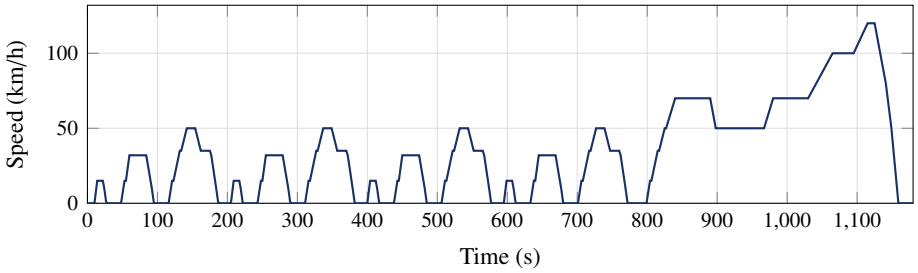


Figure 4.94: NEDC speed-time profile (official data [20]).

conditioning matters less than for engines, yet the cycle is still used in the EPA multi-cycle range test.

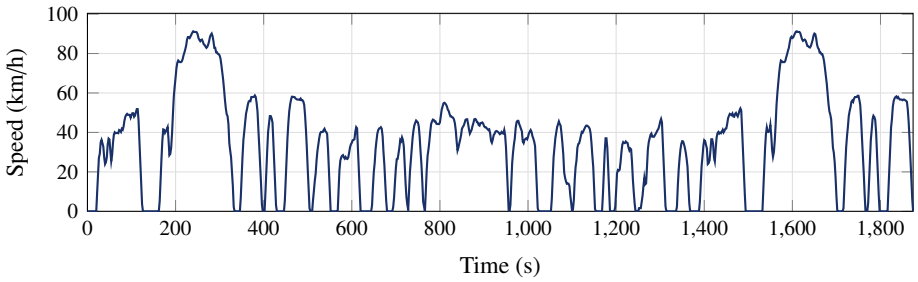


Figure 4.95: FTP-75 speed-time profile with the soak omitted (EPA data [21]).

4.12.10 Urban Dynamometer Driving Schedule (UDDS)

The UDDS represents urban driving in the USA, with 1369 s, 12.0 km and a peak of 91.2 km/h, as Figure 4.96 shows. It is transient, with many stops, and so it is a good test of regenerative braking.

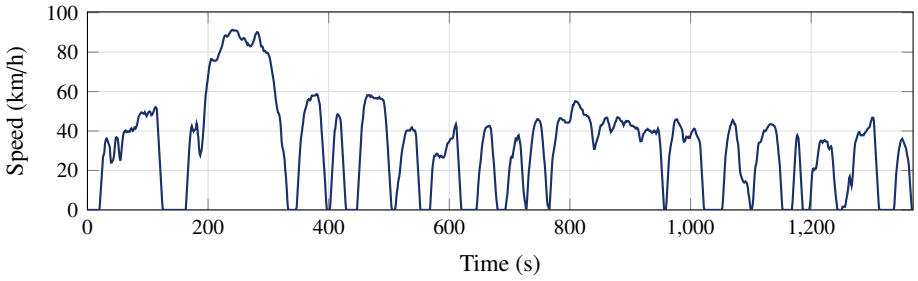


Figure 4.96: UDDS speed-time profile (EPA data [21]).

4.12.11 Highway Fuel Economy Test (HWFET)

HWFET represents highway driving below 60 mph [21]. It lasts 765 s and covers 16.5 km, with an average of 77.7 km/h and almost no idling, as Figure 4.97 shows. It is used with UDDS to compare city and highway consumption.

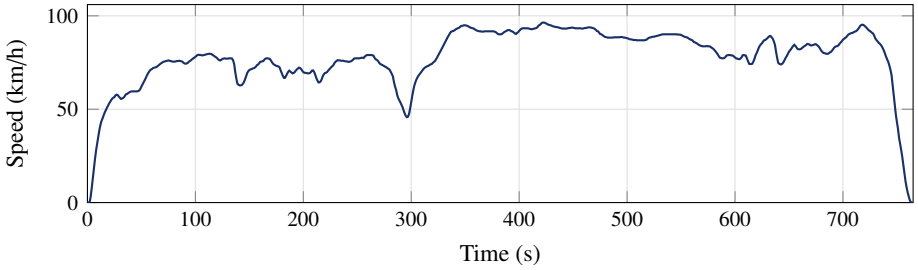


Figure 4.97: HWFET speed-time profile (EPA data [21]).

4.12.12 US06

US06 is the aggressive supplemental cycle, with hard acceleration up to 3.8 m/s^2 and a peak of 129 km/h, over 600 s and 12.9 km. Figure 4.98 shows its high-speed plateau and sharp transients. It stresses battery current and motor power, so it is a useful sizing and validation case, and it reveals power limits that gentle cycles hide.

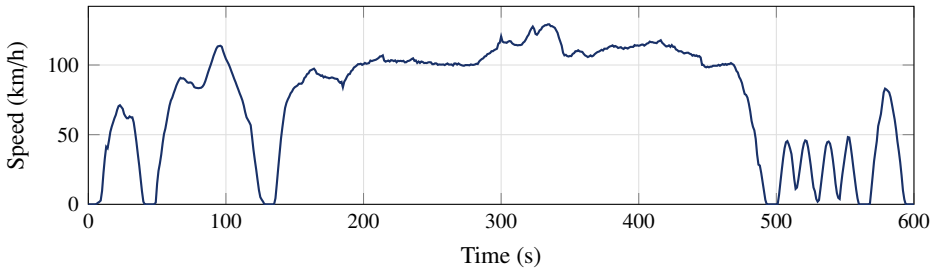


Figure 4.98: US06 speed-time profile (EPA data [21]).

4.12.13 Artemis Driving Cycles

The Artemis cycles were derived in Europe from recorded real-world driving [23]. There are three: Urban, Rural Road and Motorway, with motorway variants peaking near 130 and 150 km/h. They are not used in type approval, but are popular research cycles because they are realistic and strongly transient. Published versions differ slightly according to the start phases included. Figure 4.99 is an illustrative sketch of the three types.

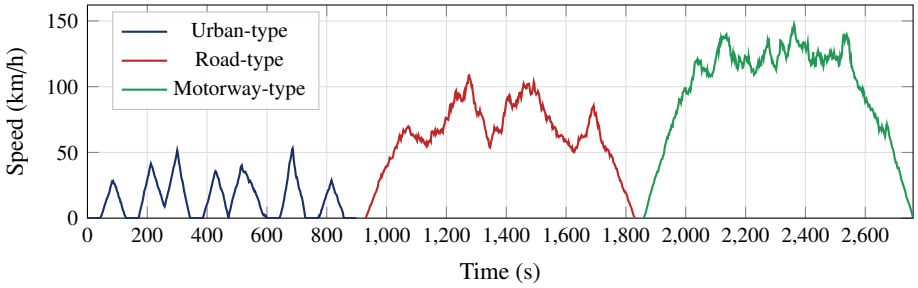


Figure 4.99: Artemis-type urban, road and motorway patterns: *randomly generated illustration, not official data*.

4.12.14 Comparison of Drive Cycles

Table 4.25: Types of drive cycles. Distance, speeds and acceleration for MIDC, WLTC, NEDC, UDDS, FTP-75, HWFET and US06 are computed from the data sets listed; IDC and Artemis values are as published.

Cycle	T (s)	D (km)	$v_{\max}$ (km/h)	v_{avg} (km/h)	$a_{\max}$ (m/s^2)	Purpose
IDC (6×108 s)	648	3.94	42	21.9	0.64	Indian 2W, 3W urban
MIDC	1180	10.67	90	32.5	1.04	Indian car approval, modal
WLTC Class 3b	1800	23.27	131.3	46.5	1.67	Global combined
NEDC	1180	11.01	120	33.6	1.04	Former EU approval, modal
FTP-75 (no soak)	1874	17.77	91.2	34.1	1.48	US city test
UDDS	1369	11.99	91.2	31.5	1.48	US urban
HWFET	765	16.51	96.4	77.7	1.43	US highway
US06	600	12.89	129.2	77.3	3.76	US aggressive
Artemis Urban	920	4.47	58	17.5	–	Real-world urban
Artemis Rural Road	1081	17.27	112	57.5	–	Real-world rural
Artemis Motorway 130	1067	28.74	132	97.0	–	Real-world motorway

Artemis values in Table 4.25 are as listed in [24], and other published lists vary somewhat. The table shows that the Indian and European approval cycles have low acceleration, while US06 reaches over three times that level. A motor sized only from NEDC or MIDC data would be far too small for real overtaking.

4.12.15 Drive-Cycle-Based Energy Consumption and Component Sizing

Cycles influence the design as follows.

- **Energy consumption.** The result of (4.17) depends on average speed, stops and acceleration. A cycle's kinematic intensity, $\sum \max(v_{k+1}^2 - v_k^2, 0)/(2D)$, measures accelerating energy per unit mass and metre, and it correlates with urban consumption.
- **Motor selection.** Plotting (T_m, ω_m) points shows the region of frequent use, and the motor's efficient zone should lie on it (Figure 4.123). Peak points check the envelope, and rms torque checks heating.
- **Battery sizing.** Energy is $E_u = eD$ for the target range, and $P_{b,\max}$ gives the C-rate. The regeneration peak checks charge acceptance.
- **Component sizing.** Inverter current and gear stress come from peak and rms values on the worst cycle.

4.12.16 Drive-Cycle Processing Algorithms

Algorithms 19 to 23 follow the flow of Figure 4.100. Data are loaded and checked; time and speed are cleaned up; and acceleration and distance are computed, leading to the cycle's energy. Each algorithm operates on arrays.

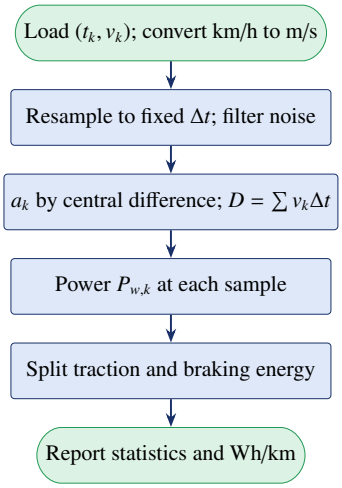


Figure 4.100: Drive-cycle data processing flowchart.

Algorithm 19 Drive-cycle data loading

Input: File with columns t, v; unit flag
Output: Arrays t_k, v_k in s and m/s

1: Read columns; reject missing or non-numeric rows
2: **if** unit is km/h **then** $v_k \leftarrow v_k/3.6$
3: **else if** unit is mph **then** $v_k \leftarrow 0.44704 v_k$
4: **end if**
5: Check t strictly increasing and $v_k \geq 0$; else stop with error

Algorithm 20 Speed-time data processing

Input: t_k, v_k, step Δt, filter window w
Output: Uniform $\tilde{t}_j$, $\tilde{v}_j$

1: $\tilde{t} \leftarrow t_1 : \Delta t : t_N$; $\tilde{v} \leftarrow$ linear interpolation of v at $\tilde{t}$
2: **if** data are measured (noisy) **then** apply moving average of width w
3: **end if**
4: Set $\tilde{v}_j \leftarrow 0$ when $\tilde{v}_j <$ threshold, to preserve idle periods

Algorithm 21 Acceleration calculation

Input: $\tilde{v}_j$, Δt
Output: a_j

1: $a_1 \leftarrow (\tilde{v}_2 - \tilde{v}_1)/\Delta t$
2: **for** $j = 2$ to $N - 1$ **do** $a_j \leftarrow (\tilde{v}_{j+1} - \tilde{v}_{j-1})/(2\Delta t)$
3: **end for**
4: $a_N \leftarrow (\tilde{v}_N - \tilde{v}_{N-1})/\Delta t$

Algorithm 22 Distance calculation

Input: $\tilde{v}_j$, Δt
Output: D, v_{avg}, idle fraction

1: $D \leftarrow \sum_{j=1}^{N-1} \frac{1}{2}(\tilde{v}_j + \tilde{v}_{j+1})\Delta t$ ▷ trapezoidal
2: $v_{\text{avg}} \leftarrow D/(t_N - t_1)$; idle fraction $\leftarrow$ share of samples with $\tilde{v}_j < 0.05$

Algorithm 23 Drive-cycle energy calculation

Input: $\tilde{v}_j$, a_j, vehicle parameters, P_{aux}, Δt
Output: E_{dis}, E_{reg}, E_{net}, e

```
for each j do
    F_t ← Algorithm 13; P_w ← F_t ṽ_j
    P_b ← (4.87) or (4.88) with limits
    if P_b > 0 then E_dis ← E_dis + P_b Δt
    else E_reg ← E_reg + |P_b| Δt
    end if
end for
E_net ← E_dis − E_reg; e ← E_net/D
```

4.12.17 MATLAB/Simulink Implementation of Drive-Cycle Analysis

Figure 4.101 and Table 4.26 show the conceptual processing chain. Powertrain Blockset also includes a `Drive Cycle Source` block with built-in standard cycles [33].

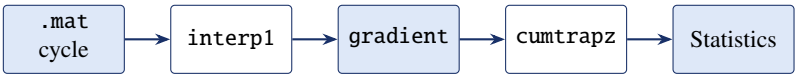


Figure 4.101: Conceptual drive-cycle analysis chain in MATLAB.

Table 4.26: Drive-cycle analysis: conceptual MATLAB implementation

Inputs	Cycle file, sample time, filter settings
Outputs	$a(t)$, D, v_{avg}, v_{max}, idle share, intensity
States	None; array calculation
Equations	(4.18) and Algorithm 22
Blocks	`readmatrix`, `interp1`, `gradient`, `cumtrapz`; or `Drive Cycle Source`
Feedback	None
Solver	Not required for array processing
Initial conditions	$x(0) = 0$
Assumptions	Linear interpolation between samples
Simulation outputs	Statistics table, speed and acceleration histograms

4.13 Types of Control Strategies

4.13.1 Introduction to EV Control Strategies

This section surveys control strategies at the vehicle level. Section 4.7 showed how a strategy fits inside the VCU; here each is described by operating principle, formulation, signals, implementation, merits, cost and application. Figure 4.102 shows the overall supervisory structure. A supervisor takes driver input and directs torque control, battery protection and thermal management, while plant measurements feed back. The specific strategies below may run at any of these locations.

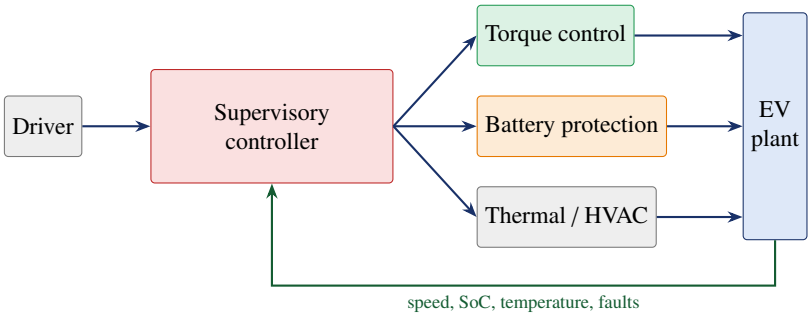


Figure 4.102: Supervisory EV control architecture.

4.13.2 Supervisory Control

Principle. A top layer coordinates subsystems and picks set-points. **Formulation.** $\mathbf{r}_k = \mathcal{S}(\mathbf{x}_k, \mathbf{d}_k)$, where $\mathcal{S}$ maps measured state and driver demand into references. **Inputs:** pedals, modes, limits. **Outputs:** torque, power and thermal set-points. **Feedback:** SoC, speed, temperature, faults. **Implementation:** periodic task on the VCU. **Advantages:** clear structure, separates concerns. **Limitations:** needs lower layers that are fast and reliable. **Computation:** low. **Applications:** all production EVs.

4.13.3 Rule-Based Control

Principle. IF-THEN rules chosen from engineering knowledge, as in Figure 4.103. Rules read the inputs, thresholds tune them and the output is a defined action. **Formulation.** $u = g_i(\mathbf{x})$ when condition $C_i(\mathbf{x})$ is true, with the first true condition winning. **Inputs:** u_a, u_b, v, z, Θ. **Outputs:** T_m^*, T_{fr}^*, flags. **Feedback:** measured states. **Implementation:** compare, branch, look up. **Advantages:** simple, transparent, cheap to verify. **Limitations:** not optimal, and tuning grows with the number of rules. **Computation:** very low. **Applications:** most mode logic and derating. Threshold rules need hysteresis, otherwise a signal near the threshold causes repeated switching.

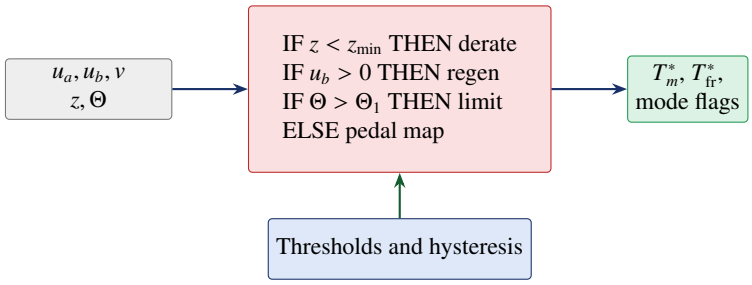


Figure 4.103: Rule-based control architecture.

4.13.4 State-Machine-Based Control

Principle. The system is in exactly one of a set of states, and defined events cause transitions, as in Figure 4.104, in which a car moves among Drive, Derated, Limp and Regen. **Formulation.** $q_{k+1} = \delta_s(q_k, \sigma_k)$, with q the state, and σ the event. **Inputs:** events derived from signals.

Outputs: mode and its control law. **Feedback:** state flags. **Implementation:** Stateflow chart, generated C code. **Advantages:** rigorous and testable; every state is defined. **Limitations:** state count can grow quickly. **Computation:** very low. **Applications:** start-up, gear selection, charge control, fault handling.

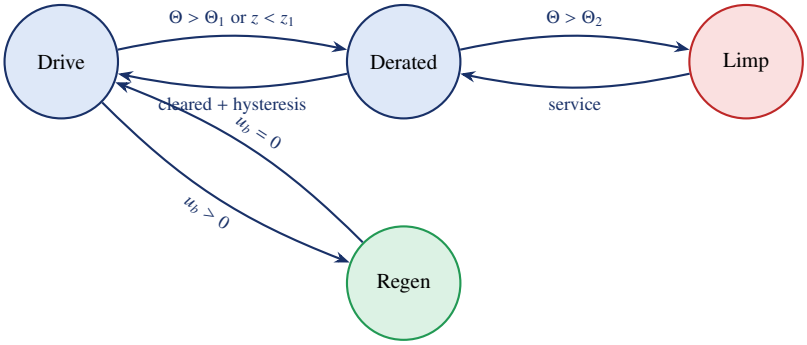


Figure 4.104: State-machine-based control with derating.

4.13.5 PI/PID Control

Principle. Corrective feedback on the error, as in Figure 4.105; terms act on present error, accumulated error and rate of change. **Formulation.** $u = K_p e + K_i \int e\,dt + K_d \dot{e}$, with limits and anti-windup. **Inputs:** reference and measured variable. **Output:** actuator demand. **Feedback:** controlled variable itself. **Implementation:** discrete $u_k = K_p e_k + \xi_k + K_d(e_k - e_{k-1})/\Delta t$, with $\xi_{k+1} = \xi_k + K_i e_k \Delta t$. **Advantages:** simple, well understood. **Limitations:** one-loop structure, fixed gains over a nonlinear plant. **Computation:** very low. **Applications:** cruise control, speed limit, current loops, thermal loops.

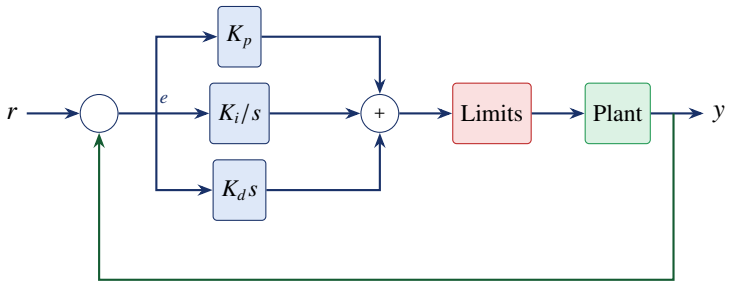


Figure 4.105: PI/PID control architecture.

4.13.6 Torque-Based, Speed-Based and Energy-Based Control

Torque-based control interprets the accelerator as a wheel torque request, $T_m^* = f(u_a, \omega_m)$, and is the standard for road EVs because it feels natural. Input: pedal and speed. Output: torque. Feedback: motor speed for limits. Advantages: direct, responsive; limitation: needs limit handling. **Speed-based control** closes a loop on ω_m or v, giving $T_m^* = C(s)(\omega^* - \omega_m)$. Used in cruise control, low-speed crawling and dynamometer motoring. Advantage: holds speed under varying load; limitation: less natural under direct driver control. **Energy-based control**

chooses commands that lower energy per km, such as an eco-mode restriction on acceleration or the dual-motor split that minimises total loss:

$$\min_{k} \frac{(kT\omega_f)}{\eta_f(kT,\omega_f)} + \frac{((1-k)T\omega_r)}{\eta_r((1-k)T,\omega_r)}. \tag{4.117}$$

Here k is the front share of total torque T, and the map is usually pre-computed offline, so that computation on the ECU is low.

4.13.7 Regenerative Braking Control

Principle. Convert braking demand to negative motor torque up to capability, as in Figure 4.106. The minimum block takes the demand and all limits, and the remainder goes to friction. **Formulation.** (4.49)–(4.51). **Inputs:** u_b, lift-off, speed, SoC, temperature. **Outputs:** T_{reg}, T_{fr}. **Feedback:** battery charge limit, wheel slip. **Implementation:** look-ups with rate limits, in the BCU or VCU. **Advantages:** recovers energy and saves pads. **Limitations:** depends on the battery state, and needs careful blending. **Computation:** low. **Applications:** all BEVs.

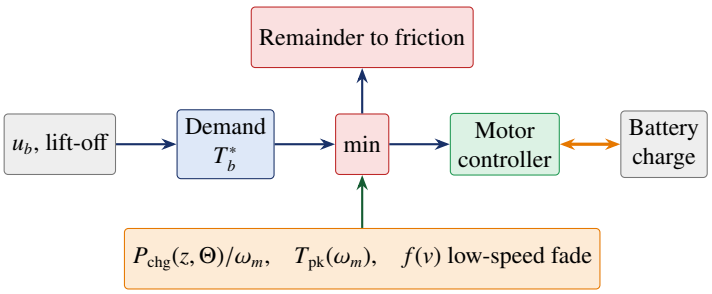


Figure 4.106: Regenerative braking controller.

4.13.8 Power Limitation, Battery Protection and Thermal Protection

These three are safety-related constraints rather than performance controllers. **Power limitation control** clamps torque to (4.58). **Battery protection control** computes P_{dis} and P_{chg} from voltage, SoC and temperature so that cell voltage stays within its limits. A simple voltage-based limit using the Rint model at minimum cell voltage $V_{\min}$ gives

$$I_{\max} = \frac{V_{oc} - V_{\min}}{R_0}, \qquad P_{\text{dis}} = V_{\min} I_{\max}. \tag{4.118}$$

Thermal protection control applies k_Θ from (4.59). Inputs are measurements of voltage, current and temperature; outputs are limits; feedback is the protected variable. Advantages are simplicity and strong protection. The limitation is that a derate reduces performance in a way the driver notices. Computation is negligible, and they are present in every EV.

4.13.9 Energy Management Strategies

For a BEV, energy management means controlling how the fixed energy store is spent. Examples are predictive pre-conditioning of the battery before DC charging, route-aware HVAC

planning, regeneration level chosen from map data, and eco-driving advice. A typical cost is total battery energy over a trip,

$$J = \int_0^T P_b(t)\, dt, \tag{4.119}$$

subject to comfort and trip-time constraints.

4.13.10 Optimisation-Based Control

Principle. Choose control inputs to minimise a cost subject to constraints. **Formulation.** $\min_{u(\cdot)} \sum_k L(x_k, u_k)$ subject to $x_{k+1} = f(x_k, u_k)$, $u \in \mathcal{U}$, $x \in \mathcal{X}$. Dynamic programming gives a global optimum when the whole cycle is known in advance. **Inputs:** model, cost, cycle. **Outputs:** optimal trajectory or map. **Feedback:** none offline; replaced by maps online. **Implementation:** offline solver, results stored as look-up tables. **Advantages:** gives the best achievable benchmark. **Limitations:** needs future knowledge, with a high computing load. **Computation:** high offline. **Applications:** benchmarks, map calibration, dual-motor split. Table 4.27 compares this with rule-based control.

Table 4.27: Rule-based versus optimisation-based control

Feature	Rule-based	Optimisation-based
Design basis	Engineering judgement	Mathematical cost function
Optimality	Near-optimal at best	Optimal for its model
Future knowledge	None	Often required
Real-time load	Very low	High unless precomputed
Certification	Easy	Harder
Robustness	High	Depends on model accuracy

The table shows why the two are combined in production: optimisation generates the calibration, and rules run on the ECU.

4.13.11 Model Predictive Control

Principle. At each sample, solve a finite-horizon optimal control problem and apply only the first move [25], as in Figure 4.107. Solving again at the next sample provides feedback. **Formulation.** Using a linear model,

$$\min_{u_0,\ldots,u_{N-1}} \sum_{i=0}^{N-1} \left[(y_i - r_i)^2 Q_y + u_i^2 R_u \right] \quad \text{s.t.} \quad x_{i+1} = Ax_i + Bu_i, \;\; u_{\min} \le u_i \le u_{\max}, \tag{4.120}$$

with horizon N, and weights Q_y and R_u. **Inputs:** state estimate, preview. **Outputs:** torque, or speed profile. **Feedback:** state at each sample. **Implementation:** quadratic programme solved in real time, or explicit MPC. **Advantages:** handles limits and preview directly. **Limitations:** depends on the model; needs a fast processor. **Computation:** high. **Applications:** eco-driving, adaptive cruise control, torque vectoring, and thermal management [26].

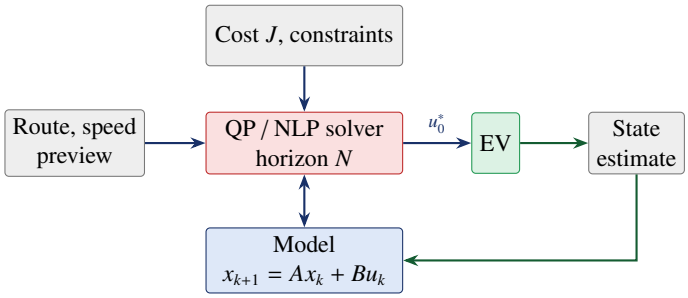


Figure 4.107: Model predictive control architecture for an EV.

4.13.12 Fuzzy Logic Control

Principle. Control knowledge is expressed in linguistic rules such as "IF SoC is high AND demand is low THEN regeneration is small" [27]. In Figure 4.108, crisp inputs are fuzzified into membership grades, rules are evaluated and the output is defuzzified. **Formulation.** Rule strength $w_i = \min(\mu_{A_i}(x_1), \mu_{B_i}(x_2))$, with centroid output $u = \sum w_i u_i / \sum w_i$. **Inputs:** e_v, z, deceleration. **Output:** T^* or κ. **Feedback:** chosen variables. **Implementation:** tables of rules and membership functions. **Advantages:** tolerates nonlinearity, requires no precise model. **Limitations:** heuristic tuning, stability is hard to prove. **Computation:** low to moderate. **Applications:** regenerative braking share, driving-style classification.

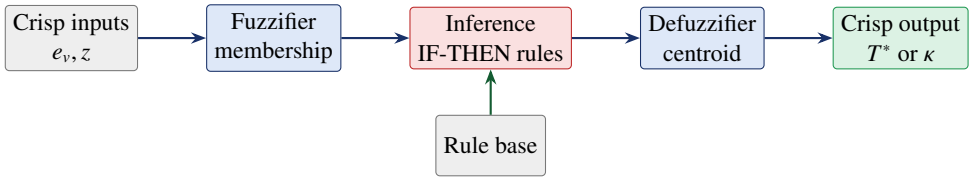


Figure 4.108: Fuzzy logic control architecture.

4.13.13 Adaptive and Intelligent Control

Adaptive control adjusts controller parameters in real time as the plant changes, for example adapting torque gain to vehicle mass, estimated from $\hat{m} = F_t/a$ during acceleration, or re-estimating R_0 as the battery ages. It has low to moderate computational load, but requires sufficient excitation to work reliably. **Intelligent control** covers neural networks, reinforcement learning and related data-driven methods. A reinforcement learning agent learns a policy $u = \pi(x)$ that maximises reward, for example low energy with good tracking. Such methods can handle complex behaviour, but require extensive data and raise verification challenges, and so most remain in research or in supervisory, non-safety roles.

4.13.14 Control Algorithms

Algorithms 24 to 29 summarise the main strategies in a compact form, for comparison with the fuller versions in Sections 4.6 and 4.7.

Algorithm 24 Rule-based EV control

Input: u_a, u_b, v, z, Θ, thresholds z_1, Θ_1, Θ_2
Output: T_m^*, derate k

$k \leftarrow 1$
if $\Theta > \Theta_2$ **then** $k \leftarrow k_{\min}$
else if $\Theta > \Theta_1$ **then** $k \leftarrow$ (4.59)
end if
if $z < z_1$ **then** $k \leftarrow \min(k, k_z)$ ▷ low-SoC derate
end if
if $u_b > 0$ **then** use Algorithm 27
else $T_m^* \leftarrow k\, u_a\, T_{\max}(\omega_m)$
end if

Algorithm 25 State-machine-based vehicle control

Input: State $q \in \{\text{Drive}, \text{Derated}, \text{Limp}, \text{Regen}\}$, signals, hysteresis h
Output: New q, control law

switch q **do**
    case Drive:
        if $\Theta > \Theta_1$ **or** $z < z_1$ **then** $q \leftarrow$ Derated
        else if $u_b > 0$ **then** $q \leftarrow$ Regen
        end if
    end case
    case Derated:
        if $\Theta > \Theta_2$ **then** $q \leftarrow$ Limp
        else if $\Theta < \Theta_1 - h$ **and** $z > z_1 + h$ **then** $q \leftarrow$ Drive
        end if
    end case
    case Regen:
        if $u_b = 0$ **then** $q \leftarrow$ Drive
        end if
    end case
    case Limp:
        Hold until service reset
    end case
end switch
Run the control law assigned to q

Algorithm 26 Torque command generation

Input: u_a, ω_m, γ, k, previous T, limits
Output: T_m^*

$T_{\text{dem}} \leftarrow k\, T_{\max}(\omega_m)\, u_a^{\gamma}$
Apply rate limit (4.56)
$T_m^* \leftarrow$ saturate with (4.60)

Algorithm 27 Regenerative braking control (compact)

Input: u_b, ω_m, v, P_{chg}
Output: T_{reg}, T_{fr}
1: $T_b^* \leftarrow u_b F_{b,\max} r_w$
2: $T_{\text{reg}} \leftarrow \min(T_b^*, GT_{\text{reg,lim}})$ using (4.49)
3: $T_{\text{fr}} \leftarrow T_b^* - T_{\text{reg}}$

Algorithm 28 Battery power limitation

Input: V_t, I, z, Θ_b, R_0, $V_{\min}$, $V_{\max}$
Output: P_{dis}, P_{chg}
1: $I_{\text{dis}} \leftarrow \min\left((V_{oc} - V_{\min})/R_0,\ I_{\text{dis,rated}}(\Theta_b)\right)$
2: $I_{\text{chg}} \leftarrow \min\left((V_{\max} - V_{oc})/R_0,\ I_{\text{chg,rated}}(\Theta_b, z)\right)$
3: $P_{\text{dis}} \leftarrow V_{\min} I_{\text{dis}}$; $P_{\text{chg}} \leftarrow V_{\max} I_{\text{chg}}$

Algorithm 29 MPC-based vehicle control overview

Input: Model (A, B), horizon N, weights, limits, preview r_i
Output: u_0^*
1: **for** each sample **do**
2: Measure or estimate x_0
3: Solve (4.120), giving $u_0^*, \ldots, u_{N-1}^*$
4: Apply u_0^* and discard the rest; shift the horizon
5: **end for**

4.13.15 Comparison of Control Strategies

Table 4.28: EV control strategies compared

Strategy	Model need	Compute	Optimality	Typical EV role
Rule-based	None	Very low	Low	Modes, derating
State machine	None	Very low	None	Sequencing, faults
PI/PID	Slight	Very low	Local	Loops
Torque-based	Slight	Low	None	Driver interpretation
Speed-based	Slight	Low	None	Cruise, crawl
Regenerative	Moderate	Low	Moderate	Braking recovery
Protection	Moderate	Very low	None	Safety
Optimisation	High	High offline	Global	Benchmark, maps
MPC	High	High	Over horizon	Predictive energy use
Fuzzy	None	Low	Heuristic	Regeneration share
Adaptive	Moderate	Moderate	Adaptive	Parameter tracking
Intelligent	Data	Moderate to high	Learned	Research, supervisory

The table shows a consistent trade-off: model and computing demands grow with achievable optimality, while ease of verification falls.

4.14 EV Performance Analysis

4.14.1 Introduction to EV Performance Analysis

Performance analysis uses the models of this unit to determine how the vehicle will behave. Figure 4.109 shows the framework. Vehicle, cycle and ambient inputs enter the simulation, which produces three groups of results: dynamic performance, energy performance and component-level detail. These then feed back into design. Table 4.29 lists the key parameters.

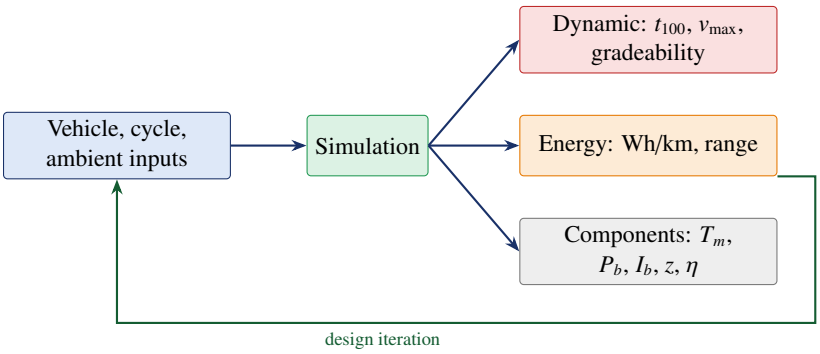


Figure 4.109: EV performance analysis framework.

Table 4.29: EV performance parameters, with results for the hypothetical reference BEV

Parameter	Equation	Reference BEV	Governed by
0–100 km/h time	(4.110)	7.5 s (no tyre limit)	Power, mass
Maximum speed	(4.121)	156 km/h	Speed limit, drag
Gradeability at 50 km/h	(4.115)	59% (motor only)	Torque, ratio
Consumption, WLTC	(4.129)	144 Wh/km	Efficiency, load
Range, WLTC	(4.130)	315 km	Usable energy
Drivetrain efficiency	(4.123)	0.875 assumed	Components

4.14.2 Vehicle Acceleration, Maximum Speed and Gradeability

Acceleration time follows from integrating (4.110) using the tractive force curve of Figure 4.87. A numerical run of the reference BEV gives 7.5 s for 0–100 km/h. The launch force of 7885 N would exceed adhesion on a driven axle at $\mu = 0.9$, so a real car with traction control takes a few tenths longer.

Maximum speed is the lowest of a power-limited value, where tractive power equals road-load power, and a motor speed limit:

$$v_{\max} = \min\!\left(v_P,\ \frac{\omega_{m,\max} r_w}{G}\right), \qquad \eta_g P_{\mathrm{pk}} = \left(C_{rr} m g + \tfrac{1}{2}\rho C_d A v_P^2\right) v_P. \tag{4.121}$$

Here $v_P = 224$ km/h, while the motor speed limit gives 156 km/h. The reference vehicle is therefore speed-limited, and its peak power has been determined by acceleration.

Gradeability and hill climbing capability follow (4.115). Peak values for the reference car are 62% at 20 km/h, 59% at 50 km/h and 24% at 100 km/h, using peak motor torque and before the tyre limit applies. A hill of length L_h and rise h also costs battery energy mgh/η_{dt}. If a grade is long, continuous power rather than peak power sets the climbing speed.

4.14.3 Tractive Effort and Vehicle Power

The **tractive effort** available at speed v is

$$F_{t,\mathrm{av}}(v) = \frac{G\eta_g}{r_w}\min\!\left(T_{\mathrm{pk}},\ \frac{P_{\mathrm{pk}} r_w}{Gv}\right), \tag{4.122}$$

and the vehicle power is $P_w = F_t v$ as in (4.81). The excess $F_{t,\mathrm{av}} - F_{\mathrm{res}}$ gives the acceleration margin at each speed.

4.14.4 Simulated Time-Domain Performance

The following figures come from the forward-looking simulation of the reference BEV over the 400 s Part Two of the official MIDC, with a starting SoC of 90% at the start of the full cycle. They are **simulated results** for a hypothetical car, not measurements.

Figure 4.110 shows vehicle speed. Since this part has clear plateaus at 70, 50 and 90 km/h, the role of each load is easy to see. Figure 4.111 overlays reference and actual speed, and the two lie almost together, with an RMS error of 0.32 km/h over the full cycle. This indicates that the car is well within its capabilities. Figure 4.112 gives acceleration: short pulses near 1 m/s^2 at launch, and the strongest deceleration, near -1.6 m/s^2, in the final stop. These are small values compared with the car's ability.

Figure 4.113 shows motor speed, which is proportional to vehicle speed since G is fixed, reaching about 6900 r/min at 90 km/h. Figure 4.114 shows motor torque. Its peak, 73 N m, is far below the 280 N m rating, and cruising needs about 10–16 N m. Figure 4.115 shows the tractive force and Figure 4.116 the wheel torque, which are scaled copies of motor torque. Negative intervals are regeneration.

Figure 4.117 gives motor mechanical power. Note how power, unlike torque, climbs during a constant-torque acceleration, since it is the product of torque and rising speed. Figure 4.118 shows battery power, shifted upward from motor power by losses and the auxiliary load; it peaks at 23.1 kW in traction, and reaches -26.3 kW during the last stop. Figure 4.119 shows battery current, with a similar shape, between 62 A and about -70 A. Figure 4.120 shows the terminal voltage, moving by a few volts about a slowly falling mean, and rising as current reverses. Figure 4.121 shows SoC falling almost linearly, with small rises where regeneration occurs. Over the whole cycle SoC falls from 90.0% to 87.6%.

Figure 4.122 shows net energy consumed versus distance over the full MIDC. Its slope is the instantaneous consumption. Urban sections have a staircase shape, the slope is steep at higher speed, and the final stop produces a small decrease due to regeneration. Figure 4.123 shows operating points of a WLTC and a US06 backward simulation on the motor torque-speed plane

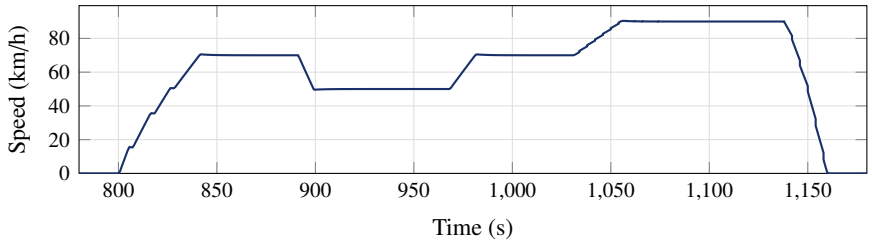


Figure 4.110: Vehicle speed versus time (simulated).

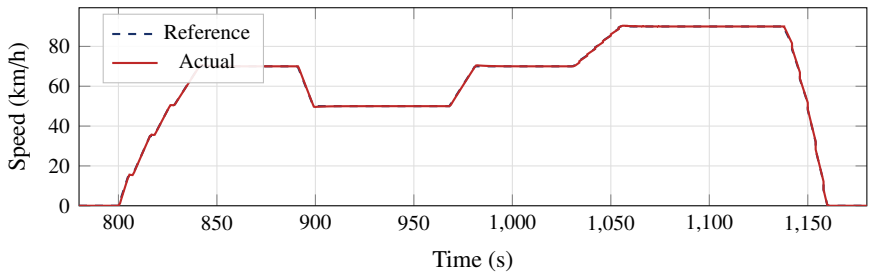


Figure 4.111: Vehicle speed tracking performance (simulated).

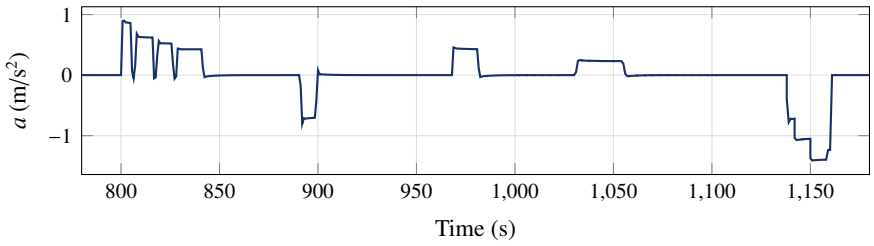


Figure 4.112: Vehicle acceleration versus time (simulated).

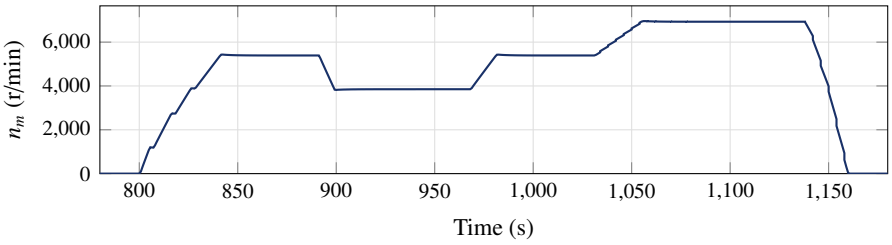


Figure 4.113: Motor speed versus time (simulated).

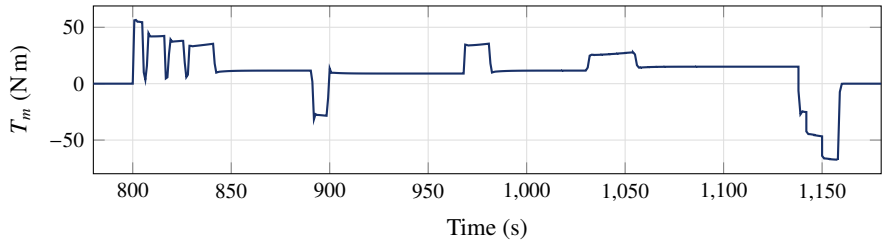


Figure 4.114: Motor torque versus time (simulated).

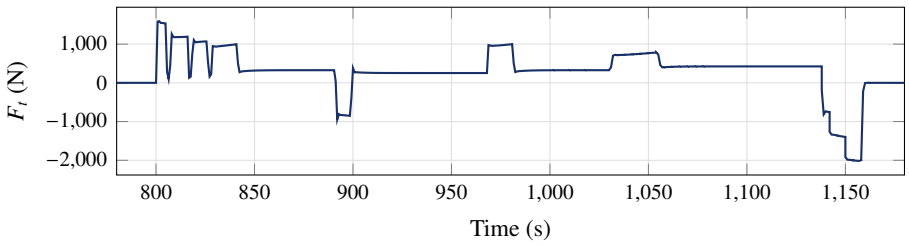


Figure 4.115: Tractive force versus time (simulated).

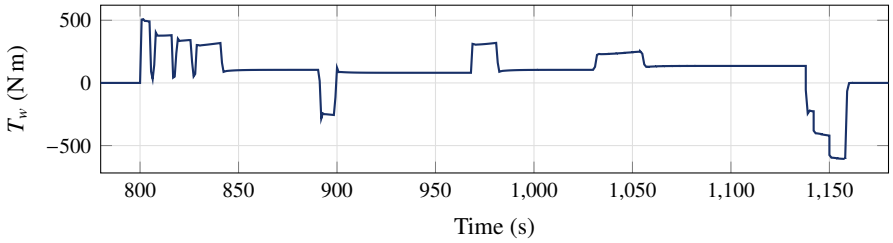


Figure 4.116: Wheel torque versus time (simulated).

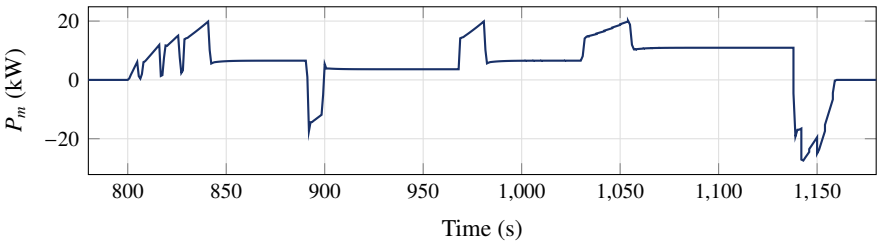


Figure 4.117: Motor power versus time (simulated).

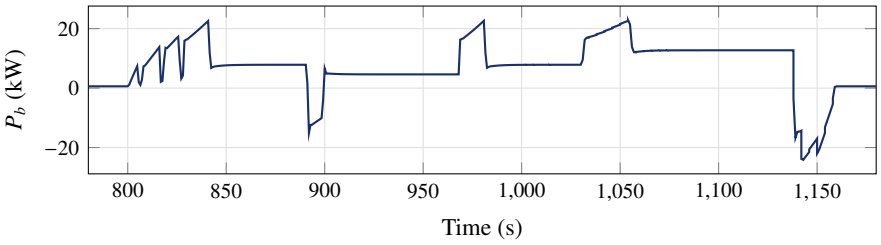


Figure 4.118: Battery power versus time (simulated).

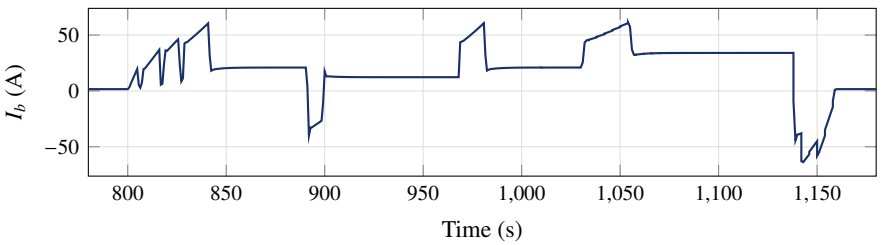


Figure 4.119: Battery current versus time (simulated).

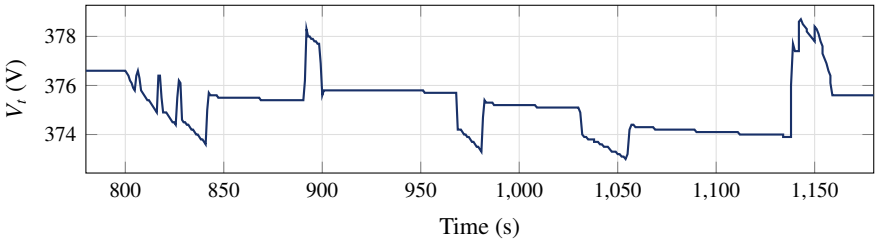


Figure 4.120: Battery terminal voltage versus time (simulated).

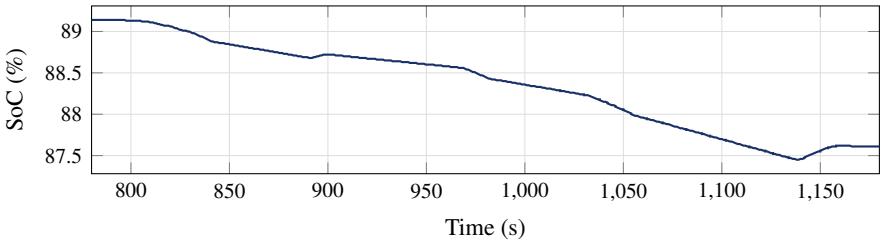


Figure 4.121: Battery SoC versus time (simulated).

(the solid curve is the motoring envelope, the dashed curve the regeneration envelope). Almost all points lie far below the peak curve, inside the continuous envelope, clustering at moderate speed and low torque. A motor should thus have its best efficiency here, not near rated torque. US06 uses a much wider area, and it reaches the highest torque on the plot.

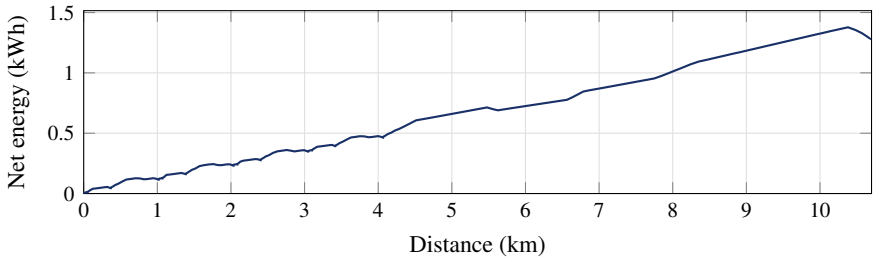


Figure 4.122: Energy consumption versus distance on MIDC (simulated).

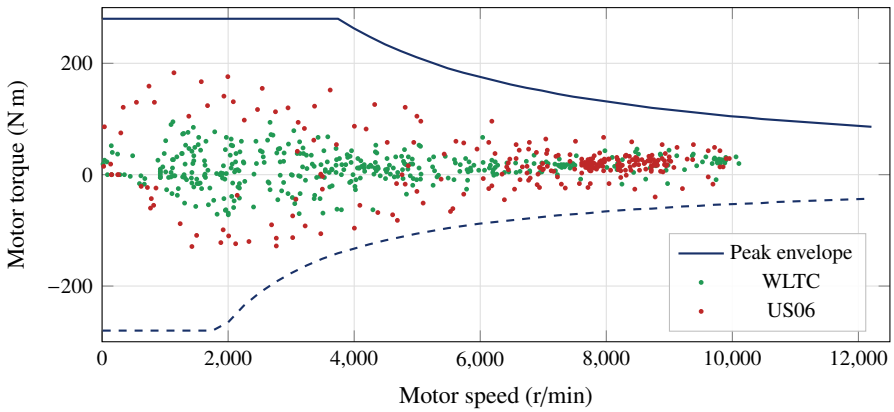


Figure 4.123: Motor torque-speed characteristics with cycle operating points (simulated).

4.14.5 Drivetrain Efficiency, Battery Energy and Regenerative Recovery

Over a cycle, the **average drivetrain efficiency** is the ratio of positive wheel energy to positive DC energy,

$$\bar{\eta}_{\mathrm{dt}} = \frac{\int_{P_w>0} P_w \, dt}{\int_{P_{\mathrm{dc}}>0} P_{\mathrm{dc}} \, dt}. \tag{4.123}$$

The regenerative energy recovery fraction is $E_{\mathrm{reg}}/E_{\mathrm{dis}}$. On WLTC, with the reference car, the wheel traction energy is 3.34 kWh, the DC energy 3.82 kWh, and the battery recovers $0.692/4.05 = 17\%$ of its discharge energy.

4.14.6 Effects of Environment, Mass, Aerodynamics and Tyres

Changing one parameter at a time in the backward model yields the sensitivities in Table 4.30. A 10% mass increase costs 6.2% on UDDS and 4.5% on HWFET. In contrast, a 10% drag reduction saves 2.2% and 4.9% respectively. A tyre with 20% lower rolling resistance saves over 8% on either cycle. Cold, dense air raises drag, and consumption increases more on the highway. In a real car, cold weather also affects tyre pressure and raises HVAC and battery heating loads, which these constant-efficiency results leave out.

Table 4.30: Simulated sensitivity of net energy consumption (hypothetical reference BEV)

Change	UDDS change (%)	HWFET change (%)
Mass +10% (1650 kg)	+6.2	+4.5
C_d −10% (0.27)	−2.2	−4.9
C_{rr} −20% (0.0096)	−8.9	−8.3
Air density +12.5% (1.35 kg/m^3, cold)	+2.8	+6.1

Example 20: Numerical performance analysis for a representative BEV

Problem. For the reference BEV, estimate the 0–100 km/h time, the speed-limited top speed and the gradeability at 100 km/h, and compare the time with one using 60 kW only.

Given. Table 4.3, peak 280 N m, 110 kW; motor limit 12 000 r/min.

Required. t_{100}, $v_{\max}$, grade at 100 km/h.

Governing equations. (4.110), (4.121), (4.115).

Solution. Numerically integrating (4.74) with $T_m = \min(280, 110\,000/\omega_m)$ and $\Delta t = 0.01$ s gives $t_{100} = 7.5$ s.
With 150 N m and 60 kW the same calculation gives 14.6 s.
Motor speed limit: $v = 1257(0.31)/9 = 43.3$ m/s, or 156 km/h.
At 100 km/h, $\omega_m = 806$ rad/s, $T = 136.4$ N m, $F_t = 3841$ N; drag = 305.6 N, so $\beta = (3841 - 306)/14\,715 = 0.2402$; (4.115) gives $\sin\theta = 0.2285$, or a grade of 23.5%.

Result. $t_{100} = 7.5$ s, $v_{max} = 156$ km/h, 23.5% gradeability at 100 km/h.

Physical interpretation. Top speed is well below the power-limited value because the motor reaches its speed limit first.

Engineering significance. Doubling peak power halves the acceleration time, roughly, but has no effect on top speed here.

4.15 Driving Range and Energy Consumption Prediction

4.15.1 Introduction to EV Range Prediction

Range is the distance a vehicle can travel on one charge. It equals usable energy divided by energy consumption per unit distance. The numerator is mostly a property of the battery, and the denominator a property of the vehicle, driver, route and weather. Range prediction therefore needs to model both, as shown in Figure 4.124. Route, speed, grade and ambient conditions go into a vehicle model to give consumption e. SoC and health determine usable energy E_u. Their ratio is shown to the driver, and the measured consumption continually updates the estimate.

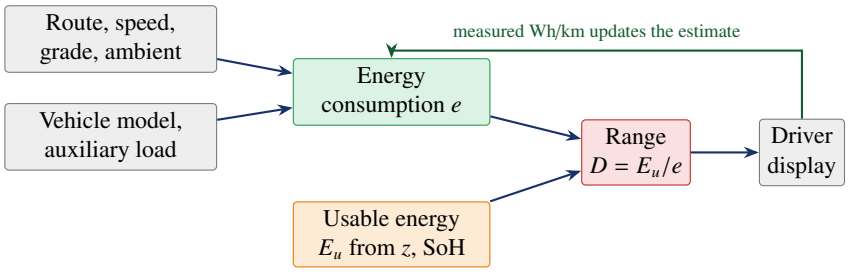


Figure 4.124: EV range prediction architecture.

4.15.2 Rated and Real-World Driving Range

Rated range comes from a standard test, such as MIDC in India, WLTP in Europe or the EPA multi-cycle procedure in the USA. Results vary among cycles, because the speed profiles differ. **Real-world range** depends on actual speed, traffic, terrain, climate and driver. The two can differ widely. A rated MIDC range is typically optimistic relative to real driving, since MIDC is gentle and is run in mild temperature with HVAC off.

4.15.3 Battery Usable, Nominal and Operating Window

Nominal energy is $E_{nom} = V_{nom} Q$. **Usable energy** is lower, because the BMS keeps a buffer at the top and bottom, within an **operating window**:

$$E_u = \mathrm{SoH} \cdot (z_{max} - z_{min})\, E_{nom}, \tag{4.124}$$

with SoH the fractional state of health. For the reference car $E_u = 0.9(50.46) = 45.4$ kWh when new. Terminal energy delivered also depends a little on discharge rate, since resistive loss rises with current.

4.15.4 Sign Conventions, Units and Assumptions

In this section:

- $P_b > 0$ means power leaves the battery; $P_b < 0$ means power enters it.
- Energy consumption is defined at the battery *terminals*, in Wh/km or kWh/100 km, with 1 kWh/100 km = 10 Wh/km.
- Time is in s, distance in m or km, and power in W unless otherwise shown. A division by 3600 converts J to Wh.
- The window is assumed fixed, and in the case studies efficiencies are assumed constant.

Consumption stated "from the wall" includes charging losses, and is therefore higher than terminal consumption.

4.15.5 Mathematical Modelling of Energy and Range

(1) Instantaneous tractive power. From (4.81), in W,

$$P_{\text{tr}}(t) = F_t(t)\, v(t). \tag{4.125}$$

(2) Instantaneous battery power. Using (4.87) and (4.88),

$$P_{\text{battery}}(t) = \begin{cases} \dfrac{P_{\text{tr}}}{\eta_{\text{dt}}} + P_{\text{aux}}, & P_{\text{tr}} \geq 0, \\ \max(\kappa P_{\text{tr}} \eta_{\text{dt}}, -P_{\text{chg}}) + P_{\text{aux}}, & P_{\text{tr}} < 0. \end{cases} \tag{4.126}$$

(3) Total battery energy consumed. Integrating over the trip of duration T,

$$E_{\text{battery}} = \int_0^T P_{\text{battery}}(t)\, dt. \tag{4.127}$$

(4) Net battery energy. Separating the two signs gives

$$E_{\text{net}} = E_{\text{dis}} - E_{\text{reg}} = E_{\text{battery}}, \tag{4.128}$$

as in (4.25); the integral of the signed power automatically gives the net value.

(5) Energy consumption per kilometre. With D the distance,

$$\text{Energy Consumption} = e = \frac{E_{\text{battery}}}{D}, \qquad D = \int_0^T v\, dt. \tag{4.129}$$

When E_{battery} is in Wh and D in km, e is in Wh/km.

(6) Driving range estimation.

$$\text{Driving Range} = D_R = \frac{E_{\text{usable}}}{\text{Energy Consumption per unit distance}} = \frac{E_u}{e}. \tag{4.130}$$

(7) Range correction based on usable energy. If the current SoC is z and the reserve is $z_{\min}$, remaining range is

$$D_{\text{rem}} = \frac{\text{SoH}\,(z - z_{\min})E_{\text{nom}}}{e}. \tag{4.131}$$

(8) Energy recovery due to regenerative braking. Let E_{brk} be wheel braking energy on a trip. The recovered energy is

$$E_{\text{reg}} = \kappa_{\text{eff}}\,\eta_{\text{dt}}\,E_{\text{brk}}, \tag{4.132}$$

with κ_{eff} the average share handled by the motor after limits. The range gain compared with no regeneration is $D_R/D_{R,0} = e_0/e$.

(9) Auxiliary load contribution. Auxiliary energy per km depends on average speed v_{avg} in km/h:

$$e_{\text{aux}} = \frac{P_{\text{aux}}}{v_{\text{avg}}}\ \text{Wh/km}, \qquad P_{\text{aux}}\ \text{in W}. \tag{4.133}$$

At 20 km/h, a 2 kW heater costs 100 Wh/km; at 100 km/h, only 20 Wh/km.

(10) Efficiency-based range prediction. A quick estimate from wheel energy per km e_w is

$$D_R = \frac{E_u}{\dfrac{e_w}{\eta_{\text{dt}}}(1 - \rho_r) + e_{\text{aux}}}, \tag{4.134}$$

where $\rho_r = E_{\text{reg}}/E_{\text{dis}}$ is the regenerated fraction of the traction energy at the battery.

Figure 4.125 and Algorithms 30 to 32 give these steps in procedure form. They reuse the backward-looking calculation, and the last adds a breakdown of braking energy.

Algorithm 30 Energy consumption calculation

Input: Cycle, vehicle data, P_{aux}, Δt
Output: E_{dis}, E_{reg}, E_{net}, D, e

1: **for** each sample k **do**
2: $P_{\text{tr}} \leftarrow F_{t,k}v_k$; $P_b \leftarrow$ (4.126)
3: Add $P_b\Delta t$ to E_{dis} if $P_b > 0$, or $|P_b|\,\Delta t$ to E_{reg} otherwise
4: $D \leftarrow D + v_k\Delta t$
5: **end for**
6: $E_{\text{net}} \leftarrow E_{\text{dis}} - E_{\text{reg}}$; $e \leftarrow E_{\text{net}}/(3.6D)$ ▷ J and m to Wh/km

4.15.6 Dependencies of Energy Consumption

- **Vehicle speed dependency.** Drag grows with v^2, whereas auxiliary energy falls as $1/v$. Figure 4.126 shows the result for steady speed: a minimum near 30–35 km/h at low auxiliary load, moving up to about 50 km/h with HVAC on, and a strong rise above that.

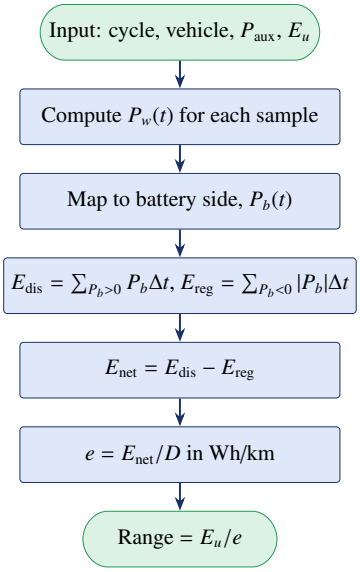


Figure 4.125: Energy consumption calculation flowchart.

Algorithm 31 Driving range prediction

Input: E_{nom}, SoH, z, $z_{\min}$, e_{model}, recent e_{meas}, weight α
Output: Remaining range D_{rem}

$E_u \leftarrow \text{SoH}(z - z_{\min}) E_{\text{nom}}$
$e \leftarrow \alpha e_{\text{meas}} + (1 - \alpha) e_{\text{model}}$ ▷ blend history and model
$D_{\text{rem}} \leftarrow E_u / e$
if HVAC changed **then** recompute e with (4.133)
end if

Algorithm 32 Regenerative energy recovery calculation

Input: $P_{w,k}$, $\eta_g, \eta_m, \eta_{\text{inv}}$, κ, P_{chg}, fade f_v
Output: E_{brk}, $E_{\text{dc,reg}}$, E_{fr}, κ_{eff}

for each sample with $P_{w,k} < 0$ **do**
    $P_s \leftarrow \min\left(\kappa f_v \left|P_{w,k}\right| \eta_g,\ P_{\text{chg}}/(\eta_m \eta_{\text{inv}})\right)$
    $E_{\text{brk}} \leftarrow E_{\text{brk}} + \left|P_{w,k}\right| \Delta t$
    $E_{\text{dc,reg}} \leftarrow E_{\text{dc,reg}} + P_s \eta_m \eta_{\text{inv}} \Delta t$
    $E_{\text{fr}} \leftarrow E_{\text{fr}} + (\left|P_{w,k}\right| - P_s/\eta_g) \Delta t$
end for
$\kappa_{\text{eff}} \leftarrow E_{\text{dc,reg}} / (\eta_g \eta_m \eta_{\text{inv}} E_{\text{brk}})$

- **Acceleration dependency.** Stronger acceleration raises peak current, loss and braking energy, as shown by the higher value for US06 in Figure 4.131.
- **Road gradient dependency.** Climbing h metres costs mgh/η_{dt}, and on the way down, at most $\kappa\eta_{\mathrm{dt}}mgh$ returns, so hills always have a net cost.
- **Ambient temperature dependency.** Cold reduces usable capacity, raises resistance and calls for heating, and heat calls for cooling. Field studies report strong regional variation in efficiency due to temperature [30].
- **HVAC and auxiliary load dependency.** See (4.133) and Figure 4.133.
- **Battery ageing dependency.** E_u falls with SoH, and resistance growth adds some loss.
- **Driving behaviour dependency.** Smooth drivers need less braking energy and use less power.
- **Regenerative braking contribution.** See Figure 4.132.
- **Drivetrain efficiency.** Consumption scales almost directly with $1/\eta_{\mathrm{dt}}$.

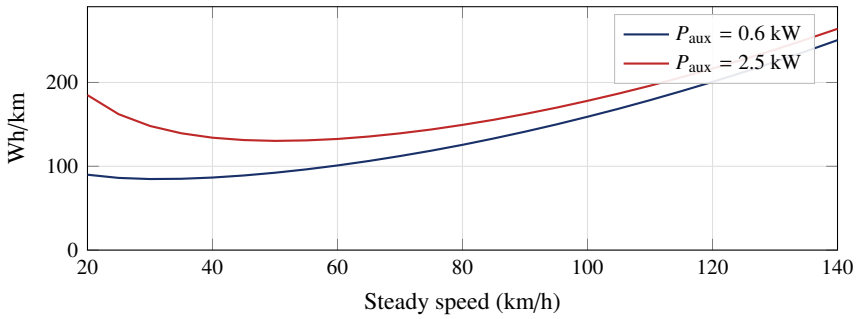


Figure 4.126: Effect of steady vehicle speed on energy consumption (calculated, hypothetical BEV).

4.15.7 Energy Consumption Prediction Models and Approaches

- **Physics-based prediction.** Uses the equations of this unit. It needs no training data and extrapolates well, but requires parameters and route information [31].
- **Drive-cycle-based prediction.** Selects or builds a representative cycle for the expected trip, then simulates it.
- **Historical energy consumption methods.** These use the average over recent kilometres, $e_{\mathrm{meas}} = \Delta E/\Delta D$. This is the simple "guess-o-meter" in many cars, which reacts slowly to route changes.
- **Statistical prediction.** Fits distributions of consumption to trip features, giving a range with confidence bounds.

- **Regression-based prediction.** Fits a function such as $e = \beta_0 + \beta_1 v^2 + \beta_2 / v + \beta_3\, s + \beta_4 P_{\text{aux}}$ by least squares. The terms are chosen from physics, which improves extrapolation.
- **Machine learning-based prediction.** Random forests and gradient boosting learn nonlinear mappings from trip features to e [29].
- **Neural network-based range prediction.** Networks, including recurrent ones, process speed and battery time series, and need a lot of data.
- **Hybrid physics and data-driven prediction.** A physics model gives the baseline e_{phys} and a learned term corrects the residual, $e = e_{\text{phys}} + g_\theta(\mathbf{x})$. This retains physical plausibility while capturing unmodelled effects.

Table 4.31 compares the two families. Physics-based models are explainable and need parameters. Data-driven models adapt to the fleet and need data. Hybrid methods try to take the best of each.

Table 4.31: Physics-based versus data-driven range prediction

Feature	Physics-based	Data-driven
Data needs	Vehicle parameters	Large trip data set
Interpretability	High	Low to moderate
Extrapolation	Good	Poor outside training data
Unmodelled effects	Missed	Captured if in data
Compute at inference	Low to moderate	Low after training
Adaptation	Parameter updates	Retraining or online learning

4.15.8 MATLAB/Simulink Implementation of Energy and Range Prediction

Figures 4.127 and 4.128 show the conceptual chains for energy accumulation and range. Table 4.32 gives the details for both. Two integrators accumulate discharge and regenerative energy separately, and the range calculation divides usable energy by a filtered consumption signal.

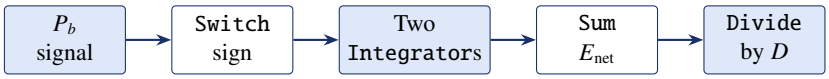


Figure 4.127: Conceptual Simulink diagram of the energy consumption calculation.

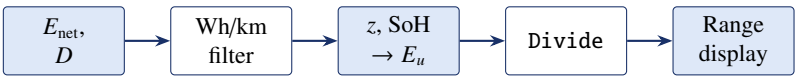


Figure 4.128: Conceptual Simulink diagram of driving range prediction.

Table 4.32: Energy consumption and range prediction: conceptual Simulink implementation

Inputs	P_b, v, z, SoH, E_{nom}, HVAC power
Outputs	E_{dis}, E_{reg}, e, D_{rem}
States	Energy integrators, distance, filter state
Equations	(4.127)–(4.131)
Blocks	`Switch`, `Integrator`, `Divide`, `Discrete Filter`, `MATLAB Function`
Feedback	Measured consumption updates predicted range
Solver	Same as vehicle model; 1 s is adequate
Initial conditions	Zero energy, $D = 0$, assumed initial e
Assumptions	Fixed window, constant SoH during trip
Simulation outputs	Wh/km, range, remaining-range trace

4.15.9 Energy and SoC versus Distance

Figures 4.129 and 4.130 repeat the WLTC until the usable energy is gone. Usable energy falls in repeating small ripples, reaching zero at about 315 km. The SoC plot shows the same run mapped to a window from 95% to 5%. The slope of either graph is the consumption, so a straight line indicates that this cycle repeats the same consumption throughout. The method is a simple form of cycle-based range test, in which voltage and resistance effects are small enough to ignore for this hypothetical pack.

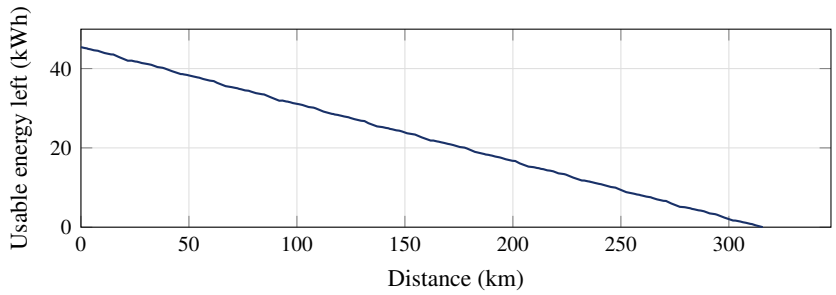


Figure 4.129: Battery energy versus driving distance on repeated WLTC (simulated).

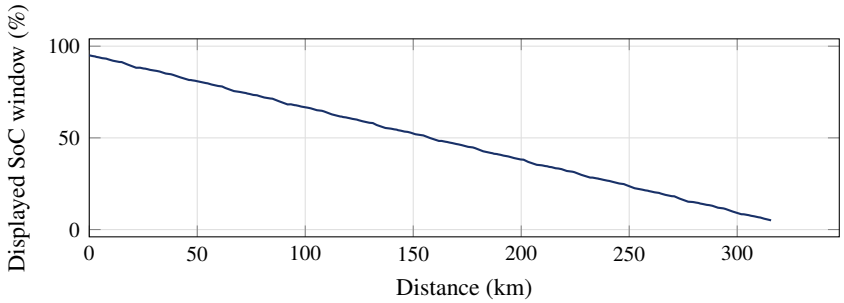


Figure 4.130: SoC versus distance on repeated WLTC (simulated).

4.15.10 Case Studies

All cases use the hypothetical reference BEV of Table 4.3, and their results come from backward-looking simulation with constant efficiencies, $\kappa = 1$ and a 50 kW charge limit. They are for teaching and are not predictions for any commercial vehicle.

Example 21: Case Study 1: Driving range from usable energy and consumption

Problem. Find the range on WLTC of the reference BEV, new and at 80% SoH, and the remaining range with 42% SoC displayed on a window of $z_{\min} = 0.05$.

Given. $E_{\text{nom}} = 50.46$ kWh, window 0.90, $e = 144.3$ Wh/km.

Required. D_R and D_{rem}.

Governing equations. (4.124), (4.130), (4.131).

Solution. New: $E_u = 0.9(50.46) = 45.41$ kWh; $D_R = 45\,410/144.3 = 315$ km.
At SoH = 0.8: $E_u = 36.33$ kWh; $D_R = 252$ km.
Remaining: $(0.42 - 0.05)(50.46) = 18.67$ kWh; $D_{\text{rem}} = 129$ km.

Result. 315 km new, 252 km aged, 129 km remaining.

Physical interpretation. Range is directly proportional to usable energy.

Engineering significance. The battery's end-of-life state must meet the warranty range, and the initial pack is often oversized for this reason. In reverse, a 300 km target at 144.3 Wh/km needs $E_u = 43.3$ kWh and a nominal 48.1 kWh at a 90% window.

Example 22: Case Study 2: Energy consumption using a drive cycle

Problem. Simulate the reference BEV over the WLTC and report an energy balance at each interface. A numerical example of a short cycle is already in Example 3.

Given. WLTC Class 3b, $D = 23.27$ km, $T = 1800$ s, $P_{\text{aux}} = 600$ W, $z_0 = 0.9$.

Required. Energy balance, e, SoC change.

Governing equations. (4.126)–(4.129).

Solution. Integrating sample by sample gives:

Quantity	kWh
Positive wheel energy, $\int_{P_w>0} P_w\,dt$	3.341
Braking energy at wheels, $\int_{P_w<0} \lvert P_w\rvert\,dt$	0.882
Friction brake energy (limits and fade)	0.013
Positive DC energy (traction)	3.818
Regenerative DC energy returned	0.760
Auxiliary energy, $600 \times 1800/3600$	0.300
Battery discharge energy, E_{dis}	4.050
Battery regenerated energy, E_{reg}	0.692
Net battery energy, E_{net}	3.358

Check: $3.818 - 0.760 + 0.300 = 3.358$ kWh. In E_{dis} and E_{reg} the auxiliary load is netted sample by sample, which explains why E_{reg} is smaller than the DC regenerative energy.
Consumption: $e = 3358/23.27 = 144.3$ Wh/km, or 14.4 kWh/100 km.
SoC falls by 6.27%.

Result. $E_{\text{net}} = 3.358$ kWh; $e = 144.3$ Wh/km.

Physical interpretation. $3.341/3.818 = 0.875$ confirms the traction efficiency assumed. Braking energy at the wheels equals 26% of the traction energy.

Engineering significance. A balance at each interface like this is the best verification of a model, as it shows at once if a loss is missing or double counted. The net energy is the correct figure for range: the separate discharge and regenerated energies are not added or used alone.

Example 23: Case Study 3: Range under urban and highway driving

Problem. Compare consumption and range across the cycles.

Given. $E_u = 45.4$ kWh, $P_{\text{aux}} = 600$ W.

Required. e and D_R for each cycle.

Governing equations. (4.129), (4.130).

Solution. Simulation gives the following:

Cycle	e (Wh/km)	D_R (km)	No-regen e (Wh/km)	Recovered (% of E_{dis})
MIDC	122.1	372	147.7	15.9
UDDS	118.6	383	161.8	24.9
FTP-75	121.2	375	163.5	24.2
WLTC	144.3	315	177.0	17.1
HWFET	131.9	344	141.4	6.3
US06	175.0	260	218.5	19.3

Result. Urban range (UDDS) is 383 km, highway range (HWFET) 344 km and aggressive range (US06) 260 km.

Physical interpretation. The EV is better on the urban cycle, since regeneration recovers a quarter of the energy and drag is low. US06 combines high speed with hard acceleration.

Engineering significance. A single quoted range is meaningless without its cycle. Figure 4.131 shows that the energy use spread is nearly 50% across these cycles for one vehicle.

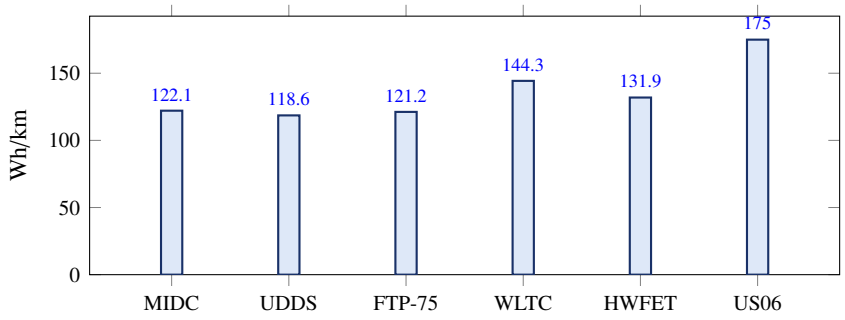


Figure 4.131: Energy consumption under different drive cycles (simulated, hypothetical BEV).

Example 24: Case Study 4: Effect of regenerative braking on net consumption

Problem. Quantify the benefit of regeneration on UDDS and on HWFET, then derive the range gains.

Given. Results in Example 23. $E_u = 45.4$ kWh.

Required. Change in consumption and range.

Governing equations. (4.132), (4.130).

Solution. UDDS: e falls from 161.8 to 118.6 Wh/km, a saving of 26.7%; range rises from 281 to 383 km, a gain of 36%.
HWFET: e falls from 141.4 to 131.9 Wh/km, a saving of 6.7%; range rises from 321 to 344 km, a gain of 7%.

Result. Regeneration adds 102 km on UDDS but only 23 km on HWFET.

Physical interpretation. On a highway cycle, there is little braking energy to recover. Most deceleration is achieved by drag.

Engineering significance. Figure 4.132 shows the comparison for all the cycles. The energy saved here assumes $\kappa = 1$, the best case; with a real blending strategy and limits the benefit is lower.

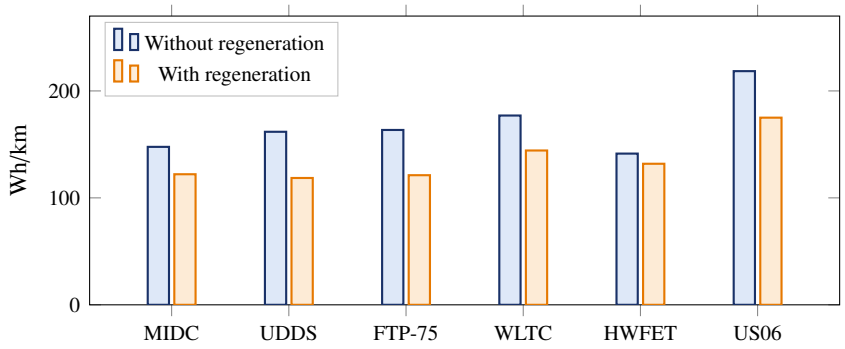


Figure 4.132: Regenerative energy recovery comparison (simulated, hypothetical BEV).

Example 25: Case Study 5: Effect of auxiliary loads on driving range

Problem. Find the range of the reference BEV on UDDS and HWFET as auxiliary power rises from 0 to 5 kW, and estimate a cold-weather WLTC range with $P_{\text{aux}} = 3$ kW and 10% less usable energy.

Given. $E_u = 45.4$ kWh; $v_{\text{avg}} = 31.5$ km/h (UDDS), 77.7 km/h (HWFET).

Required. D_R versus P_{aux}.

Governing equations. (4.133), (4.130).

Solution. At 3 kW: UDDS extra energy compared with 0.6 kW is $2400/31.5 = 76.2$ Wh/km, so $e = 194.8$ Wh/km, and $D_R = 233$ km. HWFET: $2400/77.7 = 30.9$ Wh/km, $e = 162.8$ Wh/km, $D_R = 279$ km.

P_{aux} (kW)	0	0.5	1	2	3	5
UDDS range (km)	456	393	346	279	233	176
HWFET range (km)	366	348	331	303	279	241

Cold WLTC: $e = 195.9$ Wh/km and $E_u = 40.9$ kWh, so $D_R = 209$ km, a drop of 34% from 315 km.

Result. A 3 kW auxiliary load cuts UDDS range to 233 km and HWFET range to 279 km.

Physical interpretation. Above about 1.3 kW the ordering reverses, and highway range exceeds urban range. An auxiliary load draws power for the whole trip, and a slow urban trip takes much longer per km.

Engineering significance. Figure 4.133 shows that the range loss is steepest in slow traffic, which explains why heat pumps and seat heating are so valuable in EVs, and why winter range falls most in the city. The cold figure here is a teaching estimate, not a statement about any real vehicle.

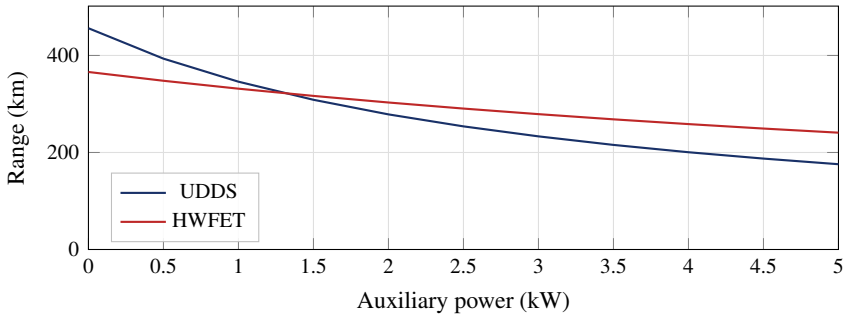


Figure 4.133: Effect of auxiliary loads on driving range (simulated, hypothetical BEV).

Bibliography

[1] D. W. Gao, C. Mi, and A. Emadi, "Modeling and simulation of electric and hybrid vehicles," *Proc. IEEE*, vol. 95, no. 4, pp. 729–745, Apr. 2007.

[2] L. Guzzella and A. Sciarretta, *Vehicle Propulsion Systems: Introduction to Modeling and Optimization*, 3rd ed. Berlin, Germany: Springer, 2013.

[3] M. Ehsani, Y. Gao, S. Longo, and K. M. Ebrahimi, *Modern Electric, Hybrid Electric, and Fuel Cell Vehicles*, 3rd ed. Boca Raton, FL, USA: CRC Press, 2018.

[4] J. Larminie and J. Lowry, *Electric Vehicle Technology Explained*, 2nd ed. Chichester, U.K.: Wiley, 2012.

[5] I. Husain, *Electric and Hybrid Vehicles: Design Fundamentals*, 3rd ed. Boca Raton, FL, USA: CRC Press, 2021.

[6] J. M. Miller, *Propulsion Systems for Hybrid Vehicles*, 2nd ed. London, U.K.: IET, 2010.

[7] T. D. Gillespie, *Fundamentals of Vehicle Dynamics*. Warrendale, PA, USA: SAE International, 1992.

[8] R. Rajamani, *Vehicle Dynamics and Control*, 2nd ed. New York, NY, USA: Springer, 2012.

[9] J. Y. Wong, *Theory of Ground Vehicles*, 4th ed. Hoboken, NJ, USA: Wiley, 2008.

[10] H. B. Pacejka, *Tire and Vehicle Dynamics*, 3rd ed. Oxford, U.K.: Butterworth-Heinemann, 2012.

[11] K. B. Wipke, M. R. Cuddy, and S. D. Burch, "ADVISOR 2.1: A user-friendly advanced powertrain simulation using a combined backward/forward approach," *IEEE Trans. Veh. Technol.*, vol. 48, no. 6, pp. 1751–1761, Nov. 1999.

[12] T. Markel *et al.*, "ADVISOR: A systems analysis tool for advanced vehicle modeling," *J. Power Sources*, vol. 110, no. 2, pp. 255–266, Aug. 2002.

[13] National Renewable Energy Laboratory, *FASTSim: Future Automotive Systems Technology Simulator*. Golden, CO, USA. [Online]. Available: https://www.nrel.gov/transportation/fastsim.html

[14] X. Hu, S. Li, and H. Peng, "A comparative study of equivalent circuit models for Li-ion batteries," *J. Power Sources*, vol. 198, pp. 359–367, Jan. 2012.

[15] G. L. Plett, *Battery Management Systems, Volume I: Battery Modeling*. Norwood, MA, USA: Artech House, 2015.

[16] Automotive Research Association of India, "Type I test," in *Document on Test Method, Testing Equipment and Related Procedures for Testing Type Approval and Conformity of Production of Vehicles (MoRTH/CMVR/TAP-115/116)*, Part IX, ch. 3. Pune, India. [Online]. Available: https://www.araiindia.com/CMVR_TAP_Documents/Part-09/Part-09_Chapter03.pdf

[17] S. Sharma, A. Goel, R. Suresh, C. S. Lakshami, R. Mhatta, and S. Sundar, *Assessment of Emission Test Driving Cycles in India: A Case for Improving Compliance*. New Delhi, India: The Energy and Resources Institute, 2013.

[18] M. Tutuianu *et al.*, "Development of the World-wide harmonized Light duty Test Cycle (WLTC) and a possible pathway for its introduction in the European legislation," *Transp. Res. Part D, Transp. Environ.*, vol. 40, pp. 61–75, Oct. 2015.

[19] United Nations Economic Commission for Europe, *Global Technical Regulation No. 15: World-wide Harmonized Light Vehicles Test Procedure (WLTP)*. Geneva, Switzerland: UNECE.

[20] European Commission Joint Research Centre, *wltp: WLTC gear-shift calculator* (Python package with GTR cycle tables). [Online]. Available: https://pypi.org/project/wltp/

[21] U.S. Environmental Protection Agency, *Dynamometer Drive Schedules*. [Online]. Available: https://www.epa.gov/vehicle-and-fuel-emissions-testing/dynamometer-drive-schedules

[22] *Code of Federal Regulations*, Title 40, Part 86, Appendix I: Dynamometer Schedules. Washington, DC, USA.

[23] M. André, "The ARTEMIS European driving cycles for measuring car pollutant emissions," *Sci. Total Environ.*, vols. 334–335, pp. 73–84, Dec. 2004.

[24] TransportPolicy.net, *EU: Light-Duty: Artemis*. [Online]. Available: https://www.transportpolicy.net/standard/eu-light-duty-artemis/

[25] J. B. Rawlings, D. Q. Mayne, and M. M. Diehl, *Model Predictive Control: Theory, Computation, and Design*, 2nd ed. Madison, WI, USA: Nob Hill, 2017.

[26] H. Borhan, A. Vahidi, A. M. Phillips, M. L. Kuang, I. V. Kolmanovsky, and S. Di Cairano, "MPC-based energy management of a power-split hybrid electric vehicle," *IEEE Trans. Control Syst. Technol.*, vol. 20, no. 3, pp. 593–603, May 2012.

[27] E. H. Mamdani and S. Assilian, "An experiment in linguistic synthesis with a fuzzy logic controller," *Int. J. Man-Mach. Stud.*, vol. 7, no. 1, pp. 1–13, Jan. 1975.

[28] K. J. Åström and T. Hägglund, *Advanced PID Control*. Research Triangle Park, NC, USA: ISA, 2006.

[29] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed. New York, NY, USA: Springer, 2009.

[30] T. Yuksel and J. J. Michalek, "Effects of regional temperature on electric vehicle efficiency, range, and emissions in the United States," *Environ. Sci. Technol.*, vol. 49, no. 6, pp. 3974–3980, Mar. 2015.

[31] C. Fiori, K. Ahn, and H. A. Rakha, "Power-based electric vehicle energy consumption model: Model development and validation," *Appl. Energy*, vol. 168, pp. 257–268, Apr. 2016.

[32] The MathWorks, Inc., *Simulink User's Guide*. Natick, MA, USA. [Online]. Available: https://www.mathworks.com/help/simulink/

[33] The MathWorks, Inc., *Powertrain Blockset User's Guide*. Natick, MA, USA. [Online]. Available: https://www.mathworks.com/help/autoblks/

[34] The MathWorks, Inc., *Simscape User's Guide*. Natick, MA, USA. [Online]. Available: https://www.mathworks.com/help/simscape/